Lecture 01

Background of Bioinformatics:

Modern Biology:

- Humans have over 25,000 genes.
- The genes may produce > 250,000 different proteins.
- Each protein may further take up multiple modifications.
- But then, there are numerous organelles & intermediary molecules as well
- The scale and speed of biomolecular interactions, reactions & transport is simply bewildering!

Experiments in Biology:

Brisk enhancements in instrumentation:

- Next Generation Sequencers
- High Resolution Mass Spectrometers
- Nuclear Magnetic Resonance Spectroscopy

Activities:

1. Developing algorithms
2. Writing software
3. Data processing pipelines
4. Statistical evaluation of data
5. Data visualization

Moving ahead...

- Learn the available bioinformatics concepts, algorithms & tools
- Advance these concepts by developing novel ideas
- Apply these ideas to forward the understanding of biology & life.

Conclusion:

- The benefits of studying bioinformatics in Pakistan
- New degree programs in several universities
- Opportunity for cutting edge discoveries in a resource-limited setting.

Digitalization of Biology:

- Modern biological experiments produce data which is stored on computers.
- The data includes text, numbers, symbols and images.

Speed of Data Growth:

- Data is being accumulated at an exponentially increasing rate
- E.g. Genome sequences in genome databases are doubling every few years!

Conclusion:

- Humans are limited in recalling information from their memory
- So, we need the help of computers to store & recall this data!
- Computers can also quickly process this information and help in its manipulation.

END

Lecture 02

Introduction to Bioinformatics:

Motivation:

1. An interdisciplinary field
2. New but rapidly developing field
3. Low requirement on infrastructure & research equipment
4. Vast opportunities for scientific discovery.

Scope of Bioinformatics:

Use of computational algorithms and techniques for:

1. Storage,
2. Organization,
3. Analysis, and
4. Representation of biological information

Lecture 03

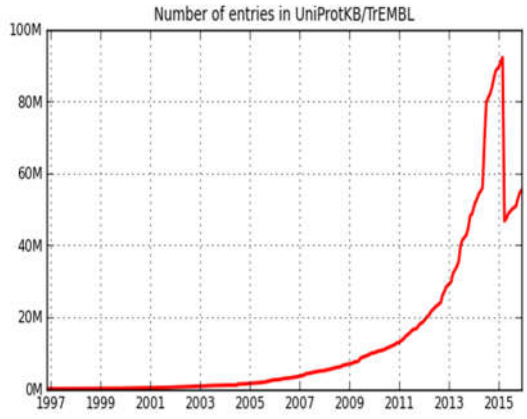
Need for Bioinformatics - I

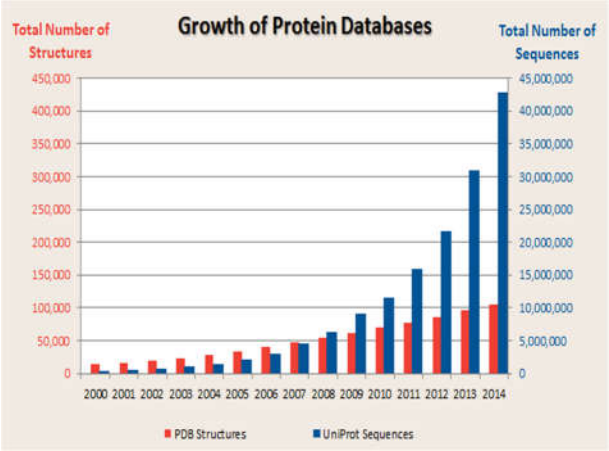
Motivation:

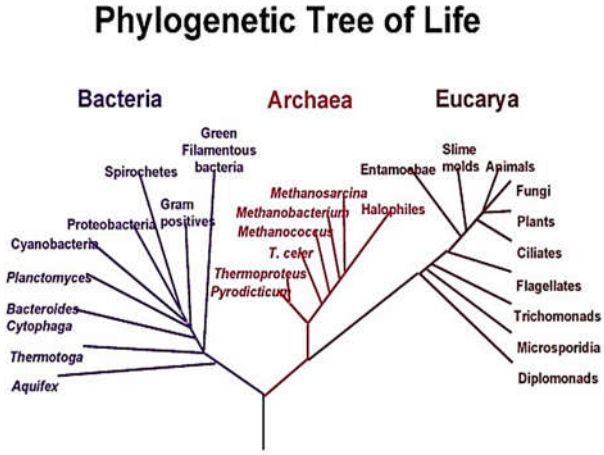
1. An interdisciplinary field
2. New but rapidly developing field
3. Low requirement on infrastructure and research equipment
4. Vast opportunities for scientific discovery

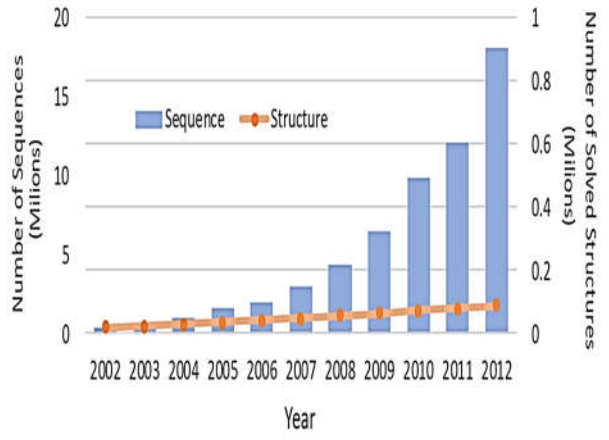
Conclusion:

- Since genome and proteome information is publically available, you can process this information after downloading it
- If you carefully search this online data, with a little bit of biology background, it is a gold mine!











Conclusion:

- From gene sequences to protein sequences Bioinformatics is the way forward.
- Protein structure, protein-protein interactions & systems biology are other research areas in bioinformatics.

Lecture 04

Need for Bioinformatics - II

Quote:

Biology easily has 500 years of exciting problems to work on.

Donald Knuth, Professor, Stanford University.

What can bioinformatics deliver?

- Life, evolution and disease can be better understood
- Drugs can be better developed after understanding molecular basis of life

Possible contributions:

- Solve important problems related to life
- Handle mammoth quantities of experimental data
- Develop efficient solutions towards these goals

Lecture 05

Applications of Bioinformatics - I

Introduction:

- **Where can bioinformatics be applied specifically?**
- **What are the avenues in biology that can benefit from this science?**
- **What benefits can it deliver to the society?**

1. Genomics:

- **DNA Sequencing**
- **Gene Finding**
- **Genome Assembly**
- **Variation in Genomes**
- **Transcription Data**
- **Databases**

2. Evolutionary Studies:
- Evolutionary relationships
- Evolutionary distances
- Phylogenetics
- Tree of life

3. Proteomics:
- Protein Sequencing
- Protein Structures
- Post-translational Modifications
- Protein-Protein Interactions
- Database Development

4. Systems Biology:
- Model protein and gene interactions
- Dynamical analysis of such models
- Understand system properties
- Predict system level behaviors

Small to Big:
- Gene finding, function prediction
- Structure prediction, modeling
- Predicting Molecular (Protein) interaction
- Biological network building & modeling
- Cell level modeling & simulation
- Cell signaling modelling & simulation
- Tissue & morphology modelling
- Biological network building & modeling
- Models integrate biological data
- Simulations help validate testable hypotheses
- Moreover, these simulations can also predict disease progression & outcomes

Conclusion: Bioinformatics not only organizes, stores and analyzes biological data, but can also validate novel hypotheses. Modern day bioinformatics also helps predict disease outcomes as well as drugs to treat them!

Conclusion:
- Genomics
- Transcriptomics
- Proteomics
- Metabolomics
- Structural Proteomics
- Drug Design
- Systems Biology
- Personalized Medicine

Lecture 07

Frontiers in Bioinformatics – I

Introduction: Bioinformatics is a relative young scientific area. It is rapidly expanding to cater for the variety and scale of biological data.
- Several unresolved challenges exist in this expansion!

Frontiers in Genomics
- Next generation sequencing of whole genomes
- Massive amounts of data (Tera byte files)
- Handling & analysis of this data
- Genome assembly & prediction

Frontiers in Transcriptomics:
- Identification of hitherto unknown types as a roles of RNA
- Understanding the regulatory dynamics of these RNA molecules
- Interactions of the RNA molecules.

Lecture 06

Applications of Bioinformatics - II

Introduction:
- Genomics
- Transcriptomics
- Proteomics
- Metabolomics
- Structural Proteomics
- Drug Design
- Systems Biology
- Personalized Medicine
- Bioinformatics techniques have enabled us to generate 'omic' data and its utilization!
- Step by step application of bioinformatics from genome level to entire biosystems.

Frontiers in Proteomics:
- Identification of low abundance proteins in patient tissue samples
- Large scale protein identification
- Identification of known as well as novel post-translational modifications.

Conclusion:
- Bioinformatics is full of challenges and opportunities.
- Amongst other frontiers in bioinformatics, there are protein structures, systems biology and personalized medicine!

Lecture 08

Frontiers in Bioinformatics - II

Frontiers in bioinformatics included:

- Next generation genomics
- Transcriptomics
- High throughput proteomics

Frontiers in Protein Structure:

- Accurate solution of protein structures is one of the toughest problems
- How to determine protein structures?
- How to predict protein structures?
- How to simulate the protein folding process?
- How to predict the protein – protein interaction?
- How to model protein-drug interaction?

Frontiers in Systems Biology:

- Cells function as a whole!
- How to integrate genes, proteins and metabolites into a unified system?
- How to simulate these models in real-time?

Frontiers in Personalized Medicine:

- Not all medicines work all the time!
- Some medicines have side effects on certain patients
- How to evaluate drugs using Bioinformatics tools?

Conclusion:

- 21st century is the century of Bioinformatics
- What will it hold for us, we can already guess that!
- An era of personalized medicines and eradication of several common diseases!

Lecture 09

Overview of Course Contents - I

Philosophy behind the Course Outlay:

1. Introduce the classical algorithms in bioinformatics
2. Link them to latest developments in the field
3. Evaluate the future applications

Chapter	Contents
Introduction	Background of Bioinformatics
	Introduce Bioinformatics
	Evaluate the need for Bioinformatics
	Study applications of Bioinformatics
	The frontiers in Bioinformatics

Chapter	Contents
Sequence Analysis	What are types of biological sequences? Where do they come from?
	How do we store sequences?
	How to visualize and plot sequences?
	How can sequences be compared/aligned?
	Various techniques for sequence alignment
	How to handle mutations?
	How to score these alignments?
	Scoring matrices

Chapter	Contents
Sequence Analysis	Global and local alignments
	Introduction to BLAST
	Introduction to FASTA
	Learn about biological databases
	Introduce Expasy, Ensemble etc.
Molecular Evolution	**Molecular evolution and phylogeny**
	Sequence Evolution
	Introduction to Unweighted Pair Group Method with Arithmetic Mean (UPGMA)
	Introduction to maximum parsimony

Chapter	Contents
RNA Secondary Structure Prediction	What are RNAs?
	What is their function and structure?
	Energy of RNA structures
	Types of RNA structures
	Representing structures
	Experimental determination of structures
RNA Secondary Structure Prediction	**Zuker's algorithm**
	Martinez algorithm
	Dynamic programming approaches
	Nussinov -Jacobson Algorithm
	Web resources for RNA structure prediction

Conclusion:

- Sequences and operations such as alignment and comparison will be covered along with phylogenetic and RNA structure modelling.
- Next up we will delve into protein sequences and structures!

Lecture 10

Overview of Course Contents - II

Summary:

- Protein sequence and structure topics will be dealt in these modules
- Next set of modules is about the homology modelling and systems biology topics!

Chapter	Contents
Protein Sequences	From DNA/RNA Sequences to Proteins
	Coding of Amino Acids
	Open Reading Frames
	Sequencing Proteins
	Application of MS in sequencing
	Bottom Up Proteomics
	Top Down Proteomics
	Protein Ionization Techniques
	MS1 and Intact Protein Mass

Chapter	Contents
Protein Sequences	Protein Sequence Database Search Algorithm
	Large Scale Proteomics
	Proteomics Data File Formats RAW and MGF
	Online Proteomics Tools Mascot, ProSight PTM
Protein Structures	**Properties of Amino Acids**
	Structural Traits of Amino Acids
	Introduction to Protein Folding
	Process of Protein Folding
	Models of Protein Folding
	Protein Structures
	Primary, Secondary, Tertiary and Quaternary Structures

Chapter	Contents
Protein Sequences	Scoring Intact Protein Mass
	Protein Fragmentation Techniques
	Tandem MS
	Measuring Experimental and Theoretical
	Fragment's Mass
	Peptide Sequence Tags
	Scoring Peptide Sequence Tags
	In silico Protein Fragmentation
	In silico Fragment Comparison and Scoring

Chapter	Contents
Protein Structures	Introduction to Protein Bond Angles
	Ramachandran Plot
	Structure Visualization
	Experimental Determination of Protein Structure
	Protein Data Bank
	Online Resources for Protein Visualization
Protein Structures	**Online Resources for Structure Comparison**
	Protein Structure Prediction
	Predicting Secondary Structures
	Introduction to Chou Fasman Algorithm

Chapter	Contents
Protein Structures	Calculating Energies of Protein Structures
	Structure Determination for Energy Calculations
	Protein Structures-Alpha Helices, Beta Sheets, Loops, Coils
	Protein Domains
	Classification Databases
	Algorithms for Structure Classification
	Protein Structure Comparison

Lecture 11
Overview of Course Contents - III

Chapter	Contents
Homology Modelling	Introduction to Homology modelling
	Need for Homology Modelling
	Seven Steps to Homology Modelling
Homology Modelling	**3D-1D Bowie Algorithm**
	Machine Learning Approaches to Structure Prediction
	Neural Networks for Structure Prediction
	PSIPRED
	Introduction to Hidden Markov Models
	Ab initio modelling

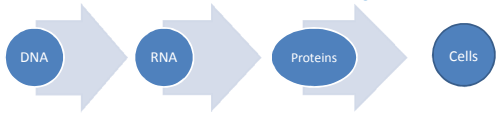
Conclusion:

- **These contents will give you an initial exposure to the variety of topics in bioinformatics**
- **After covering these topics, you should have a basic conceptual foundation for further studies into Bioinformatics!**

Chapter	Contents
Homology Modelling	Hinds and Levitt Algorithm
	Computational Assessment of Structure Prediction (CASP)
	Online Tools for Homology Modelling
Systems Biology	**Putting Proteins into Action inside a Cell**
	Divergent Spatiotemporal Biological Data
	Introduction to systems biology
	"Hallmarks of Cancer" as System Level Properties
	Integrative Biomolecular Approach
	Introduction to Bio-molecular Networks
	Networks as Graphs
	Properties & Types of Graphs
	Adjacency Descriptors of Graphs

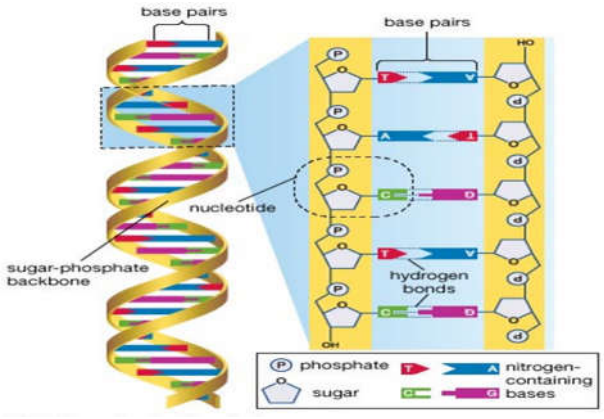
Lecture 12
Gene, mRNA and Protein Sequences

- All living things are composed of cells
- How are cells made?
- DNA provides the blueprint for building cells
- DNA dictates the production of cell's proteins, carbohydrates & vitamins
- The transformation of information in DNA into these molecules is termed the Central Dogma
- Central dogma: DNA -> RNA -> Proteins
- Proteins are then used in constructing cells

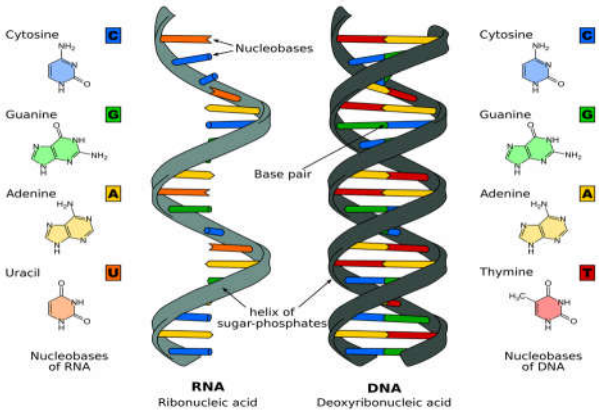


Chapter	Contents
Systems Biology	Topological Descriptors of Graphs
	Network Motifs
	Types of regulationsin Biological Networks
	Dynamic Behaviour of Networks
	Constructing Networks from Experiemental Data
Systems Biology	**Analysis of network dynamics**
	Iterative Approaches & Parameter Estimation
	Sensitivity Analysis
	Multi-scale modelling in biology
	Integrating cross scale data
	Continuous and discrete variables

DNA



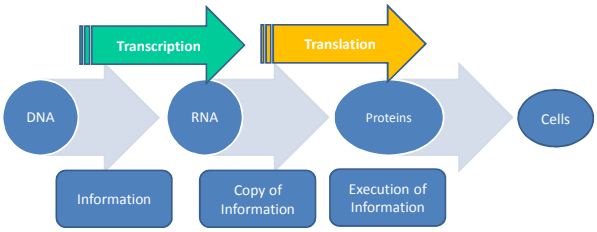
DNA -> RNA



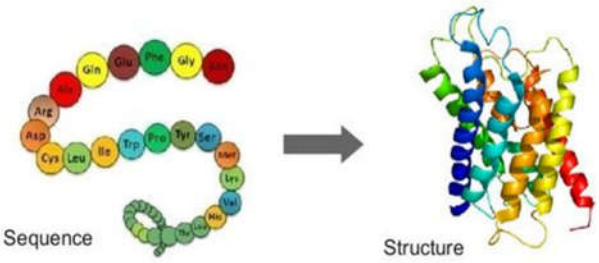
Lecture 13

Transcription:

- Cells are built up of proteins and carbohydrates
- These molecules are produced after transformation of DNA into proteins.
- What are the salient steps underlining this transformation?



RNA -> Proteins



Conclusion:

- DNA codes for an RNA
- RNA in turn helps produce Proteins
- Proteins along with some other molecules form cells including their organelles and membrane

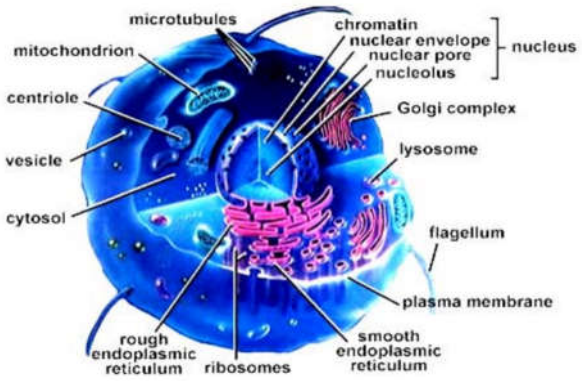
Transcription:

- **DNA is constituted by four bases**
- **These are Adenine (A), Cytosine (C), Thymine (T), and Guanine (G)**
- **DNA molecule may contain thousands of A, C, T & G.**
- **A pairs with T only and C pairs with G**

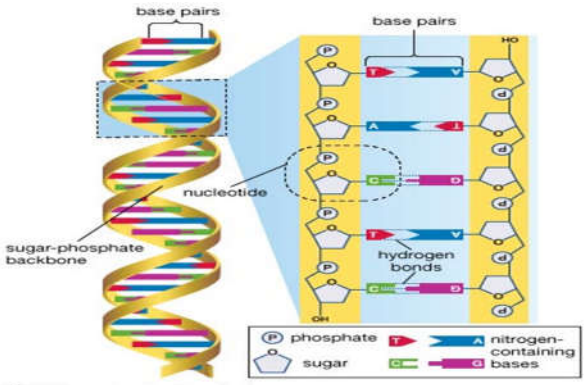
Transcription:

- **Process of transcription decodes these bases in the DNA**
- **Based on the information received after decoding DNA, an RNA molecule is encoded.**

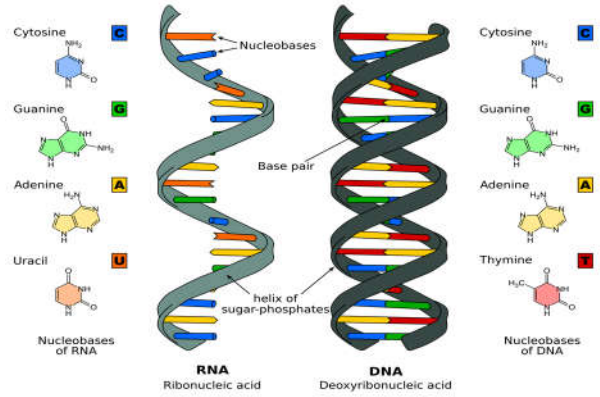
Proteins -> Cells



DNA bases, double strands & complementarity

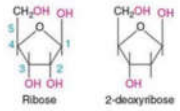


DNA -> RNA

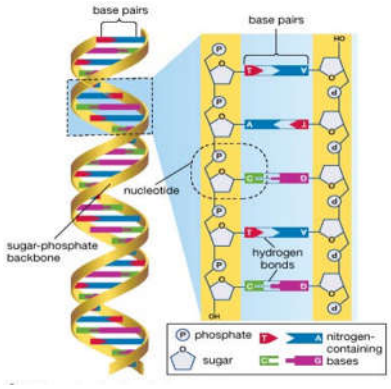


Base complementarity

In RNA In DNA



Sugars



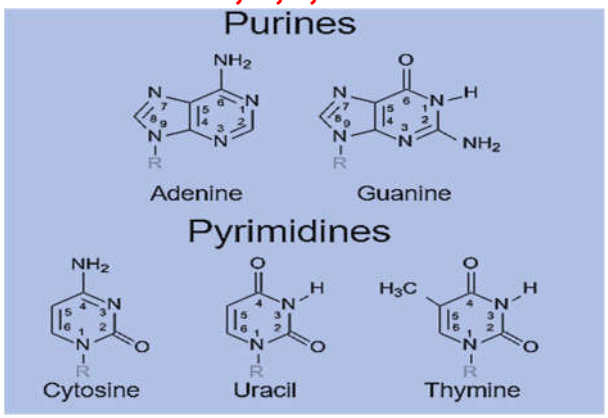
Transcription

- RNA is constituted by four bases
- These are Adenine (A), Cytosine (C), Uracil (U), and Guanine (G)
- RNA molecule may contain thousands of A, C, U and G.
- A pairs with U only and C pairs with G

Conclusion

- DNA gives rise to RNA
- DNA has A, T, C & G while RNA has A, U, G & C
- DNA is double stranded while RNA is single stranded
- RNA helps produce proteins

Structures of A, C, T, U and G



Lecture 14

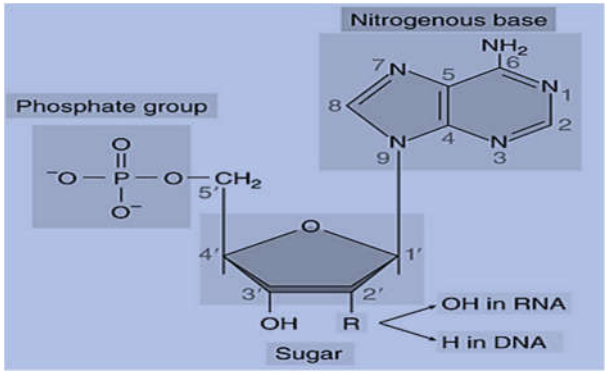
Nucleotides:

- DNA and RNA molecules are built up from four types of molecules, each
- These molecules are called nucleotides
- These are Adenine (A), Cytosine (C), Thymine (T), Uracil (U) and Guanine (G)

Conclusion:

- DNA gives rise to RNA
- DNA has A, T, C & G while RNA has A, U, G & C
- DNA is double stranded while RNA is single stranded
- RNA helps produce proteins

Phosphate, Sugar and Nitrogenous bases come together



Lecture 15

Translation:

- Cells are built up from proteins and carbohydrates
- These are produced as a result of transformation of RNA molecules
- The transformation is called translation
- Translation takes place at the ribosomes
- Ribosome reads an RNA transcript & "collects" amino acids according to that from the cell's cytosol

Mechanism

- Three nucleotides are read at a time from the RNA
- Such sets of three nucleotides are called codons
- Each codon corresponds to a specific amino acid

Amino Acid	3-Letters	1-Letter
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic Acid	Asp	D
Cysteine	Cys	C
Glutamic Acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

Second letter

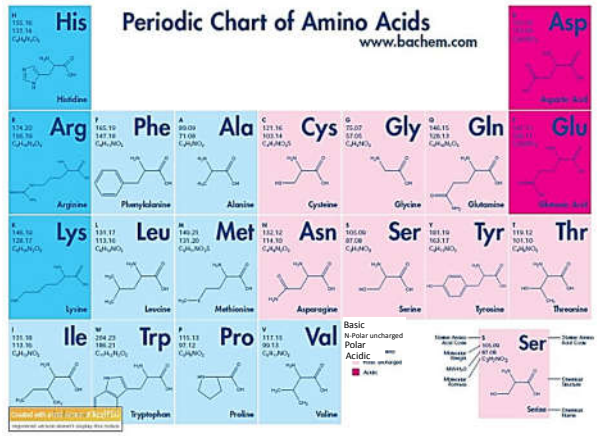
First letter	U	C	A	G	Third letter
U	UUU, UUC Phe; UUA, UUG Leu	UCU, UCC, UCA, UCG Ser	UAU, UAC Tyr; UAA Stop; UAG Stop	UGU, UGC Cys; UGA Stop; UGG Trp	U C A G
C	CUU, CUC, CUA, CUG Leu	CCU, CCC, CCA, CCG Pro	CAU, CAC His; CAA, CAG Gln	CGU, CGC, CGA, CGG Arg	U C A G
A	AUU, AUC, AUA Ile; AUG Met	ACU, ACC, ACA, ACG Thr	AAU, AAC Asn; AAA, AAG Lys	AGU, AGC Ser; AGA, AGG Arg	U C A G
G	GUU, GUC, GUA, GUG Val	GCU, GCC, GCA, GCG Ala	GAU, GAC Asp; GAA, GAG Glu	GGU, GGC, GGA, GGG Gly	U C A G

Conclusion: RNAs code for proteins. Codons of three nucleotides from the RNA molecules code one amino acid at a time. This process takes place at the ribosomes and is called translation.

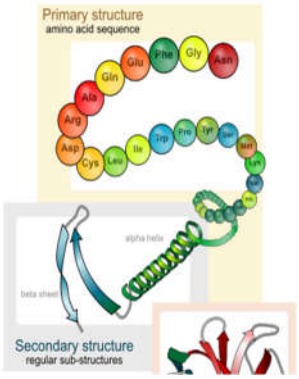
Lecture 16

Amino Acids:

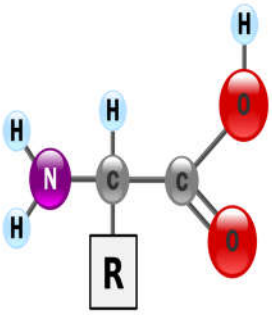
- Translation involves coding of proteins by RNAs at ribosomes.
- Codons of three nucleotides from the RNA molecules code one amino acid at a time.
- In all, 20 different amino acids can be coded by such codons during translation.
- The amino acids polymerize to form proteins.
- Next, we will study amino acids and the respective codons!



Polymerization of Amino Acids



Chemical Structure of an Amino Acid

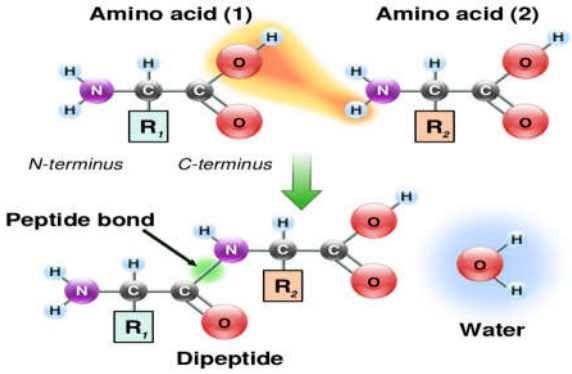


Lecture 17

Storage of Biological Sequence Information:

- DNA sequences contain A, C, T & G
- mRNA sequences contain A, C, U & G
- Protein sequences contain A, R, N, D, C, E, Q, G, H, I, L, K, M, F, P, S, T, W, Y & P
- 20 Amino Acids.
- DNA, mRNA & Proteins can be sequenced in the laboratory
- We can therefore obtain DNA, mRNA and Protein sequences which are simple lists of alphabets
- DNA sequences may contain 100 of 1000 of bases
- RNA sequences can be a little less than the DNA but they are in a much larger variety.

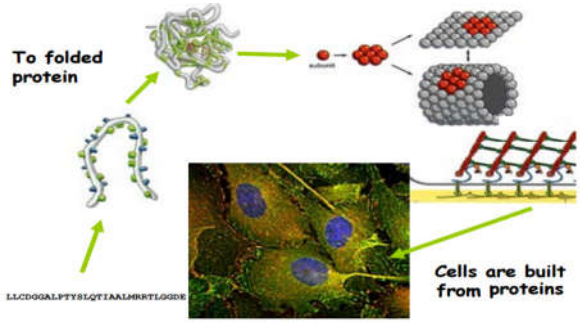
Polymerization of Amino Acids by Peptide Bond



Amino Acid	3-Letters	1-Letter
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic Acid	Asp	D
Cysteine	Cys	C
Glutamic Acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

Storing Sequences:

- Proteins have long sequences & are large in number
- A problem then arises on how to store this information?

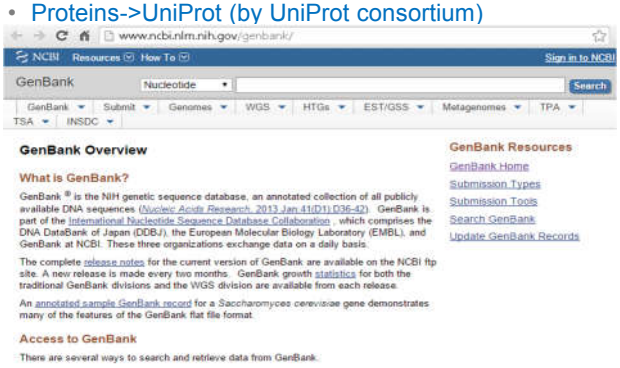


Conclusion:

- Proteins are formed after polymerization of amino acids
- Amino acids are chained together by peptide bonds
- Proteins thus formed fold to take 3D conformations!

Solution: Databases

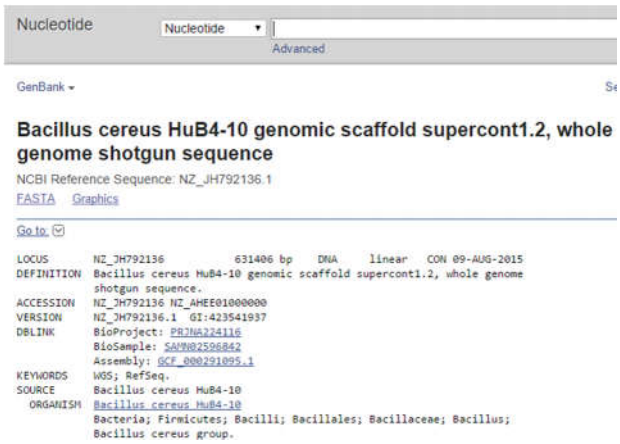
- The solution lies with public sequence databases
- DNA/RNA-> GenBank (by NIH)
- Proteins->UniProt (by UniProt consortium)





Conclusion:

- Nucleotide and amino acid sequences are stored in online databases
- DNA & RNA information is available in GenBank
- Protein sequence information is available in UniProt

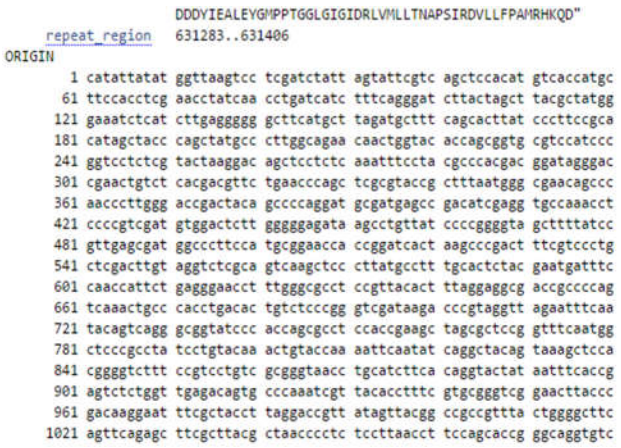


Lecture 18

Using GenBank:

- Public databases contain sequence information
- DNA/RNA-> GenBank (by NIH)
- Proteins->UniProt (by UniProt consortium)

- Searching sequence for Immunoglobulin
- Glycoprotein antibodies produced by WBC plasma cells
- Immune response to destroy antigens such as bacteria or viruses.



ncbi.nlm.nih.gov/genbank



Conclusion

GenBank can be searched by:

- **Sequence**
- **ID**
- **Name**
- **Species etc.**

Other information includes:

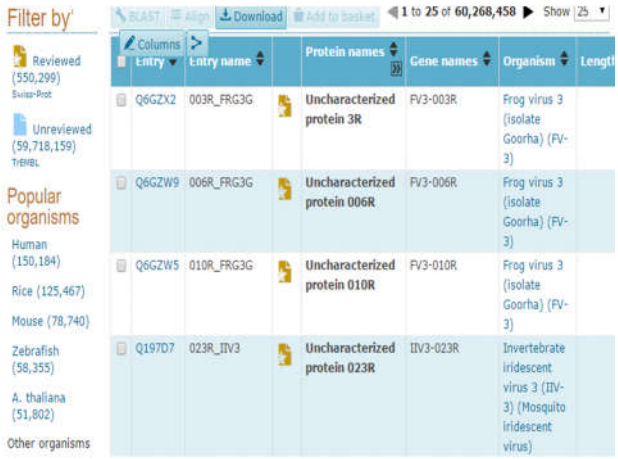
- **Locus**
- **Accession number**
- **Authors**
- **Journal etc.**

Lecture 19

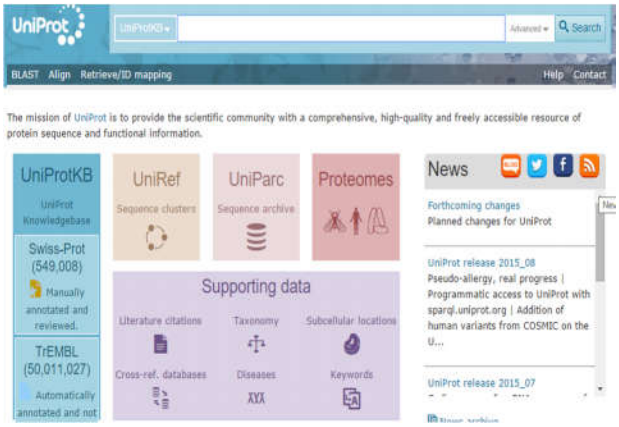
Using UniProt:

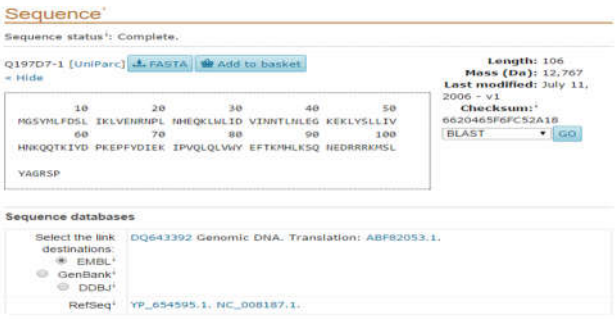
- Public sequence databases contain experimental biological sequence information
- DNA/RNA-> GenBank (by NIH)
- Proteins->UniProt (by UniProt consortium)

- Searching sequence for Ubiquitin
- An important protein for recycling
- Critical for enabling cells to recycle proteins within the cell's cytosol.



Uniprot.org





Conclusion:

UniProt can be searched by:

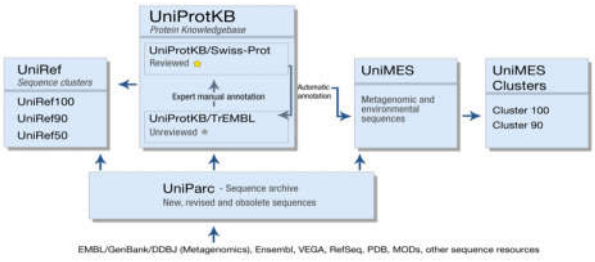
- Amino Acid Sequence , ID , Name , Species etc.

Other information includes:

- Cross reference to GenBank etc

Swiss-Prot contains human curated protein information and annotations

- **Accession number, unique identifier**
- **Sequence, Molecular mass**
- **Observed and predicted modifications**



Lecture 20

Comparing Sequences:

- A large number of sequences are available in GenBank and Uniprot, publically
- What if we try to compare them?
- What will be get from such a comparison?

By comparing sequences, we can get:

- Similarity
- Specific Differences
- Relationship
- Evolutionary Insight

DNA ACTGGTCAGTCATGTCATGTCA

RNA CAGUACGUAGCUAGGAGUAGC

Protein MQVKLFTLPHLKIHQLFTLPHLKIHDH

Conclusion:

Several conclusions can be drawn on a set of sequences in terms of:

- **Similarity**
- **Specific Differences**
- **Relationship**
- **Evolutionary Insight**

- Introduction to Computational Biology: An Evolutionary Approach By Bernhard Haubold, Thomas Wiehe.
- Mitogen-activated protein kinases (MAPK) are protein kinases that are specific to the amino acids serine, threonine, and tyrosine. MAPKs belong to the CMGC (CDK/MAPK/GSK3/CLK) kinase group.
- MAPKs are involved in directing cellular responses to a diverse array of stimuli, such as mitogens, osmotic stress, heat shock & proinflammatory cytokines. They regulate cell functions including proliferation, gene expression, differentiation, mitosis, cell survival, & apoptosis.[1]

Conclusion:

- Exact Matches include complete matching in terms of residues and their order
- In exact matching includes partial matching in terms of residues and order with room for variations in both.

Lecture 21

Similarities & Differences in Sequences

By comparing two sequences, one can study:

- Similarity
- Specific Differences
- Relationship
- Evolutionary History

Protein

M Q V K L F T ….

1. **Similarity**
2. **Specific Differences**
3. **Relationship**
4. **Evolutionary History**

Exact Matching:

Not only should the two or more sequences being compared have the exact same number of each nucleotide (for DNA /RNA) or amino acid (for Proteins), but they should be arranged in the exact same order!

Accommodating Differences – In exact Matching:

- Regular Expression
- Signature Sequences
- PROSITE Patterns Database

PROSITE Pattern for MAP Kinase

F-x(10)-R-E-x(72,86)-R-D-x-K-x(9)-C

- Hyphens separate the elements of the pattern,
- Letters refer to amino acids,
- X indicates any amino acid,
- Bracketed numbers denote the repeat length of a residue,
- If this repeat length varies, the range of this variation is quoted in the brackets.

Lecture 22

Pairwise Sequence Alignment - I

- Exact matches include complete matching in terms of residues and their order
- In exact matching includes partial matching in terms of residues and order with room for variations in both.

PROSITE Pattern for MAP Kinase

F-x(10)-R-E-x(72,86)-R-D-x-K-x(9)-C

- Hyphens separate the elements of the pattern,
- Letters refer to amino acids,
- X indicates any amino acid,
- Bracketed No. denote the repeat length of a residue,
- If this repeat length varies, the range of this variation is quoted in the brackets.

- Concept of conserved residues or nucleotides
- Before starting to matching nucleotide or amino acid sequences, we need to find location of conserved residues or nucleotides.
- The process of inexact matching while keeping in view the conserved residues is called sequence “Alignment”
- Pairwise sequence alignment is therefore alignment of a pair of two sequences.

Conclusion:

- Pairwise sequence alignment helps compare two sequences
- It takes into consideration inexact matching
- Matches are colored and missing nucleotides are denoted by ‘.’

Two Examples of Pairwise Alignment





Global alignment - maximizes the number of matches between the query and source sequences along the entire length of both the sequences.
Local alignment - gives the highest scoring local match between both query and sequences.
Optimal alignment - one that exhibits the most correspondences between the query and the source sequences. It is the alignment with the **highest score**. Biologically meaningful?

Conclusion:
- Gaps are inserted in a sequence being aligned to accommodate missing amino acids or nucleotides
- Global and Local alignment is used depending on the required type of pairwise alignment.

Lecture 23

Pairwise Sequence Alignment - II

Pairwise sequence alignment compares two amino acid or nucleotide sequences. It employs inexact matching.
- Matches are colored & missing nucleotides are denoted by '.'

Salient Points
- Sliding the sequences past each other
- Aim is to maximize matches
- Gaps ('.') could be inserted to account for insertions & deletions
- Gaps ('.') may carry a penalty for reducing scores from unreasonable alignments
- Without a gap penalty an exact match and match with gaps will get equal scores

Types of pairwise alignments

Global	Local
• Maximizing sequence matches over the entire length of two sequences, by introducing gaps.	• Find regions between two sequences that have the strongest similarity, excluding less similar regions.
• Used to determine overall similarity, conservation.	• Finding similar domains, motifs, detecting distant homology

Lecture 24

Pairwise Sequence Alignment - III

- Given a pair of sequences (strings of characters)
- Pairwise alignment tells the similarity between the sequences by maximizing the matches.
- Sequences can differ from each other in 3 different ways:
  - Substitutions ACGA → AGGA
  - Insertions ACGA → ACCGA
  - Deletions ACGA → AGA
- No change? ***match***
- Insertions or deletions ("indels") result in ***gaps*** in alignments
- Substitutions result in ***mismatches***

Optimal Alignment

Align 1 & 2

1: This Is A Rather Longer Sentence Than The Next.
2: This Is A Short Sentence.

```
1: THIS IS A RATHER LONGER SENTENCE THAN THE NEXT.
2: THIS IS A  ..........SHORT.....SENTENCE.....................

OR

1: THIS IS A RATHER LONGER SENTENCE THAN THE NEXT.
2: THIS IS A ....SHORT.....SENT..EN.....CE.....................
```

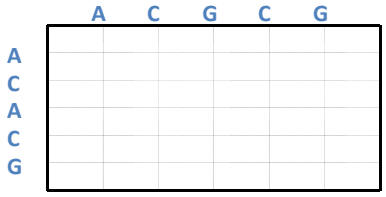
Which alignment is optimal in biological context?

Conclusion:
- Global and local pairwise alignments are two types of alignment and they can lead to different results.
- Indels lead to Gaps in alignments.
- Substitutions lead to mismatches.

Lecture 25

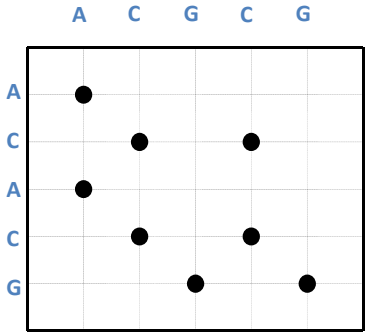
Dot Plots:

- Global and local pairwise alignments are two types of alignment and they can lead to different results.
- How can we visualize sequence alignments?
- Dot plots employ dot matrix representation for pairwise alignment and comparison.
- Sequences are written on top & left side of a dot matrix grid.



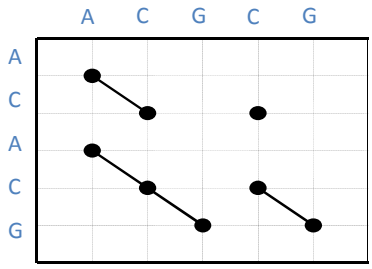
Representing Matches:

- For each nucleotide/amino acid which matches any counterpart nucleotide/amino acid, a dot is placed in the grid position across each row, one at a time!



Representing Matches:

- Similar regions are represented by a diagonal row of dots
- Non-diagonal broken dots are random matches



Conclusion

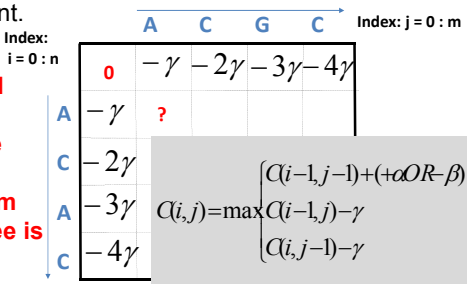
- Dot plots employ dot matrix with two sequences plotted on top & left of the matrix
- Matches are represented by dots
- Dots on diagonals are connected and represent alignments.

Lecture 26

Needleman Wunsh Algorithm - III

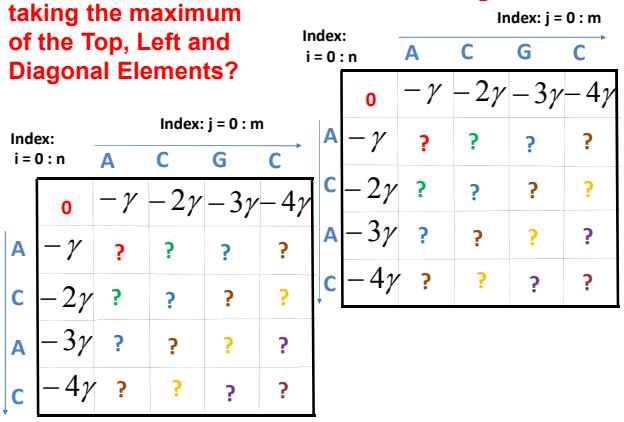
- Top, Left and Diagonal elements are considered to calculate an element in the matrix.
- The matrix is computed progressively until the bottom right element.

Left, Top and Diagonal elements are considered. The maximum from the three is selected!



What are we achieving by taking the maximum of the Top, Left and Diagonal Elements?

Order of filling the matrix



Conclusion:

- We are trying to use the best score from Top, Left and Diagonal elements.
- This strategy will be very useful in selecting the best combination of alignments!

Lecture 27

Identity vs. Similarity

- How to compare two biological sequences?
- How to measure the degree of match b/w sequences?
- Two concepts:
  - Identity
  - Similarity

Sequence Identity:

"Identity is the number of nucleotides/amino acids which match exactly b/w 2 biological sequences"

1: CATGCTT
2: CATGC

Calculate the identity between sequences 1 & 2.
Number of matches = 5
Smaller Length: Length (1) = 7, Length (2) = 5
Identity = Num. of matches/Smaller Legth * 100%=100%

- Gaps are not counted
- Identity measurement is made on the shorter of the two sequences

1: CA T GC T . C
2: CA . G . TG C

Sequence Similarity:
"Result of matching and transforming one sequence to the other by finding the smallest set of edit operations (insertions, deletions, and substitutions)."

Can be calculated by pairwise sequence alignment.

For computing similarity b/w sequences, you will need to 1st align the 2 sequences using pairwise sequence alignment!

Conclusion:

- Identity is the count of exact matches between two sequences.
- Gaps are excluded
- Similarity is the comparison between sequences calculated by using alignment approach.

Lecture 28

Introduction to Alignment Approaches

- Sequences vary due to insertions, deletions and substitutions.
- To calculate the similarity between two sequences, we need to align the sequences.
- What are the approaches to aligning sequences
- Two approaches are:
  - Global Alignment
  - Local Alignment

Local Alignment
"Match one sequence with a portion of 2nd sequence"

```
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
        |||| |||||| ||||||||||||||||
     5' TACTCACGGATGAGGTACTTTAGAGGC 3'
```

Global Alignment
"An end to end match of one sequence with the 2nd sequence"

```
5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'
   ||||||||||    |||||||  |||||||||||||| |||||||
5' ACTACTAGATT----ACGGATC--GTACTTTAGAGGCTAGCAACCA 3'
```

Conclusion:

- Local alignment focuses on the highly matching sub regions within the sequences.
- Global alignment creates an end-to-end matching between two sequences.

Lecture 29

Why Local Alignments?

- Sequences vary due to insertions, deletions & substitutions.
- To calculate the similarity between two sequences, we need to align the sequences.
- Local alignments have the power to detect small regions of high similarity between two sequences.
- Such matches may be "domains" or "motifs" in case of proteins.

Domain Shuffling:

- Aligned portions of sequence can be considered in varying orders & the process is called domain shuffling.

Domain Shuffling



Other Advantages:

- Different length sequences can be compared
- Conserved domains can be elicited from proteins. belonging to different families.
- Common functional features can thus be identified.

Conclusion:

- Local alignment focuses on the highly matching sub regions within the sequences.
- Common functional features can be compared b/w proteins

Lecture 30

Aligning Insertions & Deletions:

Insertions are:

- Addition of amino acids into protein sequences
- Addition of nucleotides into DNA or RNA sequences

Deletions are:

- Removal of amino acids from protein sequences
- Removal of nucleotides from DNA or RNA sequences

Aligning Insertions:

"Insert gaps in template(2) at positions where nucleotides/amino acids are inserted in target(1) "

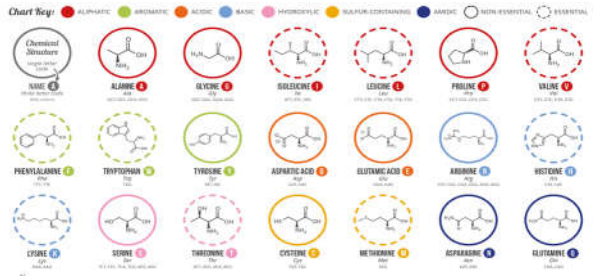
1: A C T G A C T G
2: A C G A C T G

1: A C T G A C T G
2: A C G A C T G

1: A C T G A C T G
2: A C . G A C T G

Scoring Mutations:

Example: Acidic AA replaced with an other acidic AA



www.compoundchem.com

Conclusion

- Indels are treated using gaps and gap penalties
- Mutations are treated with substitution penalties
- Subst. penalties vary depending on the substitution

Aligning deletions

"Insert gaps in target(2) at positions where nucleotides/amino acids are deleted from template(1)"

1: A C C T G A A
2: A C T G A A

1: A C C T G A A
2: A C T G A A

1: A C C T G A A
2: A C . T G A A

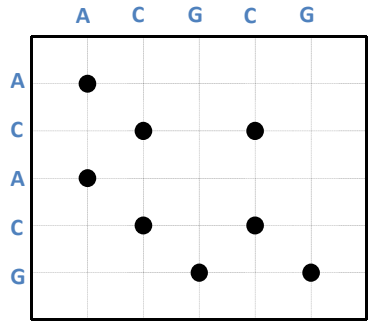
Conclusion:

- Alignment of insertions and deletions can be performed by insertion of gaps in either the template or the target sequence.
- Such an insertion is scored with a –ve score/"penalty"

Lecture 32

Introduction to Dynamic Programming

- Pairwise alignment for comparison of two sequences
- Sequences are slid across each other
- Indels and gaps
- Global versus Local alignment.
- To perform an alignment by sliding sequences across each other, we used dot plots to find matching nucleotides and amino acids.



Lecture 31

Aligning Mutations in Sequences:

- Indels were addition or removal of amino acids from protein sequences OR nucleotides from DNA or RNA sequences.
- What about Mutations?

Mutations are:

- Replacement of amino acids with certain other amino acids in proteins.
- Replacement of nucleotides with certain other nucleotides in DNA/RNA.

Aligning Mutations

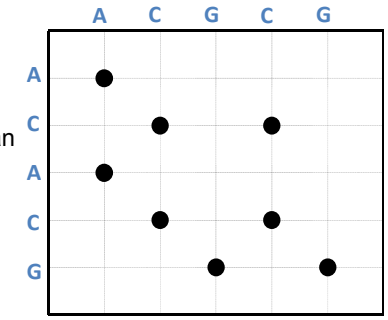
"No gap is inserted in template or target "

1: A C T G A C T
2: A C G G A C T

1: A C T G A C T
2: A C G G A C T

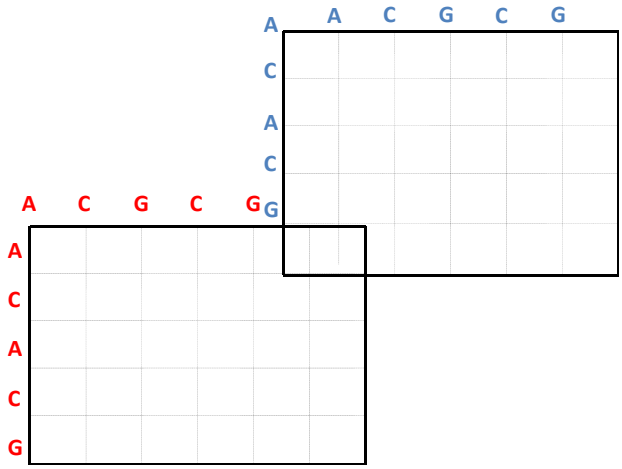
Where are the indels and gaps?

- Dot plots cannot capture insertions, deletions and gap indications. How can we deal with them?
- Modify the dot plot!



Conclusion:

- Matches are labelled by +1 instead of a dot
- Indels (gaps) and substitutions (mutations) can be included in the dot plot as -1



Lecture 33

Dynamic Programming - Essentials

- Why do we need "Dynamic Programming"?
- Comparison of two sequences is a slow and "computationally expensive" process!

Compare "A" with "B"
Compare "AB" with "BC"
Compare "ABC" with "DEF"
Compare "ABCD" with "EFGH"

Order of the algorithm:

- Calculation steps determine the time taken by an algorithm
- Comparing two sequences of length 'n' requires n^2
- Hence its order is $O(n^2)$

	A	C	G	C	G
A	1	-1	-1	-1	-1
C	-1	1	-1	1	-1
A	1	-1	-1	-1	-1
C	-1	1	-1	1	-1
G	-1	-1	1	-1	1

Conclusion:

- Sequence comparison is a costly process
- Dynamic programming helps reduce this cost
- How, we will see in the next modules!

Lecture 34

Dynamic Programming Methodology

- Dynamic programming (DP) helps reduce the large computational cost associated with sequence comparisons
- Let's learn the DP methodology

DP uses a "scoring function" to deal with matches, mismatches and gaps!

An example Scoring Scheme:

Match:	**+a**
Mismatch:	**- b**
Gap:	**- c**

Score = #matches + #mismatches + #gaps

DP uses a "scoring function" to deal with matches, mismatches and gaps!

Match Rewards:	10
Mismatch penalty:	2
Gap penalty:	5

C	T	G	T	C	G	–	C	T	G	C
-	T	G	C	–	C	G	–	T	G	-
-5	10	10	-2	-5	-2	-5	-5	10	10	-5

Total = 11

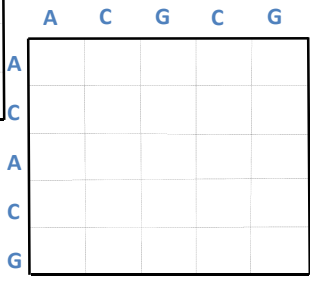
Conclusion:

- Alignments are represented by diagonals in the dot matrix plot.
- Total score can be computed for each possible alignment.
- The best alignment is then selected.

Coming back to the dot plot version

	A	C	G	C	G
A	1	-1	-1	-1	-1
C	-1	1	-1	1	-1
A	1	-1	-1	-1	-1
C	-1	1	-1	1	-1
G	-1	-1	1	-1	1

Match Rewards: 10
Mismatch penalty: 2



Lecture 35

Needleman Wunsh Algorithm - Example

- Top, Left and Diagonal elements are considered to calculate an element in the matrix.
- Match, mismatch and gap penalty is used to compute score from each position!

Conclusion:

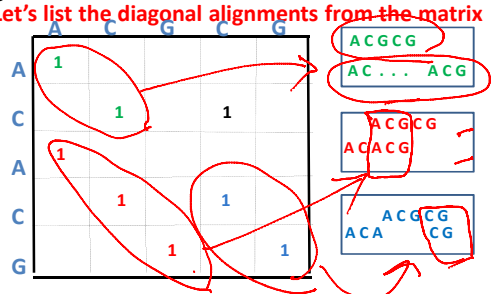
- DNA, RNA and Protein sequences can be aligned using Needleman Wunsh algorithm.
- The optimal combination of aligned regions can be extracted from the completed matrix.

Lecture 36

Needleman Wunsh Algorithm - II

- Alignments are represented by unbroken diagonals in the dot matrix plot. Numerous combinations of such alignments can be created.

Let's list the diagonal alignments from the matrix

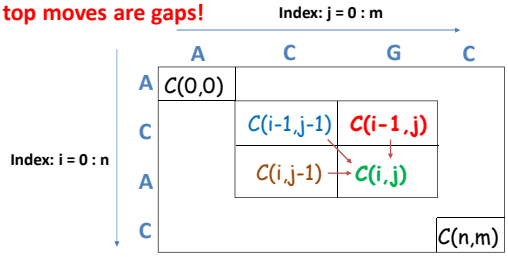


+10 for match, -2 for mismatch, -5 for space

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5						
A	-10								
T	-15								
T	-20								
C	-25								
A	-30								
C	-35								

- If the sequence is long, there can be several possible diagonal alignments
- We would like to select the best combination of these alignments
- Answer: Needleman Wunsh Algorithm

Diagonal move is a match and Left and top moves are gaps!



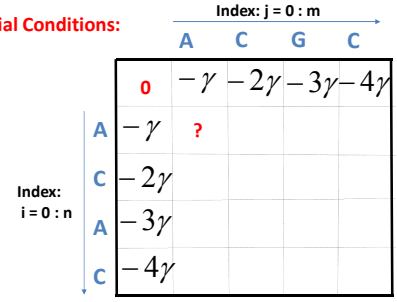
	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

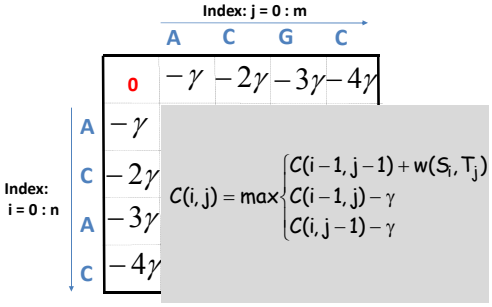
Conclusion

- Top, Left and Diagonal elements are considered to calculate an element in the matrix.
- The matrix is computed progressively until the bottom right element.

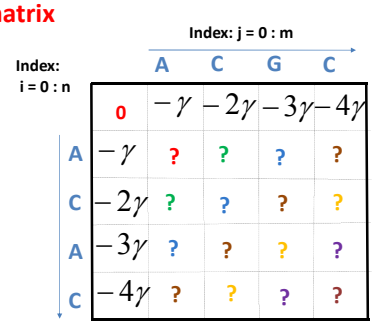
Initial Conditions: $C(i,0) = -i \cdot \gamma$ $C(0,j) = -j \cdot \gamma$



Left, Top and Diagonal elements are considered. The maximum from the three is selected!



Order of filling the matrix



Conclusion:

- We are trying to use the best score from Top, Left and Diagonal elements
- This strategy will be very useful in selecting the best combination of alignments!

Defining a few terms!

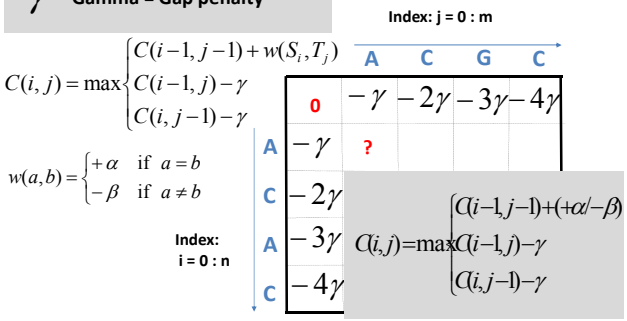
α Alpha = Match reward

β Beta = Mismatch penalty

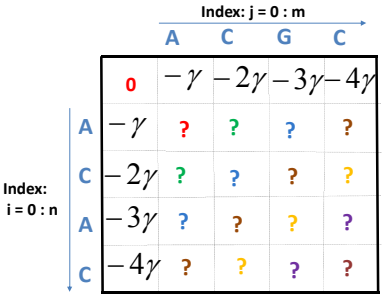
γ Gamma = Gap penalty

$$w(a,b) = \begin{cases} +\alpha & \text{if } a = b \\ -\beta & \text{if } a \neq b \end{cases}$$

Left, Top and Diagonal elements are considered. The maximum from the three is selected!



What are we achieving by taking the maximum of the Top, Left and Diagonal Elements?

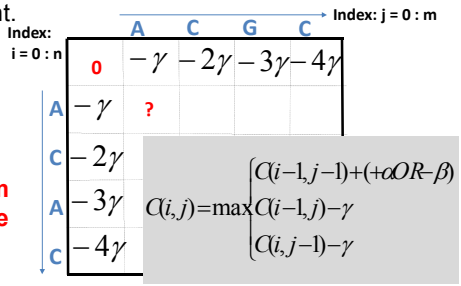


Lecture 37

Needleman Wunsh Algorithm - III

- Top, Left and Diagonal elements are considered to calculate an element in the matrix.
- The matrix is computed progressively until the bottom right element.

Left, Top and Diagonal elements are considered. The maximum from the three is selected!



Lecture 38

Needleman Wunsh Algorithm - Example

- **Top, Left and Diagonal elements are considered to calculate an element in the matrix.**
- **Match, mismatch and gap penalty is used to compute score from each position!**

Conclusion

- **DNA, RNA and Protein sequences can be aligned using Needleman Wunsh algorithm.**
- **The optimal combination of aligned regions can be extracted from the completed matrix.**

+10 for match, -2 for mismatch, -5 for space

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5						
A	-10								
T	-15								
T	-20								
C	-25								
A	-30								
C	-35								

+10 for match, -2 for mismatch, -5 for space

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5						
A	-10								
T	-15								
T	-20								
C	-25								
A	-30								
C	-35								

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

Lecture 39

Backtracking Alignments

- Top, Left and Diagonal elements are considered to calculate an element in the matrix.
- Match, mismatch and gap penalty is used to compute score from each position!
- How can we use the Needleman Wunsh algorithm to find the optimal alignment?
- The solution lies in a method called the “Traceback”
- Let’s see how a traceback works.

Conclusion: After completely calculating the matrix, we need to do a “traceback”. Traceback is such that we start from the bottom right and select the element from top, left or diagonal which led to the starting element.

Lecture 40

Revisiting Global and Local Alignments

- **We have filled the alignment matrix using Needleman Wunsh Algorithm**
- **Traceback allows us to extract the optimal alignment**
- **But, is it the local or global alignment?**

The difference?

- **Needleman Wunsh algorithm traceback begins from the bottom right element**
- **Proceeds progressively until we reach the top left**
- **This provides us with a global alignment.**

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

Revisiting Global and Local Alignments

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

Lecture 41

Overlap Matches

- Dot plots can assist in finding residues in two sequence.
- Needleman Wunsch algorithms can be used to find global alignment & score each match/mismatch & gap.
- What if a sequence "contains" an other sequence but has trailing or leading unmatched regions.
- Solution: Global alignment, but one that does not penalize leading or trailing ends.

Conclusion:

- Trace back can also be started from high score on the right column and bottom row.
- Such trace backs helps find overlap matches containing leading or trailing sequence.

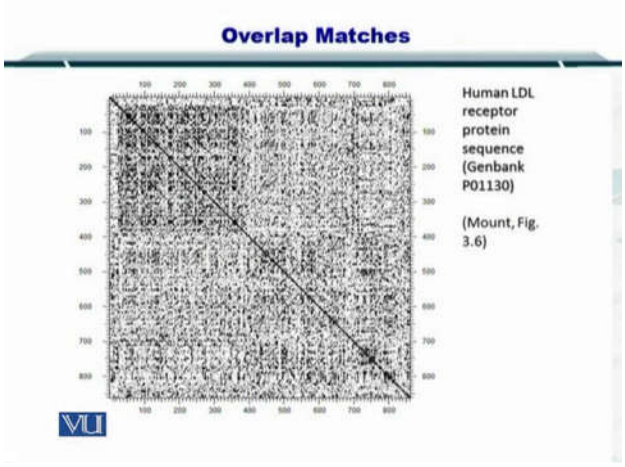
Skipping the gaps

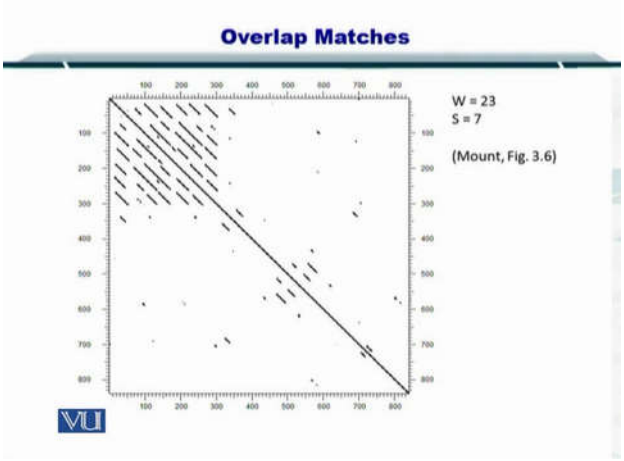
- What if we have a different traceback strategy?
- Can we start from any position within the alignment matrix to avoid gap regions?
- Yes, we can and that will give us a local alignment.

Conclusion:

- Traceback strategy allows us to differentiate between a local and a global alignment.
- The Smith Waterman Algorithm allows us to elicit local alignments and we will look at it later.

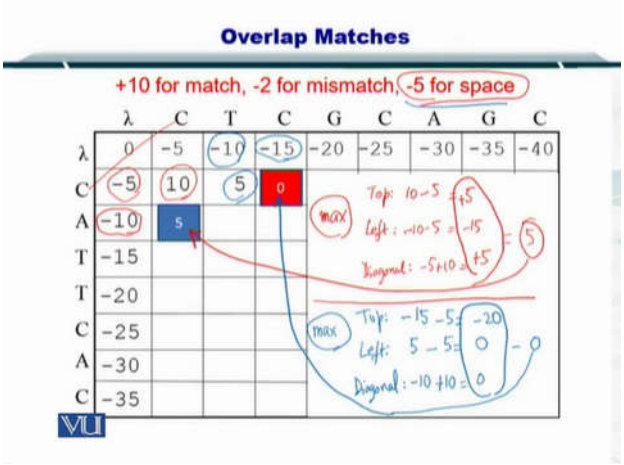
Overlap Matches

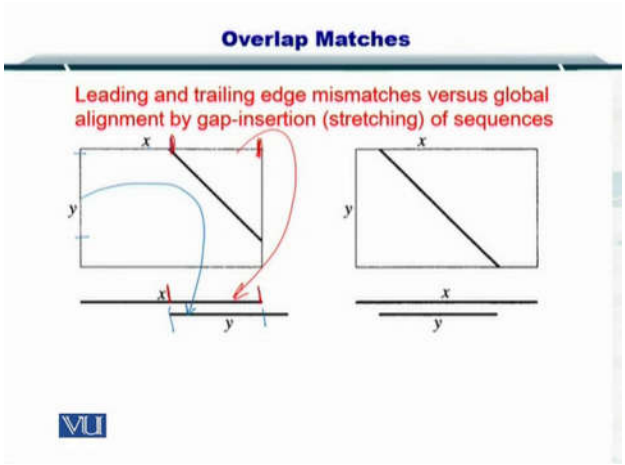




	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space





	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

Lecture 42

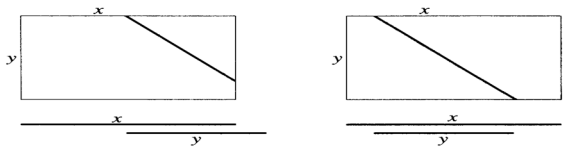
Example of Overlap Matches:

- In traceback for global alignments, we started from bottom-right corner of the alignment matrix
- Modification in traceback strategy can help find overlap matches containing leading or trailing sequence.

Conclusion:

- We have seen how a slight variation in the traceback can help us find overlap matches.
- There are several other interesting applications of modification in the traceback strategy.

Leading and trailing edge mismatches versus global alignment by gap-insertion (stretching) of sequences



		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0										
A	0										
W	0										
H	0										
E	0										
A	0										
E	0										

For $1 \le i \le n$, $1 \le j \le m$:

$$C(i,j) = \max \begin{cases} C(i-1,j-1) + w(S_i, T_j) \\ C(i-1,j) - \gamma \\ C(i,j-1) - \gamma \end{cases}$$

where $$w(a,b) = \begin{cases} +\alpha & \text{if } a = b \\ -\beta & \text{if } a \neq b \end{cases}$$

Match = +2. Mismatch = -1, Gap = -2

Global Alignment

Overlap Matches

Local Alignment

Conclusion:

- Local Alignments can identify exons which are present in both sequences
- Proteins of different kind and of different species often exhibit local similarities
- Hence, local similarities may indicate ”functional subunits”

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	−2	−1	−1	**2**	−1	−4	−2	−2	−1	−1
A	0	0	−2	−2	4	−1	**3**	−4	−4	−4	−3	−2
W	0	0	−3	−5	−4	1	−4	**18**	**10**	2	6	−6
H	0	0	10	2	6	−6	−1	10	16	**20**	12	4
E	0	0	2	16	8	0	7	2	8	16	**26**	18
A	0	0	−2	8	21	13	5	3	2	8	18	**25**
E	0	0	0	4	13	18	12	4	4	2	14	24

Match = +2. Mismatch = -1, Gap = -2

GAWGHEE
PAW-HEA

Lecture 44

Smith Waterman Algorithm:

- Exons in DNA tend to be more conserved as compared with introns
- Functional domains in proteins tend to be more conserved than the hydrophobic core
- Top row and left column are set to zero
- Alignment can end anywhere in matrix
- The trace back begins from the highest-scoring cell and ends where we encounter 0.

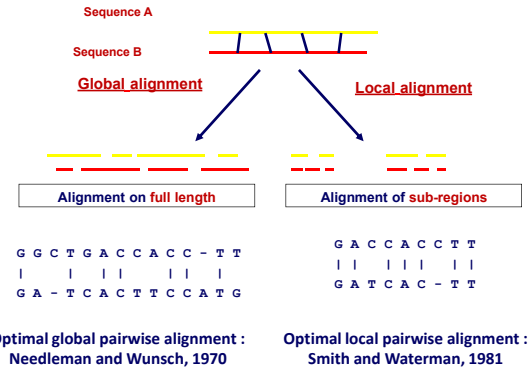
Conclusion:

- Local Alignments can identify coding portions in a DNA
- Such an alignment strategy can also help elicit functional domains from protein sequences.

Lecture 43

Moving from Global to Local Alignment

- DNA has coding and non-coding regions.
- Coding regions (Exons) are expressed as proteins.
- Exons tend to be more conserved due to their role in making functional proteins.
- Hence, non-coding regions (introns) are more likely to be subjected to mutations than coding regions (exons).
- High degree of alignment can be expected between two exon sequences.
- Local alignment may then need to be computed between two such sequences.
- Let’s take a look at why local alignments are useful in real life biology.



$$C[i,j] = \max \begin{cases} C[i-1,j-1] + score(i,j) \\ C[i-1,j] - \gamma \\ C[i,j-1] - \gamma \\ 0 \end{cases}$$

Difference between Needleman Wunsch and Smith Waterman is that '0' is also placed in the relationship!

	λ	C	T	C	G	C	A	G	C
λ	0	0	0	0	0	0	0	0	0
C	0								
A	0								
T	0								
T	0								
C	0								
A	0								
C	0								

+1 for a match, -1 for a mismatch, -5 for a space

Lecture 45

Example of Smith Waterman Algorithm

- Exons in DNA tend to be more conserved as compared with introns
- Functional domains in proteins tend to be more conserved than the hydrophobic core

Conclusion

- Local alignments can be performed using Smith Waterman algorithm
- Traceback can start from any position in the scoring matrix
- Local alignments can be extracted by starting from a high score till reaching '0'

	λ	C	T	C	G	C	A	G	C
λ	0	0	0	0	0	0	0	0	0
C	0	1	0	1	0	1	0	0	1
A	0	0	0	0	0	0	2	0	0
T	0	0	1	0	0	0	0	1	0
T	0	0	1	0	0	0	0	0	0
C	0	1	0	2	0	1	0	0	1
A	0	0	0	0	1	0	2	0	0
C	0	1	0	1	0	2	0	1	1

+1 for a match, -1 for a mismatch, -5 for a space

	λ	C	T	C	G	C	A	G	C
λ									
C									
A									
T									
T									
C									
A									
C									

+1 for a match, -1 for a mismatch, -5 for a space

Lecture 46

Repeated Alignments:

- Smith Waterman helped find best local alignment
- Since exons or domains are conserved and may be repeated, can we find repeated matches of one sequence in an other?

- Yes, we can find such matches
- How? We fill in the alignment matrix and devise a new traceback strategy
- Let's try to understand how this scheme works?

Standard Smith Waterman Algorithm

	λ	C	T	C	G	C	A	G	C
λ	0	0	0	0	0	0	0	0	0
C	0	1	0	1	0	1	0	0	1
A	0	0	0	0	0	0	2	0	0
T	0	0	1	0	0	0	0	1	0
T	0	0	1	0	0	0	0	0	0
C	0	1	0	2	0	1	0	0	1
A	0	0	0	0	1	0	2	0	0
C	0	1	0	1	0	2	0	1	1

+1 for a match, -1 for a mismatch, -5 for a space

F (0,0) = 0 ;

$$F(i,0) = \max \begin{cases} F(i-1,0), \\ F(i-1,j)-T, \end{cases} \quad j = 1,\ldots,m;$$

- **Unmatched regions and ends of matches, only allowing matches to end when they have score of at least T**

$$F(i,j) = \max \begin{cases} F(i,0), \\ F(i-1,j-1)+s(x_i,y_j), \\ F(i-1,j)-d, \\ F(i,j-1)-d. \end{cases}$$

- **Fills rest of the matrix and deals with start & extensions of matches**

Keeps track of which of the alternatives produced the maximum!

Finding one sequence in an other!

What if we try to find a portion repeatedly?

Two modifications:

- Use a score threshold 'T' above which the matches will be considered.
- This will help avoid low scoring local alignments
- Traceback should find multiple aligned regions by multiple traceback steps.

Conclusion:

- It will be very useful to search for repeated DNA coding regions i.e. Exons as well as functional units in proteins i.e. domains
- Slight modification to Smith Waterman can help us achieve that!

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	1	1	1	1	1	3	9 ← 9
P	0	0	0	0	1	1	1	1	1	3	9
A	0	0	0	5	1	6	1	1	1	3	9
W	0	0	0	0	2	1	21 ←	13 ←	5	3	9
H	0	10 ←	2	0	1	1	13	19	23 ←	15	9
E	0	2	16 ←	8	1	1	5	11	19	29	21
A	0	0	8	21 ←	13	6	1	5	11	21	28
E	0	0	6	13	18	12 ←	4	1	5	17	27

Match = +2. Mismatch = -1, Gap = -2, T = 1

HEAGAWGHEE
HEA.AW-HE.

Lecture 47

Example of Repeated Alignments

- Finding repeated DNA coding regions i.e. Exons as well as functional units in proteins i.e. domains
- Slight modification to Smith Waterman can help us achieve that!

Traceback - I

- The trace back starts at the last element of the first row.
- Trace back as usual, until you reach the top row, again.
- Next, move to the highest score in the next column.

Traceback - II

- Continue to traceback until you reach the top row again
- Repeat the previous two steps
- Stop the trace back if you reach '0' at (0,0).

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	1	1	1	1	1	3	9 ← 9
P	0	0	0	0	1	1	1	1	1	3	9
A	0	0	0	5	1	6	1	1	1	3	9
W	0	0	0	0	2	1	21 ←	13 ←	5	3	9
H	0	10 ←	2	0	1	1	13	19	23 ←	15	9
E	0	2	16 ←	8	1	1	5	11	19	29	21
A	0	0	8	21 ←	13	6	1	5	11	21	28
E	0	0	6	13	18	12 ←	4	1	5	17	27

HEAGAWGHEE
HEA.AW-HE.

Match = +2. Mismatch = -1, Gap = -2, T = 1

Conclusion:

- Repeat sequences can be found by using the modified recurrence relation and the new traceback strategy
- Threshold 'T' helps filter out low scoring i.e. possibly random local alignments.

Lecture 48

Review of Traceback Strategies:

- We have seen how biological sequences can be searched and compared using various recurrence relations and traceback strategies.
- Let's review them!

Conclusion

- Slight modification in the recurrence relation and changes to the traceback strategy can help compare sequences in a variety of ways
- DynamicPrograming solves this problem in quick time!

Smith Waterman Algorithm

	λ	C	T	C	G	C	A	G	C
λ	0	0	0	0	0	0	0	0	0
C	0	1	0	1	0	1	0	0	1
A	0	0	0	0	0	0	2	0	0
T	0	0	1	0	0	0	0	1	0
T	0	0	1	0	0	0	0	0	0
C	0	1	0	2	0	1	0	0	1
A	0	0	0	0	1	0	2	0	0
C	0	1	0	1	0	2	0	1	1

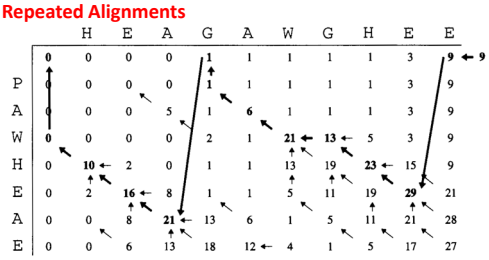
+1 for a match, -1 for a mismatch, -5 for a space

Needleman Wunsch Algorithm

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

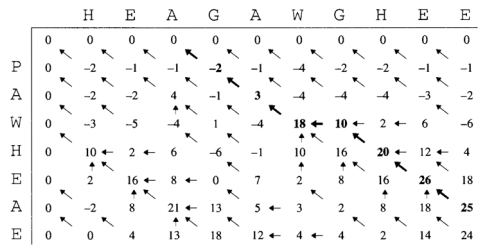
Repeated Alignments



Match = +2. Mismatch = -1, Gap = -2, T = 1

```
HEAGAWGHEE
HEA.AW-HE.
```

Overlap Matches



Match = +2. Mismatch = -1, Gap = -2

```
GAWGHEE
PAW-HEA
```

Lecture 49

Introduction to Scoring Alignments

- Optimal Alignment vs. Best Alignment
- Scoring scheme has crucial role in producing optimal alignments.
- What may be the elements of an optimal scoring scheme?

An optimal scoring scheme should:

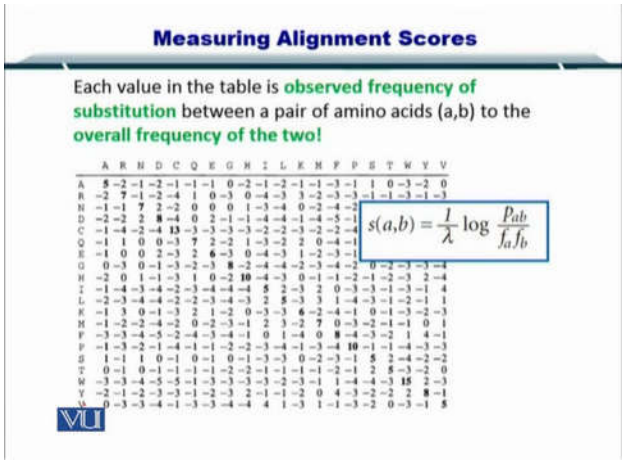
- Appropriately reward or penalize matches & mismatches/gap.
- Give due consideration to more abundant nucleotides or amino acids found in biological sequences.
- Identify pairs of symbols which appear more frequently aligned in a family of aligned sequences.
- Scoring schemes is essentially a combination of the above quantities.

Points to Consider:

- AA Substitutions have a pattern
- Specific type of amino acid is more likely substituted by another amino acid of the same type as
- Why? Because it can help preserve function of the protein.
- Earlier, we were giving the same match score +2 or +1 to all matches
- Can we develop the substitution matrices based on experimental data of substitutions?

Solution:

- Simply count how frequently each amino acid occurs in related protein sequences?
- Then count how often one amino acid is substituted by another within these sequences.



Conclusion:

- Statistics of amino acid & nucleotide occurrences can help better assign match & mismatch scores
- Optimal gap penalties for insertions & deletions can also be statistically computed.

Lecture 50

Measuring Alignment Scores

- Score of match or mismatch should be equal to observed/overall occurrence of an amino acid or nucleotide
- Let's consider an example of how these scores can be measured!

Points to Consider:

- The matrix has positive and negative scores, both.
- Matches and Mismatches are therefore all considered!
- It's a diagonal matrix!

Conclusion:

- If we can somehow build such "scoring matrices" with match and mismatch scores, we can align biological sequences in accordance with the real life substitutions etc.

Needleman Wunsch Algorithm

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

Lecture 51

Scoring Matrices

- Alignment is used to compare biological sequences
- Amino acids / nucleotides are more easily substituted based on their chemical similarity
- Since amino acids are substituted with different probabilities, we need a flexible match and mismatch scoring strategy

"Scoring Matrices" contain such flexible scores for use during alignment

How to build scoring matrices?

- We analyze the observed frequency with which each amino acid / nucleotide is substituted by an other one in similar proteins/genes

Scores in the Scoring Matrices:

- Scoring matrices may contain +ve and -ve values as well as 0.
- +ve values are for more frequent matches / mismatches
- -ve values are for unlikely ones

Conclusion:

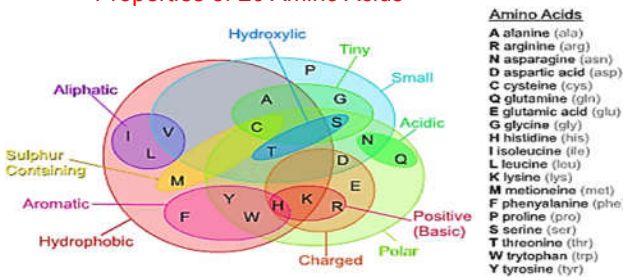
- Scoring matrices accurately reward and penalize matches and mismatches
- Several types of scoring matrices can be developed based on the underlying strategy for building them.

Lecture 52

Deriving Scoring Matrices:

- Alignment is used to compare biological sequences
- Amino acids / nucleotides are more easily substituted based on their chemical similarity.

Properties of 20 Amino Acids



The example of Ubiquitin

Human	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	T	S	G	T	S	P	V	H
Chimp	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	T	S	G	T	S	P	V	H
Orangutan	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	T	S	G	T	S	P	V	H
Macaque	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	T	S	G	T	S	P	V	H
Mouse	V	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	A	S	G	T	S	P	V	H
GuineaPig	V	A	Q	E	L	L	K	Q	G	A	S	P	N	V	Q	D	A	S	G	T	S	P	L	H
Dog	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	A	T	G	T	T	P	A	H
Horse	I	A	L	E	L	L	K	Q	G	A	N	P	N	I	Q	D	A	S	G	T	T	P	A	H
Cow	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	A	S	G	T	T	P	A	H
Opossum	I	A	L	E	L	L	K	Q	G	A	S	P	N	V	Q	D	G	S	G	T	S	P	A	H
Platypus	I	A	L	E	L	L	K	Q	G	A	N	P	N	V	Q	D	A	T	G	T	T	P	V	H
Frog	V	A	S	E	L	L	K	Q	G	A	S	P	N	I	Q	D	A	Y	G	T	T	P	A	H
Stickleback	I	A	G	L	L	L	E	K	G	A	D	P	N	V	Q	D	N	H	G	I	A	P	V	H
Fugu	I	A	S	L	L	L	E	K	G	A	D	P	N	V	Q	D	K	H	G	I	A	P	V	H
Tetraodon	I	A	S	L	L	L	E	K	G	A	D	P	N	V	Q	D	K	H	G	I	A	P	V	H
Medaka	I	A	S	L	L	L	E	K	G	A	D	A	N	V	Q	D	K	H	G	I	A	P	V	H

$$s(a,b) = \frac{1}{\lambda} \log \frac{p_{ab}}{f_a f_b}$$

Frequency of 20 Amino Acids

Table 7.1 The average amino acid frequencies reported by Robinson and Robinson in [173].

Amino acid	Freqency	Amino acid	Freqency	Amino acid	Freqency	Amino acid	Freqency
Ala	0.078	Gln	0.043	Leu	0.090	Ser	0.071
Arg	0.051	Glu	0.063	Lys	0.057	Thr	0.058
Asn	0.045	Gly	0.074	Met	0.022	Trp	0.013
Asp	0.054	His	0.022	Phe	0.039	Tyr	0.032
Cys	0.019	Ile	0.051	Pro	0.052	Val	0.064

Let's compute a scoring matrix from 6 sequences of comprising of 3 alphabets A, B and C

B	A	B	A
A	A	C	A
A	A	C	C
A	A	B	A
A	A	B	C
A	A	B	B

aligned pairs	*observed frequencies* f_{ij}
A to A	28/60
A to B	8 /60
A to C	6/60
B to B	6/60
B to C	10/60
C to C	2/60

amino acids	*observed frequencies* f_i
A	14/24
B	6/24
C	4/24

Resulting Scoring Matrix

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
A	**5**	−2	−1	−2	−1	−1	−1	0	−2	−1	−2	−1	−1	−3	−1	1	0	−3	−2	0
R	−2	**7**	−1	−2	−4	1	0	−3	0	−4	−3	3	−2	−3	−3	−1	−1	−3	−1	−3
N	−1	−1	**7**	2	−2	0	0	0	1	−3	−4	0	−2	−4	−2	1	0	−4	−2	−3
D	−2	−2	2	**8**	−4	0	2	−1	−1	−4	−4	−1	−4	−5	−1	0	−1	−5	−3	−4
C	−1	−4	−2	−4	**13**	−3	−3	−3	−3	−2	−2	−3	−2	−2	−4	−1	−1	−5	−3	−1
Q	−1	1	0	0	−3	**7**	2	−2	1	−3	−2	2	0	−4	−1	0	−1	−1	−1	−3
E	−1	0	0	2	−3	2	**6**	−3	0	−4	−3	1	−2	−3	−1	−1	−1	−3	−2	−3
G	0	−3	0	−1	−3	−2	−3	**8**	−2	−4	−4	−2	−3	−4	−2	0	−2	−3	−3	−4
H	−2	0	1	−1	−3	1	0	−2	**10**	−4	−3	0	−1	−1	−2	−1	−2	−3	2	−4
I	−1	−4	−3	−4	−2	−3	−4	−4	−4	**5**	2	−3	2	0	−3	−3	−1	−3	−1	4
L	−2	−3	−4	−4	−2	−2	−3	−4	−3	2	**5**	−3	3	1	−4	−3	−1	−2	−1	1
K	−1	3	0	−1	−3	2	1	−2	0	−3	−3	**6**	−2	−4	−1	0	−1	−3	−2	−3
M	−1	−2	−2	−4	−2	0	−2	−3	−1	2	3	−2	**7**	0	−3	−2	−1	−1	0	1
F	−3	−3	−4	−5	−2	−4	−3	−4	−1	0	1	−4	0	**8**	−4	−3	−2	1	4	−1
P	−1	−3	−2	−1	−4	−1	−1	−2	−2	−3	−4	−1	−3	−4	**10**	−1	−1	−4	−3	−3
S	1	−1	1	0	−1	0	−1	0	−1	−3	−3	0	−2	−3	−1	**5**	2	−4	−2	−2
T	0	−1	0	−1	−1	−1	−1	−2	−2	−1	−1	−1	−1	−2	−1	2	**5**	−3	−2	0
W	−3	−3	−4	−5	−5	−1	−3	−3	−3	−3	−2	−3	−1	1	−4	−4	−3	**15**	2	−3
Y	−2	−1	−2	−3	−3	−1	−2	−3	2	−1	−1	−2	0	4	−3	−2	−2	2	**8**	−1
V	0	−3	−3	−4	−1	−3	−3	−4	−4	4	1	−3	1	−1	−3	−2	0	−3	−1	**5**

$$(i,j)\text{-entry of matrix is } s_{ij} = \log_2 \frac{f_{ij}}{p_{ij}}, \quad p_{ij} = \begin{cases} f_i f_i & i = j \\ 2 f_i f_j & i \neq j. \end{cases}$$

aligned pairs	*observed frequencies* f_{ij}	*expected frequencies* p_{ij}	$2s_{ij}$
A to A	28/60	196/576	0.91
A to B	8/60	168/576	-2.26
A to C	6/60	112/576	-1.92
B to B	6/60	36/576	1.35
B to C	10/60	48/576	2.00
C to C	2/60	16/576	0.52

amino acids	*observed frequencies* f_i
A	14/24
B	6/24
C	4/24

	A	B	C
A	1	-2	-2
B		1	2
C			1

Conclusion:

- Based on specific sequence samples, we can build scoring matrices for DNA, RNA and proteins.
- We will see how variations can be introduced into computing these scoring matrices.

Lecture 53

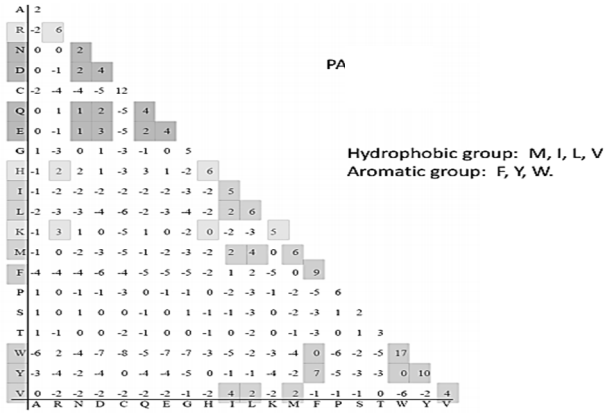
PAM Matrices:

- Alignment scoring matrices are very useful in giving suitable scores to matches and mismatches
- There are 2 types of scoring matrices i.e. PAM and BLOSUM
- PAM means “Point Accepted Mutations”
- Point accepted mutation is a substitution of one amino acid by another such that the protein functions stays conserved.

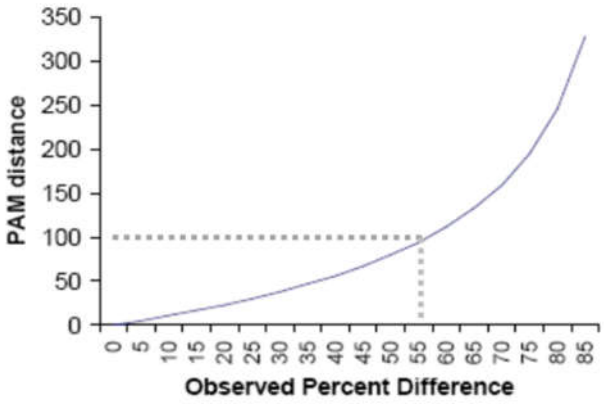
PAM Unit

- PAM unit is a time in which about 1% of amino acids in a sequence undergo accepted mutations.
- Important: If two sequences diverge by 100 PAM units, it does not imply that they are totally different in each position.

PAM 250



Sequence Difference vs. PAM Distance



Lecture 54

BLOSUM Matrices:

- BLOSUM matrices were first proposed in 1992 by Henikoff et al.
- Protein sequences can be aligned using BLOSUM scoring matrices.
- BLOSUM or Block Substitution Matrix
- A block is an a multiple sequence alignment without any gaps
- Let’s take a look at a block of protein sub-sequences

Conclusions - I

- Selection of scoring matrix helps find different types of alignments.
- Typically used matrices:BLOSUM62 or PAM120
- In PAMx, larger x detects more divergent sequences

Steps to compute PAM matrices

Step 1: Align proteins sequences which are 1-PAM unit diverged

Step 2: Let $A_{i,j}$ be the number of times A_i is substituted by A_j

Step 3: Compute the frequency f_i of amino acid A_i

Then, PAM1 $= p_{ij} = \frac{A_{ij}}{\sum_k A_{ik}}$

PAM’n’ = (PAM1) ^{n}

Conclusion:

- PAM matrices are scoring matrices that are useful in computing sequence alignment scores.
- PAM1, PAM2 …PAM250 can be computed
- PAM120 is considered optimal scoring matrix!

A block of aligned sequences with out any gaps

```
WWYIRCASILRKIYIYGPVGVSRLRT
WHYVRCASILRHLYHRSPAGVGSITK
WFYTRAASTARHLYLRGGAGVGSMTK
WWYVRAAALLRRVYIDGPVGVNSLRT
```

Note that there are some mismatches in the alignment

Computing BLOSUM-x matrices

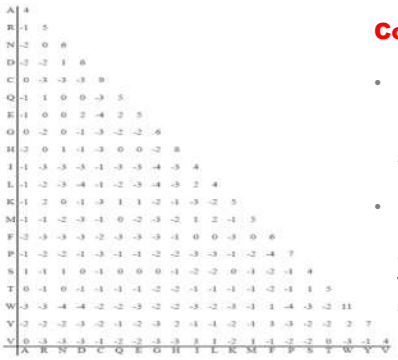
Step 1: Eliminate sequences that are identical in x% positions

Step 2: Compute observed frequency $f_{i,j}$ of aligned pair A_i to A_j. Hence, $f_{i,j}$ becomes the probability of aligning A_i and A_j in the selected blocks.

Step 3: Compute f_i which is the frequency of observing A_i in the entire block

$$s_{ij} = \log_2 \frac{f_{ij}}{p_{ij}}, \quad p_{ij} = \begin{cases} f_i f_i & i = j \\ 2 f_i f_j & i \neq j. \end{cases}$$

BLOSUM62



Conclusions - II

- **For BLOSUMn, a lower n detects more divergent sequences**
- **BLOSUM62 and PAM120 are considered optimal for use in protein sequence alignment**

Lecture 55

Multiple Sequence Alignment

- Pairwise sequence alignment involved comparison i.e. alignment, of two sequences
- Scoring matrices were used to score and rank the sequences.
- Multiple sequence alignment involves comparison of three of more sequences
- Pairwise alignment was either local or global
- Multiple alignment is mostly global.

MSA vs Pairwise

- For pairwise alignments, we used Dynamic Programming
- For MSA, Dynamic Programming may get very expensive, computationally.
- Solution: Progressive Alignment

A block of multiply aligned sequences

M Q V K L F T P L H D K S D H G K Y H
M Q V K I F T P L H D K S - H G K S H
M Q V H L Y - P L H D K S - T G K S H
M Q V H L F - P L H D K S D T G K S H
M Q V K L Y T P L H D K S D H G K Y H

Conclusion:

- MSA can help align multiple sequences
- Dynamic Programming cannot be used as it becomes expensive
- Progressive alignment can help perform MSA!
- CLUSTAL – Online tool

Lecture 56

More on Multiple Sequence Alignment

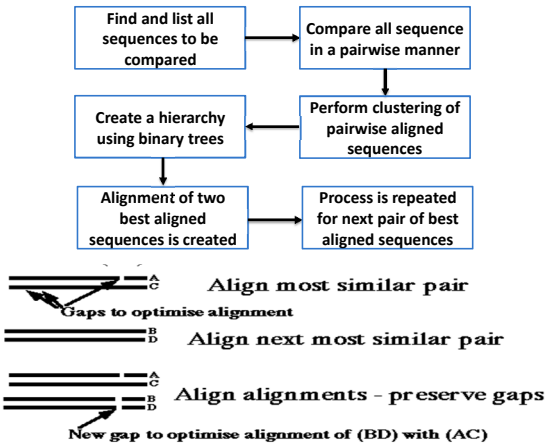
- MSA helps compare several sequences by aligning them
- MSA can extract consensus sequences from several aligned sequences
- Characterize protein families based on homologous regions

Application of MSA:

- Predict secondary and tertiary structures of new protein sequences
- Evaluate evolutionary order of species or "Phylogeny"

Methodology:

- Pairwise alignment is the alignment of two sequences
- MSA can be performed by repeated application of pairwise alignment



Conclusion:

- MSA can help align multiple sequences
- Progressive alignment can help perform MSA!
- Need to remove sequences with >80% similarity
- CLUSTAL – Online tool

Lecture 57

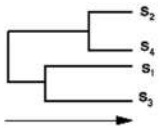
Progressive Alignment for MSA

- MSA involves progressive alignment of sequences
- Doing so many progressive alignments can be slow
- How do progressive alignments work?

Shortcomings of this approach

- Dependence upon initial alignments. If sequences are dissimilar, errors in alignment are propagated
- Solution: Begin by using an initial alignment, and refine it repeatedly

- Step 1 : Pairwise Alignment of all sequences
  Example: S_1, S_2, S_3, S_4 , so that is 6 pairwise comparisons!
- Step 2: Construct a Guide Tree (Dendogram) using a *Distance Matrix*

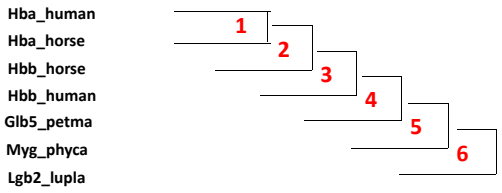


$$Distance = 1 - \frac{Matches}{Comparisons}$$

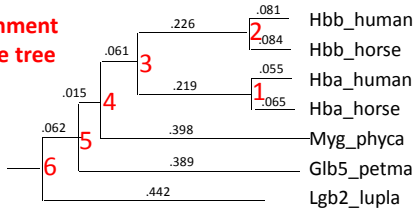
$$Similarity = \frac{Matches}{Comparisons}$$

- Step 3: Progressive alignment following branching order in tree

Progressive alignment using sequential branching

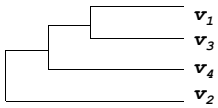


Progressive alignment following a guide tree



Similarity Matrix

$$Similarity = \frac{Matches}{Comparisons}$$

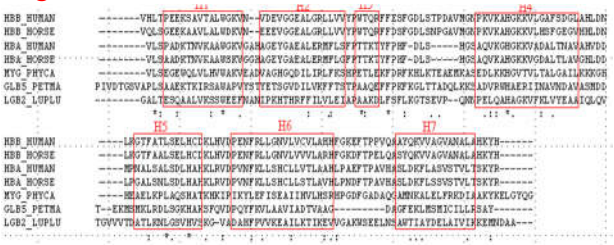


	v_1	v_2	v_3	v_4
v_1	-			
v_2	.17	-		
v_3	.87	.28	-	
v_4	.59	.33	.62	-

Conclusion

- Progressive alignments are used in aligning multiple sequences
- Iterative approaches can help refine results from progressive alignments

Alignment results:



Conclusion:

- MSA can be better performed using clustering strategies followed by alignment of the alignments later
- CLUSTAL is a free online tool that does all of this for us!

Lecture 58

MSA - Example

- MSA involves progressive alignment of sequences
- Doing so many progressive alignments can be slow
- Let's take a look at an example

MSA on globin sequences

distance between 2 sequences = 1- $\frac{\text{No. identical residues}}{\text{No. aligned residues}}$

	Hbb_human	Hbb_horse	Hba_human	Hba_horse	Myg_phyca	Glb5_petma	Lgb2_lupla
1	-						
2	.17	-					
3	.59	.60	-				
4	.59	.59	.13	-			
5	.77	.77	.75	.75	-		
6	.81	.82	.73	.74	.80	-	
7	.87	.86	.86	.88	.93	.90	-

Lecture 59

CLUSTALW

- MSA involves progressive alignment of sequences
- Doing so many progressive alignments can be slow
- CLUSTALW is an online tool to perform MSA
- Developed by European Molecular Biology Laboratory & European Bioinformatics Institute
- Performs alignment in:
  - slow/accurate
  - fast/approximate

Scope:

CLUSTALW can:

- create multiple alignments,
- optimize existing alignments,
- profile analysis & create phylogenetic trees

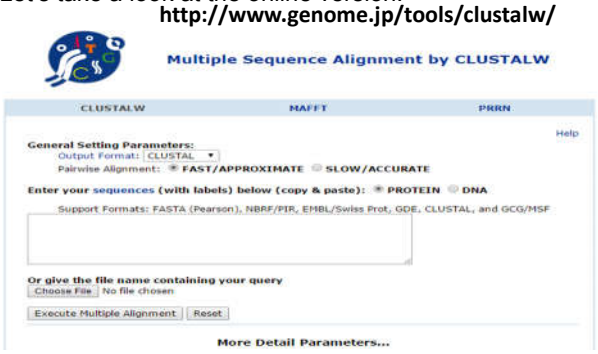
Using CLUSTALW:

CLUSTALW can use multiple file formats including:

- EMBL/SwissProt, Pearson (Fasta) etc

Let's take a look at the online version.

http://www.genome.jp/tools/clustalw/



Conclusion

- CLUSTALW can be used to perform MSA
- Fast and slow modes of alignment are available
- CLUSTAL Omega is now also available which includes several upgrades!

Lecture 60

Introduction to BLAST - I

- National Center for the Biotechnology Information (NCBI) - USA
- BLAST developed in 1990
- "Basic Local Alignment Search Tool"
- Searches databases for query protein and nucleotide sequences
- Also searches for translational products etc.
- Online availability --blast.ncbi.nlm.nih.gov/Blast.cgi



http://blast.ncbi.nlm.nih.gov/Blast.cgi

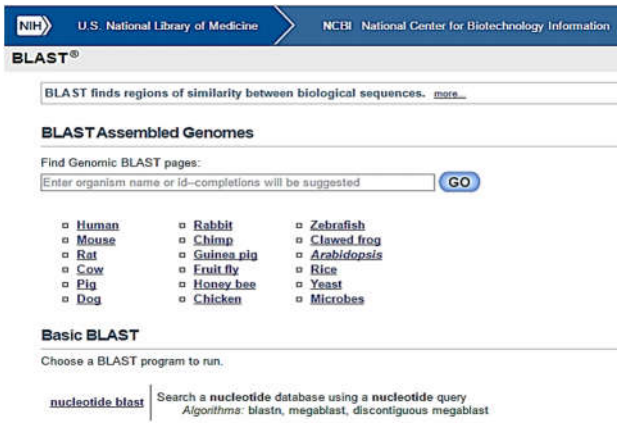
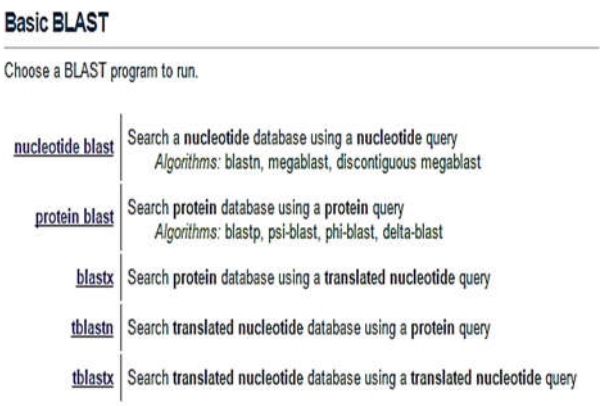


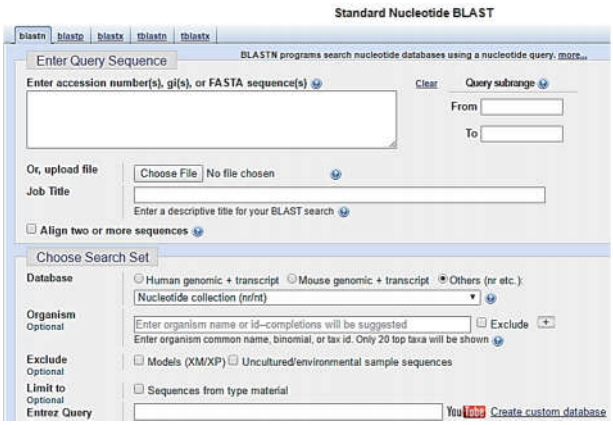
Table 1. Some recent and less recent available methods for MSAs.		
Name	**Algorithm**	**URL**
MSA	Exact	http://www.ibc.wustl.edu/ibc/msa.html
DCA	Exact (requires MSA)	http://bibiserv.techfak.uni-bielefeld.de/dca
OMA	Iterative DCA	http://bibiserv.techfak.uni-bielefeld.de/oma
ClustalW, ClustalX	Progressive	ftp://ftp-igbmc.u-strasbg.fr/pub/clustalW or clustalX
MultAlin	Progressive	http://www.toulouse.inra.fr/multalin.html
DiAlign	Consistency-based	http://www.gsf.de/biodv/dialign.html
ComAlign	Consistency-based	http://www.daimi.au.dk/~ ocaprani
T-Coffee	Consistency-based/progressive	http://igs-server.cnrs-mrs.fr/~ cnotred
Praline	Iterative/progressive	jhering@nimr.mrc.ac.uk
IterAlign	Iterative	http://giotto.Stanford.edu/~ luciano/iteralign.html
Prrp	Iterative/Stochastic	ftp://ftp.genome.ad.jp/pub/genome/saitama-cc/
SAM	Iterative/Stochastic/HMM	rph@cse.ucsc.edu
HMMER	Iterative/Stochastic/HMM	http://hmmer.wustl.edu/
SAGA	Iterative/Stochastic/GA	http://igs-server.cnrs-mrs.fr/~ cnotred
GA	Iterative/Stochastic/GA	czhang@watnow.uwaterloo.ca

C. Notredame, "Recent progresses in multiple alignment: a survey", *Pharmacogenomics* (2002) 3(1)

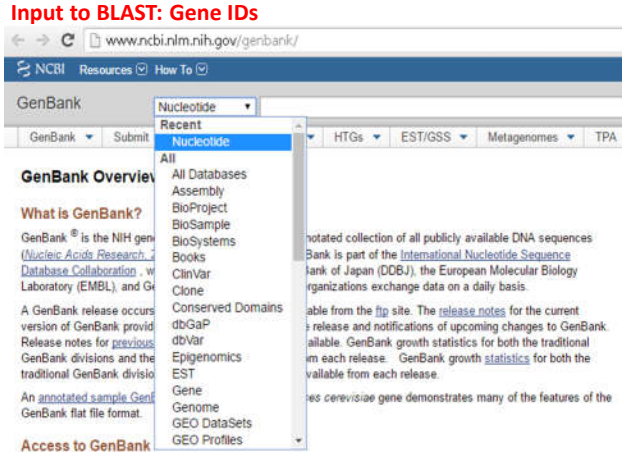
http://blast.ncbi.nlm.nih.gov/Blast.cgi



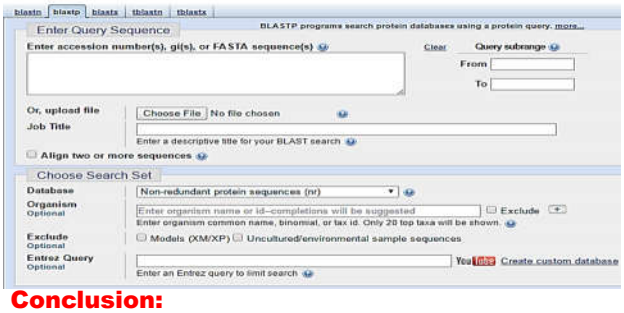
Nucleotide BLAST 'blastn'



Input to BLAST: Gene IDs



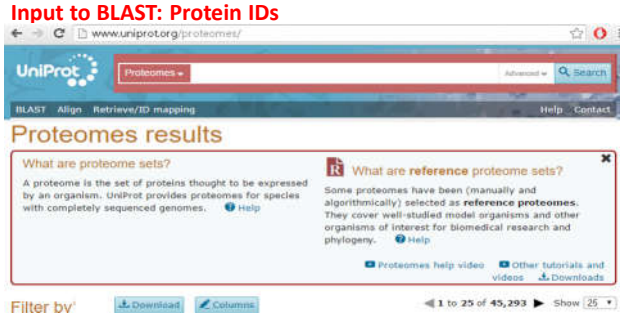
Protein BLAST 'blastp'



Conclusion:

- BLAST can be used to search for local alignment of protein and nucleotide sequences
- It is available online
- Can perform searches across species and organisms

Input to BLAST: Protein IDs



Conclusion:

- Purpose of BLAST
- BLAST vs. Smith Waterman
- Inputs and outputs from BLAST
- But what are the internal algorithms & workings of BLAST?

Lecture 61

Introduction to BLAST - II

- Smith Waterman can align complete sequences
- BLAST does it in an approximate way
- Hence, BLAST is faster BUT does not ensure optimal alignment.

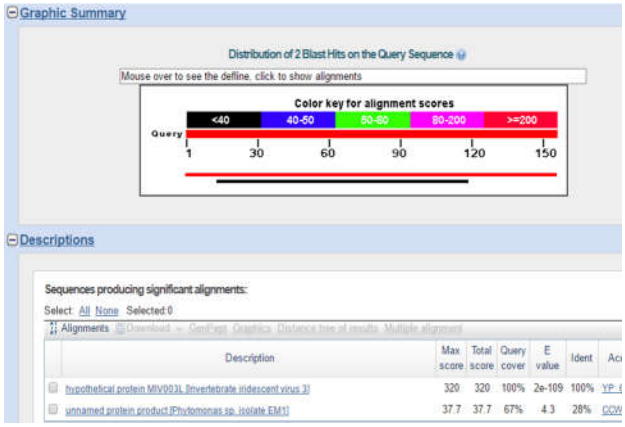
Use Cases of BLAST

- BLAST provides for approximate sequence matching
- Input to BLAST is a FASTA formatted sequence and a set of search parameters

Output of BLAST:

- Results are shown in HTML, plain text, and XML formats
- A table lists the sequence hits found along with scores
- Users can read this table off and evaluate results

Results from BLAST



Lecture 62

BLAST Algorithm

- BLAST can search sequence databases and identify unknown sequences by comparing them to the known sequences. This can help identify the parent organism, function and evolutionary history.

Query sequence: PQGELV--- Make list of all possible words (length 3 for proteins)
PQG (score 15)
QGE (score 9)
GEL (score 12)
ELV (score 10)
Assign scores from Blosum62, use those with score> 11 :
PQG & GEL--- Mutate words such that score still > 11
PQG (score 15) similar to PEG (score 13) At the end, we get: PQG, GEL and PEG

Find all database sequences that have at least 2 matches among our 3 words: PQG, GEL & PEG
Find database hits and extend alignment (High-scoring Segment Pair):

Query: M E T P Q G I A V
Database: - - - P Q G E L V
8 5 5 2 0 8

High Scoring Pair: PQGI (score 8+5+5+2)
If 2 HSP in query sequence are < 40 positions away
Full dynamic alignment on query and hit sequences

Conclusion:

- BLAST performs quick alignments on sequences
- The results are tabulated with alignment regions overlapping each other
- Statistical evaluation is also provided alongside

Lecture 63

Types of BLAST

- BLAST can search sequence databases and identify unknown sequences by comparing them to the known sequences
- BLAST performs searches by quick alignments on sequences.

Two Main Types of BLAST

Nucleotides

- Blastn: Compares a nucleotide query sequence against a nucleotide database.

Proteins

- Blastp: Compares an amino acid query sequence against a protein database.

Other Types of BLAST

Blastx: Compares a nucleotide query sequence against a protein sequence database.
Helps find potential translation products of unknown nucleotide sequences.

Reverse of Blastx -tblastn: Compares a protein query sequence against a nucleotide sequence database.
Nucleotide sequence dynamically translated into all reading frames.
tblastx: Compares the six-frame translated proteins of a nucleotide query sequence against the six-frame translated proteins of a nucleotide sequence database.

Conclusion: BLAST performs quick alignments on biological sequences. Several types of BLAST exist which can assist in comparing nucleotide sequences with amino acids & vice versa.

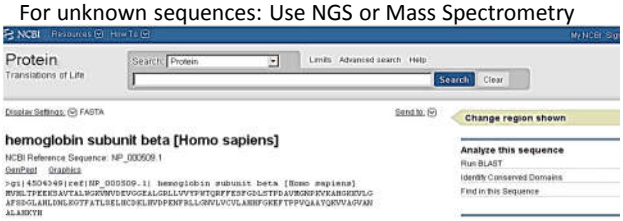
Lecture 64

Summary of BLAST

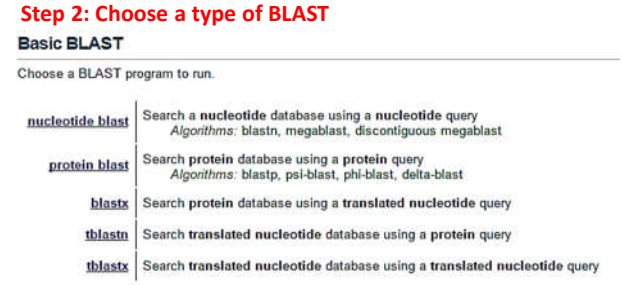
- BLAST can quickly search sequence databases if given a query sequence.
- Multiple types of BLAST exist which assist in aligning nucleotide and amino acid sequences.

Step 1: Obtain a query sequence

For known sequences: Use NCBI, UCSC etc..
For unknown sequences: Use NGS or Mass Spectrometry

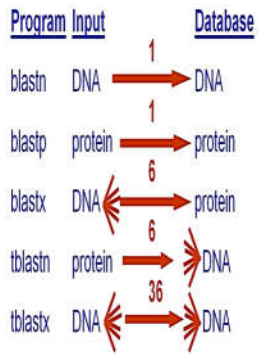


Step 2: Choose a type of BLAST

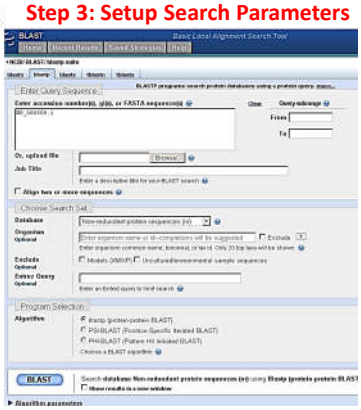


- **blastn (nucleotide BLAST)**
- **blastp (protein BLAST)**
- **blastx (translated BLAST)**
- **tblastn (translated BLAST)**
- **tblastx (translated BLAST)**

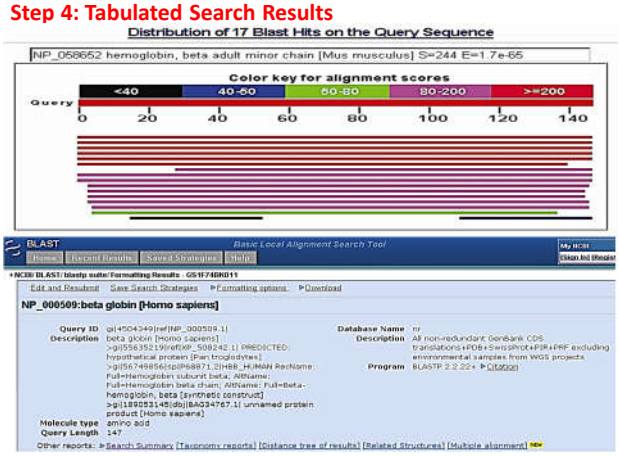
Step 2: Choose a type of BLAST



Step 3: Setup Search Parameters



Step 4: Tabulated Search Results



Conclusion:

- BLAST is an online tool which performs quick alignments on biological sequences.
- Depending on your need you can choose a specific type of BLAST to compare and score alignments.

Lecture 65

Introduction to FASTA - I

- Comparison between two sequences can be performed by pairwise sequence alignment.
- For comparing multiple sequences, multiple sequence alignment (MSA) is used!
- MSA is a progressive pairwise alignment so as to handle multiple sequences
- Smith-Waterman performs local alignment
- Needleman-Wunsch calculates global alignment
- The local and global alignment algorithms use Dynamic Programming Approaches.
- But, this takes time if there are several sequences that need comparison!

Solution?

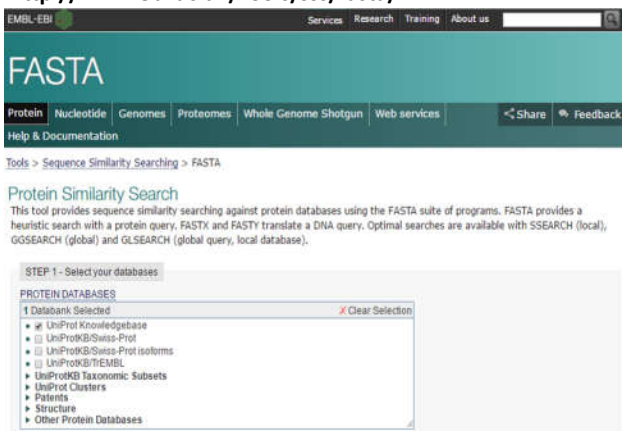
- We used BLAST which is an approximate local alignment search tool
- BLAST compares a large number of sequences, quickly!
- FASTA took a similar approach!

FASTA

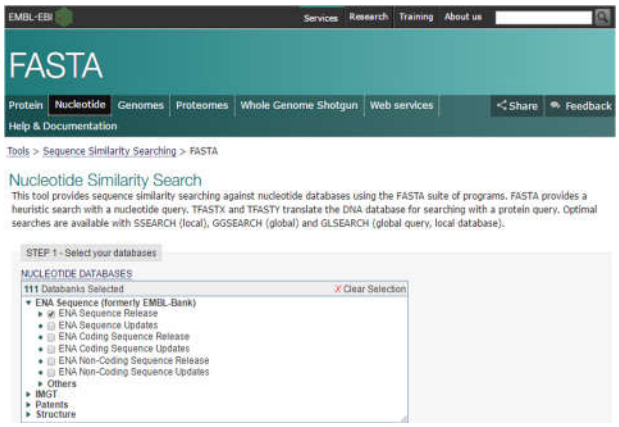
- Developed in 1988
- "Fast Alignment"
- Searches databases for query protein and nucleotide sequences
- Was later improved upon in BLAST



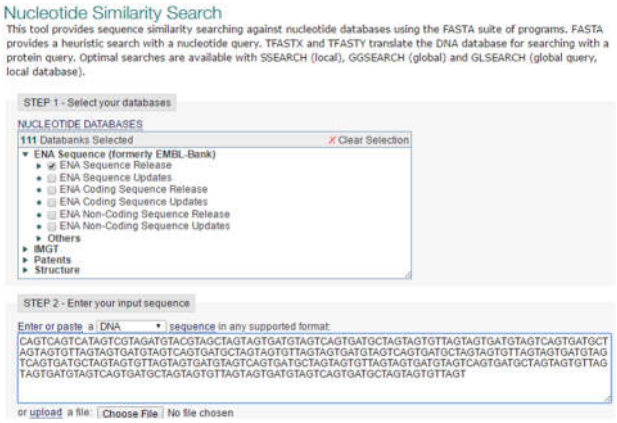
http://www.ebi.ac.uk/Tools/sss/fasta/



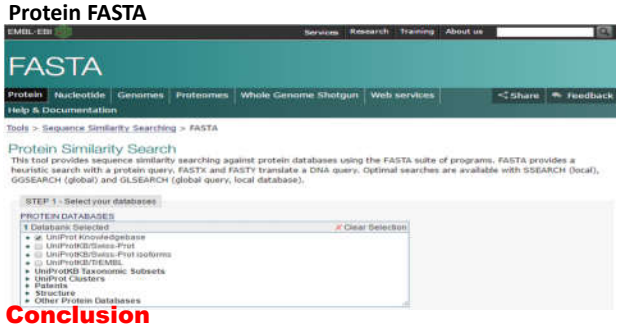
Nucleotide FASTA



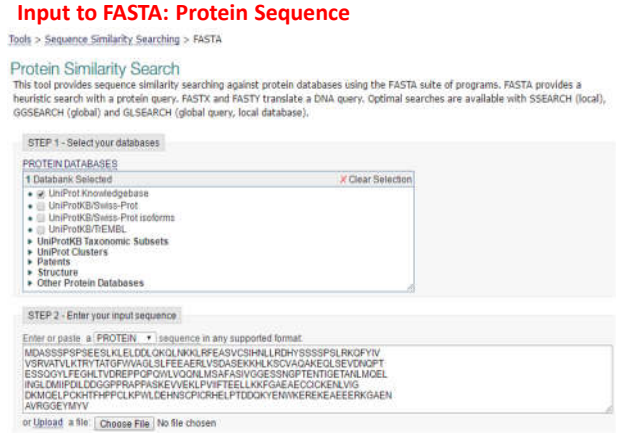
Input to FASTA: Gene IDs



Protein FASTA



Input to FASTA: Protein Sequence



Conclusion

- FASTA can perform quick comparison of protein and nucleotide sequences.
- Can also perform genome and proteome similarity search. It is available online.

Lecture 66

Introduction to FASTA - II

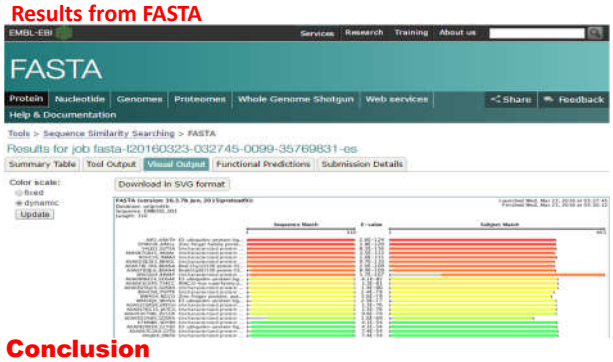
- FASTA – Fast Alignment Algorithm
- Classical global and local alignment algorithms are time consuming
- FASTA achieves alignment by using short lengths of exact matches.

Use Cases of FASTA

- FASTA relies on aligning subsequences of absolute identity
- Input to FASTA search can be in FASTA, EMBL, GenBank, PIR, NBRF, PHYLIP or UniProt formats

Output of BLAST: Results are output in visual format along with functional prediction. A table lists the sequence hits found along with scores. Users can click on each reported match to look at the details.

Results from FASTA



Conclusion

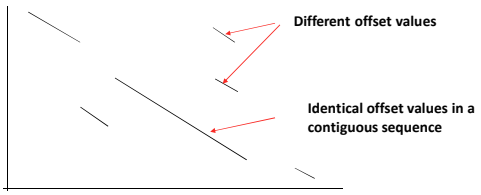
- Purpose of FASTA. FASTA vs. Smith Waterman
- Inputs and outputs from FASTA
- Internal working of FASTA?

Lecture 67

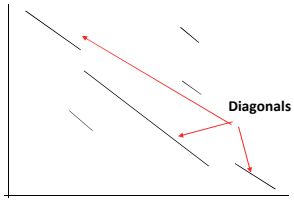
FASTA Algorithm

- FASTA can search sequence databases and identify unknown sequences by comparing them to the known sequence databases.
- This can help obtain information on the parent organism, function and evolutionary history.

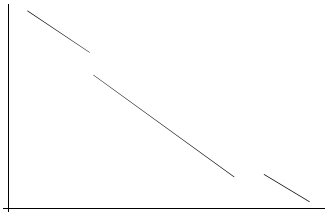
Step 1: Local regions of identity are found



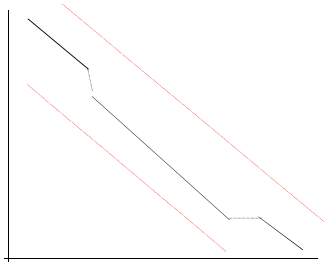
Step 2:
Rescore the local regions using PAM or BLOSUM matrix



Step 3:
Eliminate short diagonals below a cutoff score



Step 4:
Create a gapped alignment in a narrow segment and then perform Smith Watermann alignment



Conclusion:

- FASTA performs preliminary alignments quickly followed by Smith-Waterman
- Results are output in visual format along with statistical scores.

Lecture 68

Types of FASTA

- FASTA can search sequence databases and compare sequences to the known sequences in DNA/RNA/Protein Databases
- It performs search on matching words followed by application Smith-Waterman

Types of FASTA

Fasta35
- Scan a protein or DNA sequence library for similar sequences

Fastx35
- Compare a translated DNA sequence (6 ORFs) to a protein sequence database

tfastx35
- Compare a protein sequence to a DNA sequence database (6 ORFs)

fasty35
- Compare a DNA sequence (6ORFs) to a protein sequence database

fasts35
- Compare unordered peptides to a protein sequence database

fastm35
- Compare ordered peptides (or short DNA sequences) to a protein (DNA) sequence database

Conclusion

- FASTA performs quick alignments on biological sequences
- Several types of FASTA exist which can assist in comparing DNA/RNA/Protein sequences with each other

Lecture 69

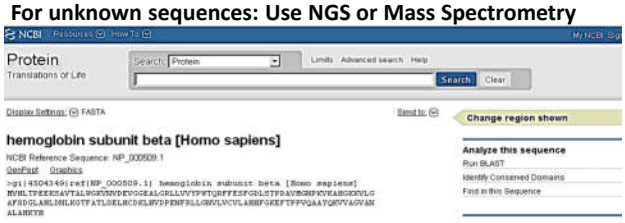
Summary of FASTA:

- FASTA can briskly perform sequence search databases if given a query sequence
- Multiple types of FASTA exist which assist in aligning DNA/RNA/Protein sequences

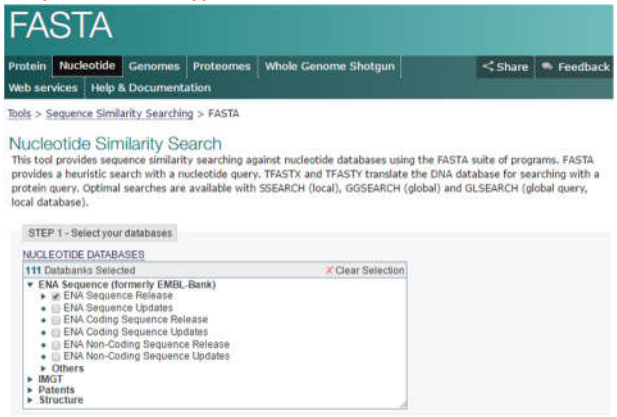
Step 1: Obtain a query sequence

For known sequences: Use NCBI, UCSC etc..

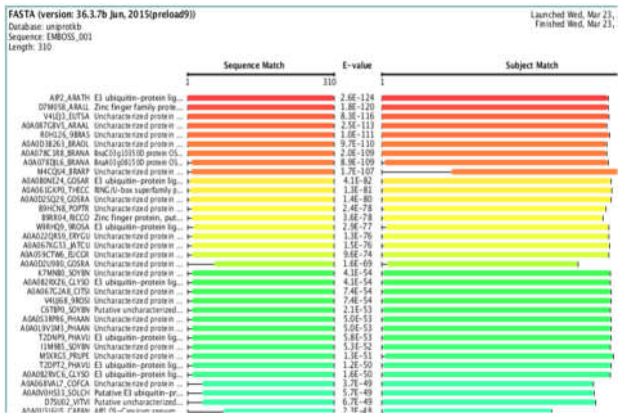
For unknown sequences: Use NGS or Mass Spectrometry



Step 2: Choose a type of FASTA



Step 4: Tabulated Search Results



Step 2: Type of FASTA

http://fasta.bioch.virginia.edu/fasta_docs/fasta35.shtml

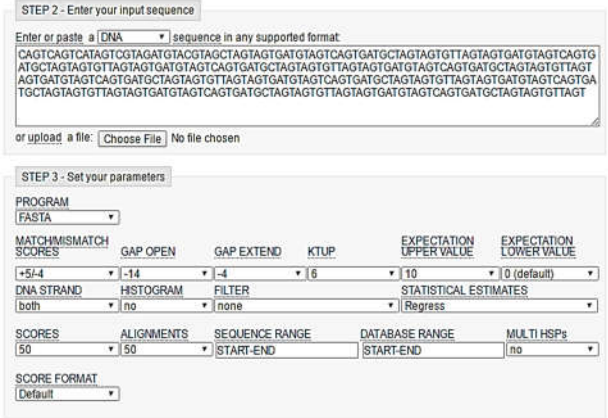
fasta35, fasta35_t*	scan a protein or DNA sequence library for similar sequences
fastx35, fastx35_t	compare a DNA sequence to a protein sequence database, comparing the translated DNA sequence in forward and reverse frames.
tfastx35, tfastx35_t	compare a protein sequence to a DNA sequence database, calculating similarities with frameshifts to the forward and reverse orientations.
fasty35, fasty35_t	compare a DNA sequence to a protein sequence database, comparing the translated DNA sequence in forward and reverse frames.
tfasty35, tfasty35_t	compare a protein sequence to a DNA sequence database, calculating similarities with frameshifts to the forward and reverse orientations.
fasts35, fasts35_t	compare unordered peptides to a protein sequence database
fastn35, fastn35_t	compare ordered peptides (or short DNA sequences) to a protein (DNA) sequence database
tfasts35, tfasts35_t	compare unordered peptides to a translated DNA sequence database
fastf35, fastf35_t	compare mixed peptides to a protein sequence database
tfastf35, tfastf35_t	compare mixed peptides to a translated DNA sequence database
ssearch35, ssearch35_t	compare a protein or DNA sequence to a sequence database using the Smith-Waterman algorithm.

Align	DB:ID	Source	Length	Score (Bits)	Identities %	Positives %	E()
1	SP:AIP2_ARATH	E3 ubiquitin-protein ligase AIP2 OS=Arabidopsis thaliana GN=AIP2 PE=1 SV=1	310	453.0	100.0	100.0	2.6E-124
2	TR:D7M058_ARALL	Zinc finger family protein OS=Arabidopsis lyrata subsp. lyrata GN=ARALYDRAFT_488997 PE=4 SV=1	310	440.3	96.5	99.7	1.8E-120

Conclusion: FASTA is an online tool which performs quick alignments on biological sequences.

- Depending on your need you can choose a specific type of FASTA to compare and score alignments.

Step 3: Setup Search Parameters



Lecture 70

Biological Databases & Online Tools

- Biomolecular sequences (DNA, RNA, Proteins)
- Biomolecular Structures (RNA, Proteins, Drugs)
- Biomolecular Interaction
- This information needs to be stored and retrieved!
- Sequences are obtained from genome sequencing and mass spectrometry.
- Structures are obtained from X-Ray Crystallography, Atomic Force Microscopy & Nuclear Magnetic Resonance Spectroscopy.

Storage & Sharing

- This sequence and structure data is very useful for researchers
- Vast amounts of such data exists
- Moreover, this data is rapidly accumulating

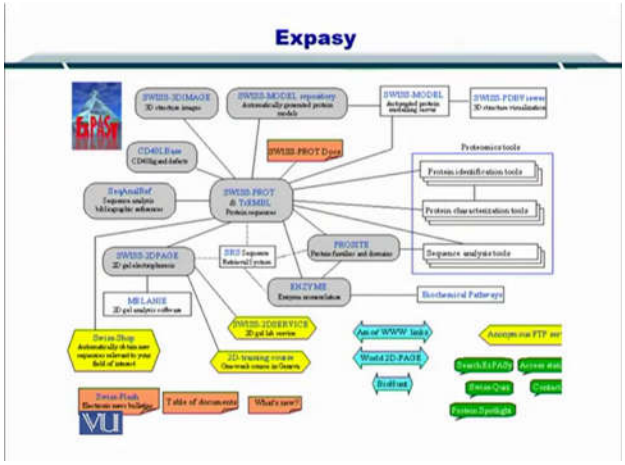
- So how do we store this data
- And how do we share it with others?
- Databases are formed to store and share this data.

Objective:

- Make biological data available to scientists in computer-readable form
- For handling, sharing and analysis of the data
- The best way to share is to keep this data on the web!

Conclusion:

- Several sequence, structure and molecular interaction databases exist
- These are available online on the web
- Users can freely access and download such data.



Lecture 71

Expasy

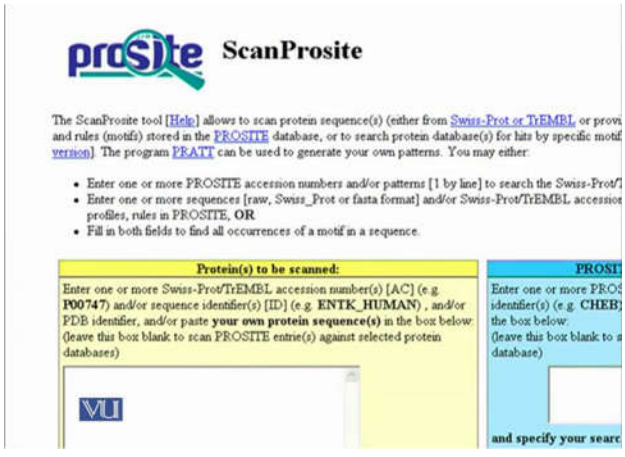
- Developed by Swiss Bioinformatics Institute (SIB)
- Website provides access to databases and tools
- Proteomics, Genomics, Phylogeny, Systems biology, Population genetics, transcriptomics etc.

Conclusion:

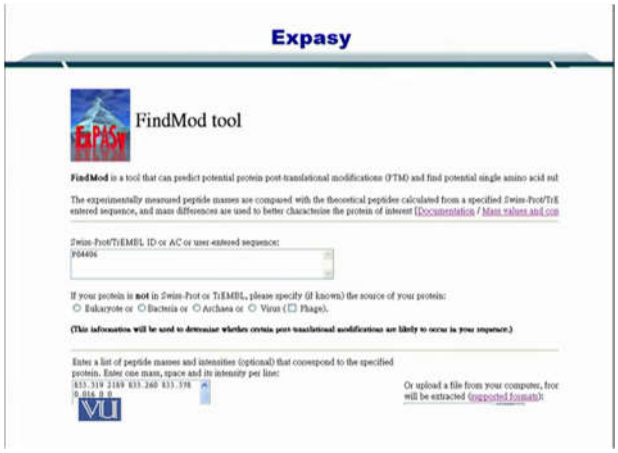
- Expasy provides access to a variety of online databases and tools.
- Depending upon your requirement, you find sequence information from Expasy.

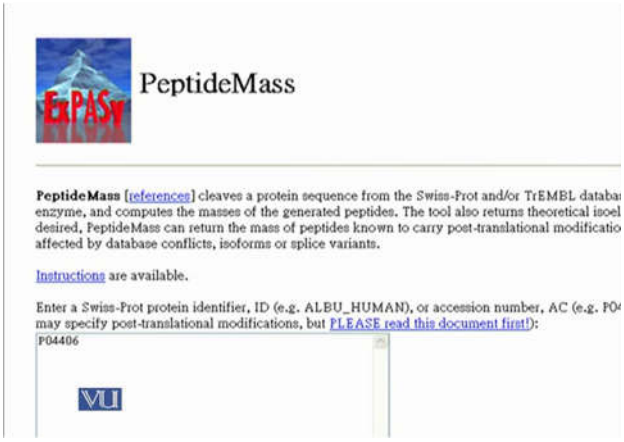








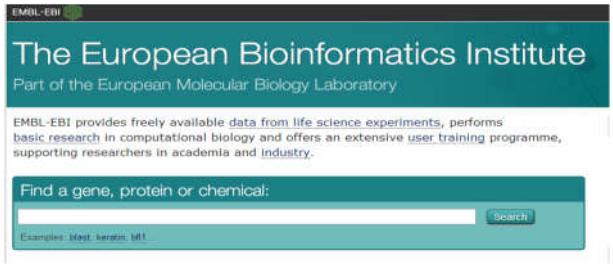


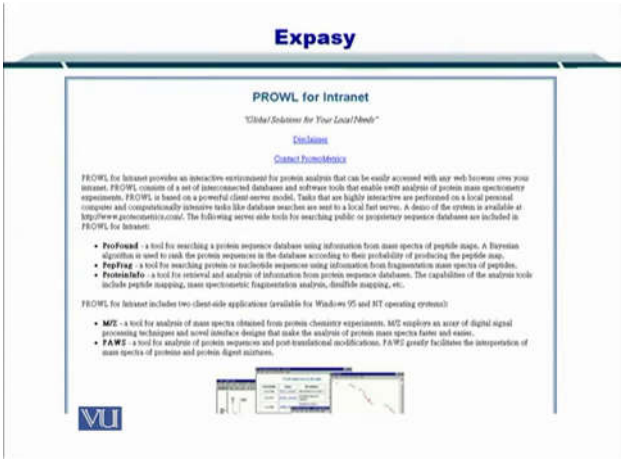


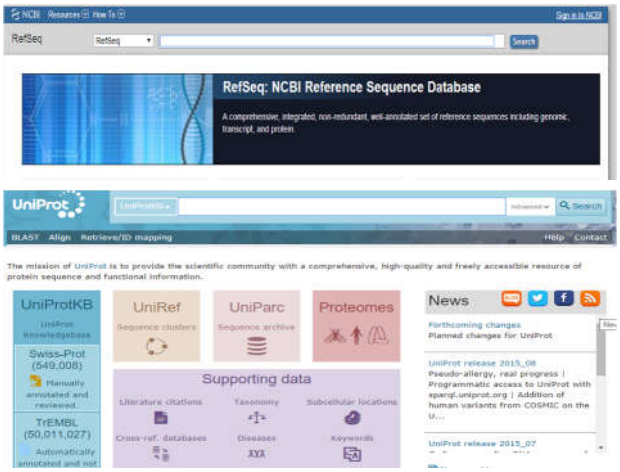
Lecture 72

UniProt & SwissProt:

- Protein Sequences are reported from sequencing experiments
- This data is stored in protein sequence databases
- The famous ones include UniProt & SwissProt

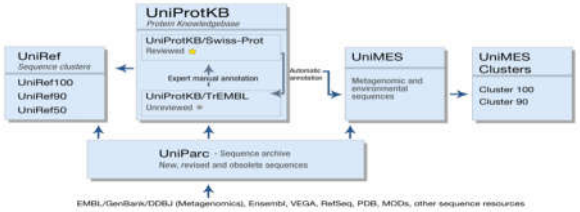






Swiss-Prot contains human curated protein information

- Accession number, unique identifier
- The sequence
- Molecular mass
- Observed and predicted modifications



Conclusion: Protein sequences from various species and organisms can be found in uniprot. SwissProt is the manually annotated version of the UniProt Database

P0CG47 - UBB_HUMAN



Lecture 73

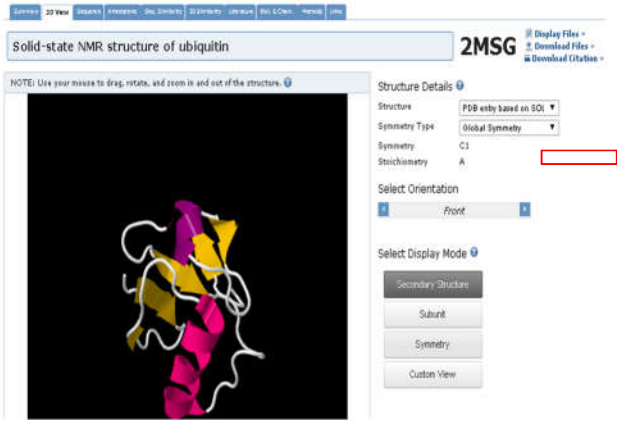
Protein Databank

- Protein Data Bank is the premier resource of protein structures
- These structures have been determined using experimental techniques
- Open & Free

Conclusion:

- Protein Data Bank provides Cartesian coordinates of each atom in the protein structure.
- Over 50,000 protein structures are reported and present in this database.

P0CG47 - UBB_HUMAN






```
ATOM      1  N   MET A   1     102.329 111.862  92.452  1.00 78.64           N
ATOM      2  CA  MET A   1     103.332 112.165  93.516  1.00 77.39           C
ATOM      3  C   MET A   1     103.877 113.584  93.255  1.00 76.87           C
ATOM      4  O   MET A   1     103.802 114.075  92.129  1.00 78.54           O
ATOM      5  CB  MET A   1     104.437 111.099  93.495  1.00 78.51           C
ATOM      6  CG  MET A   1     105.176 110.881  94.812  1.00 76.25           C
ATOM      7  SD  MET A   1     106.505 112.076  95.116  1.00 76.95           S
ATOM      8  CE  MET A   1     107.077 111.551  96.763  1.00 73.31           C
ATOM      9  N   GLU A   2     104.404 114.235  94.292  1.00 74.31           N
ATOM     10  CA  GLU A   2     104.917 115.609  94.211  1.00 70.70           C
ATOM     11  C   GLU A   2     105.963 115.909  93.143  1.00 68.10           C
ATOM     12  O   GLU A   2     106.048 117.044  92.683  1.00 67.95           O
ATOM     13  CB  GLU A   2     105.464 116.053  95.574  1.00 75.97           C
ATOM     14  CG  GLU A   2     106.692 115.246  96.029  1.00 82.57           C
ATOM     15  CD  GLU A   2     107.263 115.682  97.378  1.00 83.76           C
ATOM     16  OE1 GLU A   2     106.611 115.396  98.412  1.00 86.90           O
ATOM     17  OE2 GLU A   2     108.373 116.276  97.401  1.00 81.23           O
ATOM     18  N   ASN A   3     106.789 114.924  92.784  1.00 64.50           N
ATOM     19  CA  ASN A   3     107.834 115.134  91.773  1.00 59.93           C
ATOM     20  C   ASN A   3     107.381 114.767  90.340  1.00 55.41           C
ATOM     21  O   ASN A   3     108.159 114.856  89.416  1.00 53.33           O
ATOM     22  CB  ASN A   3     109.086 114.308  92.095  1.00 62.56           C
ATOM     23  CG  ASN A   3     109.531 114.441  93.535  1.00 62.42           C
ATOM     24  OD1 ASN A   3     109.045 113.724  94.408  1.00 63.06           O
ATOM     25  ND2 ASN A   3     110.484 115.326  93.787  1.00 62.31           N
ATOM     26  N   PHE A   4     106.131 114.348  90.224  1.00 48.13           N
ATOM     27  CA  PHE A   4     105.613 113.946  88.941  1.00 44.38           C
ATOM     28  C   PHE A   4     104.326 114.601  88.521  1.00 48.72           C
```

Lecture 74

Review of Sequence Alignment

- Biological Sequence Information from DNA, RNA & Proteins
- How do we obtain this information?
- How do we store this information?
- How do we compare sequences?

Obtaining Sequences:

- Gene Sequencing
  - Next generation sequencing or Whole genome sequencing
- Protein Sequencing
  - Mass Spectrometry
  - Edman Degradation

Sequence Storage:

- Sequence information is stored digitally
- Databases are designed to store sequence data
- Several databases exist depending on the type of sequence data

Sharing & Access to Sequence Data

- Sequence databases are shared via online websites
- Access to several such websites is free
- Data can be downloaded or searched on these website

Usage of Sequence Data

Sequence data can be used to obtain:

- Similarity of sequences
- Evolutionary History
- Predict the function of molecules

Conclusion: Sequence information can help us obtain critical insights into life.

- This is undertaken by sequence alignment process
- Local and Global alignment can be performed to comparing sequences

Lecture 75

GenBank

- GenBank: Public database of nucleotide sequences for over 200,000 organisms
- Growing exponentially
- http://www.ncbi.nlm.nih.gov/genbank/

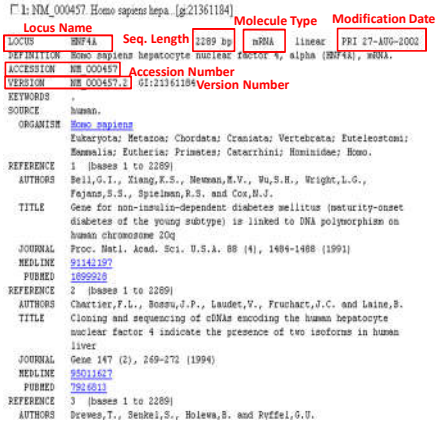
Collaboratively maintained by:

- DNA DataBank of Japan (DDBJ)
- European Molecular Biology Laboratory (EMBL)
- GenBank at NCBI

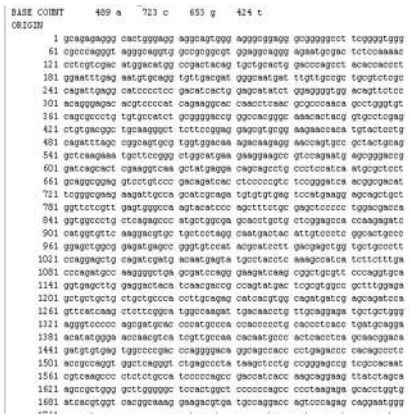
Data in GenBank

- Entries can be a contiguous stretch of DNA or RNA sequences and their annotations
- Entries are updated every two months

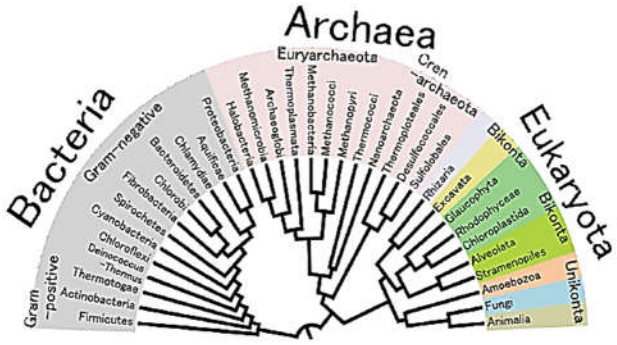




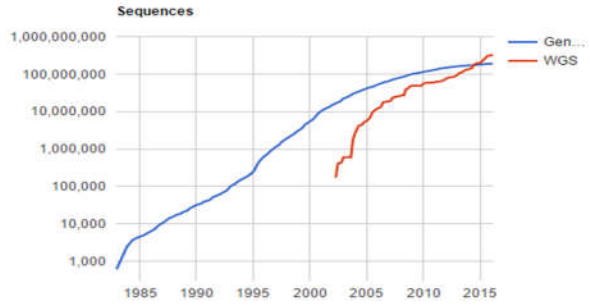




Phylogenetic Trees

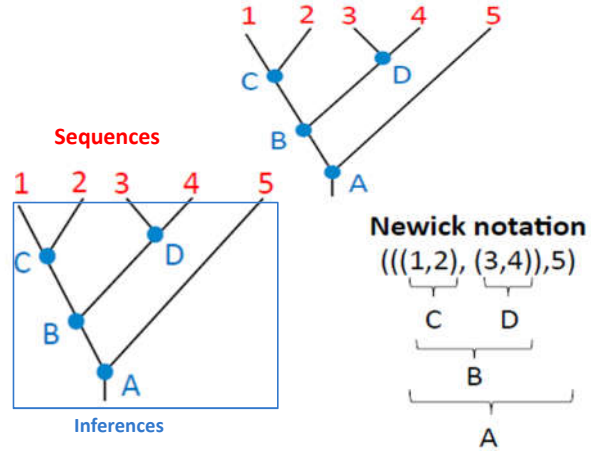


http://bacterialphylogeny.info/overview.html



Conclusion:

- All known nucleotide sequences can be obtained from GenBank.
- Thousands of genomes are available online on GenBank, for free and are searchable.



Lecture 76

Molecular Evolution & Phylogeny

- Molecular evolution helps species adapt to their environmental stresses
- Genes and Proteins are modified in this process
- All molecules have an evolutionary history
- Phylogenetics is the science of studying evolutionary relationships. Phylogenetics has led to the creation of relationship trees b/w various species of Bacteria, Archaea, & Eukaryota

Types of Phylogenetic Trees

Scaled Trees: Branch lengths are equal to the magnitude of change in the nodes

Unscaled Trees: Only representing the relationship between sequences

Conclusion:

- Phylogenetics is the study of extracting evolutionary relationships between species
- Sequence information from each species is used to measure the difference between the species

Lecture 77

Evolution of Sequences

- DNA as cellular memory
- Proteins as DNA translation products
- Evolution as a requirement to survive in diverse environments.

Method of Change: DNA gets modified by:

- Mutation & Substitution
- Insertion
- Deletion

Discussion:

- Over time, species evolve to adapt to their circumstances
- Since the environment and circumstances may be different for each species, they evolve uniquely
- Unique evolutionary pressures may be encountered by each cell
- However, in which sequence they are presented to the cells is also unique
- Combinations of evolutionary factors

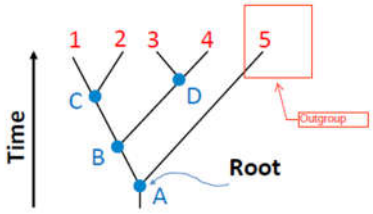
Conclusion:

- The evolutionary events and their combination impart relationships between sequences
- These relationships are explored in Phylogenetics
- Several algorithms exist for finding such relationships

Root node is the ancestor of all other nodes!

The direction of evolution cannot be inferred from unrooted trees!

Rooted Trees



Root node is the ancestor of all other nodes!

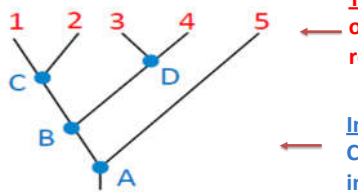
The direction of evolution is from ancestor to the terminal nodes!

Lecture 78

Concepts and Terminologies - I

- Phylogenetics involves processing sequence information from different species to find evolutionary relationships
- Output from such studies include Phylogenetic Trees

Phylogenetic Trees



Terminal Nodes: Genes or Proteins from related Organisms

Internal Nodes: Common ancestor inferred by processing sequence data

Conclusion

- Phylogenetics specifies evolutionary relationship with the help of trees
- Trees can be rooted or unrooted
- Rooted trees can show temporal evolutionary direction

Lecture 79

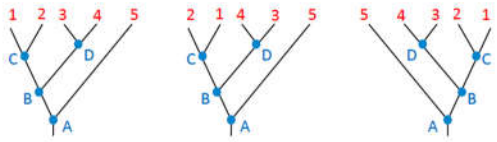
Concepts and Terminologies - II

- Rooted and Unrooted trees can be used to show phylogenetic relationships between sequences
- Let's examine the properties of these trees further!

Conclusion:

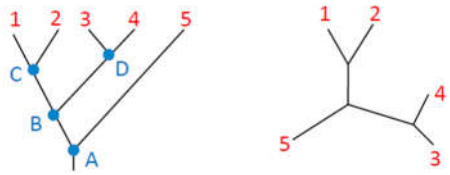
- Rooted and Unrooted trees have their own advantages and disadvantages.
- Depending on our requirement, we can choose between them!

Layout of Trees

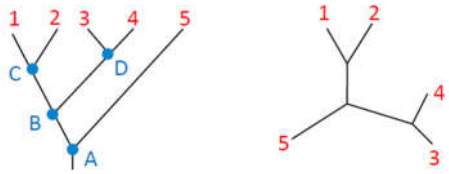


All trees have the same meaning!

Rooted Trees vs. Unrooted Trees



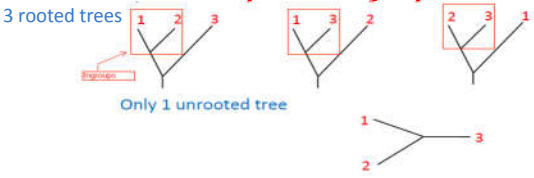
Rooted Trees vs. Unrooted Trees



Apparently, we should always use rooted trees!

BUT, what if we do not have an out group?

Rooted trees are computationally expensive



Rooted trees are computationally expensive

Number of Sequences	Number of Rooted Trees	Number of Unrooted Trees
3	3	1
4	15	3
5	105	15
10	34, 459,425	2,027,025
15	213,458,046,767,875	7,905,853,580,625
...	...	...

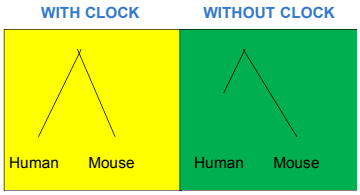
How many trees?

Rooted: $(2n-3)! / 2^{n-2}(n-2)!$

Unrooted: $(2n-5)! / 2^{n-3}(n-3)!$

Edge length with and without a clock!

Since the rate of evolution in species is different, it needs to be considered!



Least Squares Distance Method

Branch lengths, represent the "observed" distances between sequences (*i* & *j*)

Observation

D(Human,Chimp) = 0.3

D(Human,Gorilla) = 0.4

D(Chimp, Gorilla) = 0.5

Find X, Y and Z such that *D (i , j)* are conserved ?



Conclusion: Several methods exist for constructing phylogenetic trees. Broadly, they belong to objective methods or clustering methods.

- We will study UPGMA and Distance Methods.

Lecture 80

Algorithms and Techniques

- Rooted and Unrooted trees can be used to show phylogenetic relationships between sequences
- Several types of algorithms exist which are divided into two classes.

Methods for constructing trees

Clustering Approach	Objective based Methods
UPGMA	Least Square Distances
WPGMA	Maximum Likelihood
Neighbor Joining	Maximum Parsimony
Single Linkage	
Complete Linkage	

Lecture 81

Introduction to UPGMA

- Phylogenetic trees can be used to show phylogenetic relationships between sequences
- To construct these trees, several types of algorithms exist which are divided into two classes

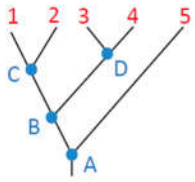
Methods for constructing trees

Clustering Approach	Objective based Methods
UPGMA	Least Square Distances
WPGMA	Maximum Likelihood
Neighbor Joining	Maximum Parsimony
Single Linkage	
Complete Linkage	

UPGMA: Unweighted Pair – Group Method using arithmetic Averages

UPGMA: Unweighted Pair – Group Method using arithmetic Averages

- Two sequences with with the **shortest evolutionary distance** between them are considered.
- These sequences will be **the las to diverge**, and **represented by the most recent internal node.**



Methods for constructing trees

Clustering Approach	Objective based Methods
UPGMA	Least Square Distances
WPGMA	Maximum Likelihood
Neighbor Joining	Maximum Parsimony
Single Linkage	
Complete Linkage	

Calculating distance between two clusters:

Cluster X + Cluster Y = Cluster Z

Calculate the distance of a cluster (e.g. W) to the new cluster Z

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

N_x is the number of sequences in cluster x

Assume we have N sequences

Cluster X has N_X sequences, cluster Y has N_Y sequences

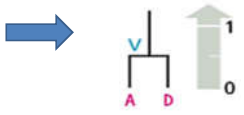
d_{XY} : the evlotionary distance between X and Y

$$d_{XY} = \frac{1}{N_X N_Y} \sum_{i \in X, j \in Y} d_{ij}$$

Methods for constructing trees

The distance matrix is obtained using pairwise sequence alignment.

d_{ij}	A	B	C	D	E	F
A	–	6	8	1	2	6
B		–	8	6	6	4
C			–	8	8	8
D				–	2	6
E					–	6



A – D becomes a new cluster lets say V. We have to modify the distance matrix!. What are the distances between:

- V and B (Calculate),
- V and C,
- V and E,
- V and F.

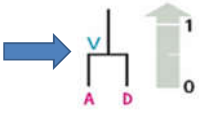
d_{ij}	A	B	C	D	E	F
A	–	6	8	1	2	6
B		–	8	6	6	4
C			–	8	8	8
D				–	2	6
E					–	6

Methods for constructing trees

The distance matrix is obtained using pairwise sequence alignment.

$$d_{XY} = \frac{1}{N_X N_Y} \sum_{i \in X, j \in Y} d_{ij}$$

d_{ij}	A	B	C	D	E	F
A	–	6	8	1	2	6
B		–	8	6	6	4
C			–	8	8	8
D				–	2	6
E					–	6



A – D becomes a new cluster lets say V. We have to modify the distance matrix!. What are the distances between:

- V and B (Calculate),
- V and C,
- V and E,
- V and F.

d_{ij}	A	B	C	D	E	F
A	–	6	8	1	2	6
B		–	8	6	6	4
C			–	8	8	8
D				–	2	6
E					–	6

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

$$d_{VB} = \frac{N_A d_{AB} + N_D d_{DB}}{N_A + N_D} = \frac{1*6 + 1*6}{1+1} = 6$$

Conclusion:

- UPGMA is a clustering algorithm which can help us compute phylogenetic trees.
- We will see the detailed working of this approach in later modules.

What are the distances b/w :**V & B (Calculate)!**

d_{ij}	A	B	C	D	E	F
A	–	6	8	1	2	6
B		–	8	6	6	4
C			–	8	8	8
D				–	2	6
E					–	6

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

$$d_{VB} = \frac{N_A d_{AB} + N_D d_{DB}}{N_A + N_D} = \frac{1*6 + 1*6}{1+1} = 6$$

What are the distances between: **V and C (Calculate)!**

$$d_{VC} = \frac{N_A d_{AC} + N_D d_{DC}}{N_A + N_D} = \frac{1*8 + 1*8}{1+1} = 8$$

Lecture 82

UPGMA - I

- UPGMA has two components to it. These include distance calculations b/w two clusters & b/w two trees.

Building trees using UPGMA

Combining Clusters: Cluster X + Cluster Y = Cluster Z

Calculate the distance of each cluster (e.g. W) to the new cluster Z

N_x is the number of sequences in cluster x

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

Calculating the distance b/w two trees:

Assume we have N sequences

Cluster X has N_X sequences, cluster Y has N_Y sequences

d_{XY} : the evlotionary distance between X and Y

What are the distances between: **V and E (Calculate)!**

$$d_{VE} = \frac{N_A d_{AE} + N_D d_{DE}}{N_A + N_D} = \frac{1*2 + 1*2}{1+1} = 2$$

What are the distances between: **V and F (Calculate)!**

$$d_{VF} = \frac{N_A d_{AF} + N_D d_{DF}}{N_A + N_D} = \frac{1*6 + 1*6}{1+1} = 6$$

Conclusion:

- UPGMA starts with creating clusters of sequences which are the closest
- Next, distance is computed between the new cluster and the remaining sequences
- The process is repeated for all sequences

Lecture 83

UPGMA - II

- UPGMA steps include distance calculations between two clusters and between two trees
- We formed clusters from sequences which had the shortest distance.

Building trees using UPGMA

Combining Clusters: Cluster X + Cluster Y = Cluster Z

Calculate the distance of each cluster (e.g. W) to the new cluster Z

N_x is the number of sequences in cluster x

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

Calculating the distance b/w two trees:

Assume we have N sequences. Cluster X has N_X sequences, cluster Y has N_Y sequences

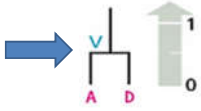
d_{XY} : the evlotionary distance between X and Y

Methods for constructing trees

The distance matrix is obtained using pairwise sequence

$$d_{XY} = \frac{1}{N_X N_Y} \sum_{i \in X, j \in Y} d_{ij}$$

d_{ij}	A	B	C	D	E	F
A	-	6	8	1	2	6
B		-	8	6	6	4
C			-	8	8	8
D				-	2	6
E					-	6



d_{ij}	A	B	C	D	E	F
A	-	6	8	1	2	6
B		-	8	6	6	4
C			-	8	8	8
D				-	2	6
E					-	6

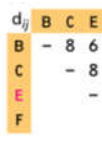
$$d_{VB} = \frac{N_A d_{AB} + N_D d_{DB}}{N_A + N_D} = \frac{1*6 + 1*6}{1+1} = 6$$

$$d_{VC} = \frac{N_A d_{AC} + N_D d_{DC}}{N_A + N_D} = \frac{1*8 + 1*8}{1+1} = 8$$

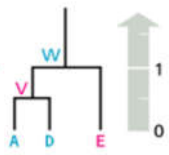
$$d_{VE} = \frac{N_A d_{AE} + N_D d_{DE}}{N_A + N_D} = \frac{1*2 + 1*2}{1+1} = 2$$

$$d_{VF} = \frac{N_A d_{AF} + N_D d_{DF}}{N_A + N_D} = \frac{1*6 + 1*6}{1+1} = 6$$

d_{ij}	A	B	C	D	E	F
A	-	6	8	1	2	6
B		-	8	6	6	4
C			-	8	8	8
D				-	2	6
E					-	6



d_{ij}	B	C	E	F	V
B	-	8	6	4	6
C		-	8	8	8
E			-	6	2
F				-	6



- V – E becomes a new cluster lets say W
- Now we have to modify the distance matrix again.
- What are the distances between:
  - **W and B,**
  - **W and C,**
  - **W and F.**

Conclusion: Once a cluster is selected and its distance is computed with all other sequences, we update the distance matrix. Next, we select the shortest distance from the new matrix and repeat the process.

Lecture 84

UPGMA - III

- UPGMA has two components to it. These include progressive distance calculations b/w two clusters or b/w two trees

Building trees using UPGMA

Combining Clusters: Cluster X + Cluster Y = Cluster Z

Calculate the distance of each cluster (e.g. W) to the new cluster Z

N_x is the number of sequences in cluster x

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

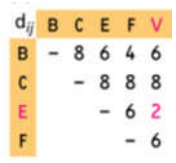
Calculating the distance b/w two trees:

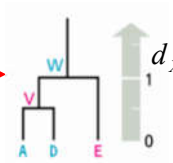
Assume we have N sequences

Cluster X has N_X sequences, cluster Y has N_Y sequences

d_{XY} : the evlotionary distance between X and Y

d_{ij}	B	C	E	F	V
B	-	8	6	4	6
C		-	8	8	8
E			-	6	2
F				-	6





$$d_{XY} = \frac{1}{N_X N_Y} \sum_{i \in X, j \in Y} d_{ij}$$

- V – E becomes a new cluster lets say W
- Now we have to modify the distance matrix again.
- What are the distances between:
  - **W and B,**
  - **W and C,**
  - **W and F.**

$$d_{WB} = \frac{N_V d_{VB} + N_E d_{EB}}{N_V + N_E} = \frac{2*6 + 1*6}{2+1} = 6$$

$$d_{WC} = \frac{N_V d_{VC} + N_E d_{EC}}{N_V + N_E} = \frac{2*8 + 1*8}{2+1} = 8$$

$$d_{WF} = \frac{N_V d_{VF} + N_E d_{EF}}{N_V + N_E} = \frac{2*6 + 1*6}{2+1} = 6$$

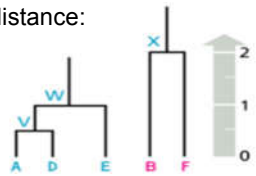
- New matrix:

d_{ij}	B	C	E	F	V
B	-	8	6	4	6
C		-	8	8	8
E			-	6	2
F				-	6

d_{ij}	B	C	F	W
B	-	8	4	6
C		-	8	8
F			-	6

- Cluster according to min distance:

d_{ij}	B	C	F	W
B	-	8	4	6
C		-	8	8
F			-	6



Conclusion: Now we have formed three clusters

- Also, two separate trees have been formed. Next, we need to join these trees to create a complete tree.

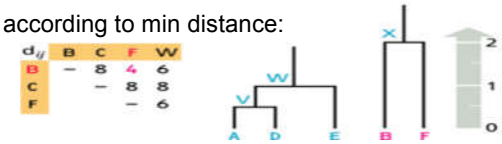
Lecture 85

UPGMA – IV

- Application of UPGMA resulted in formation of two sub-trees
- The need now was to join them into a single tree
- Let's see how that is done
- New matrix:



- Cluster according to min distance:

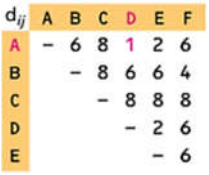


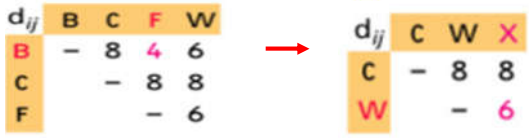
- F – B becomes a new cluster lets say X
- We have to modify the distance matrix yet again
- What is the distance between trees: **W and X.**

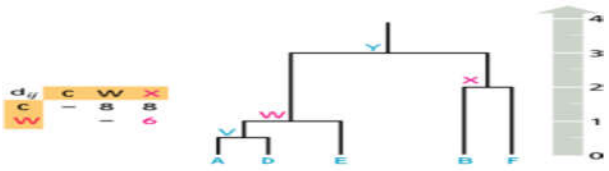
$$d_{WX} = \frac{1}{N_W N_X} \sum_{i \in W, j \in X} d_{ij} =$$

$$\frac{1}{N_W N_X}(d_{AB} + d_{AF} + d_{DB} + d_{DF} + d_{EB} + d_{EF}) =$$

$$\frac{1}{3*2}*(6+6+6+6+6+6) = 6$$







X – W becomes a new cluster lets say Y

We have to modify the distance matrix

What is the distance between: **Y and C.**

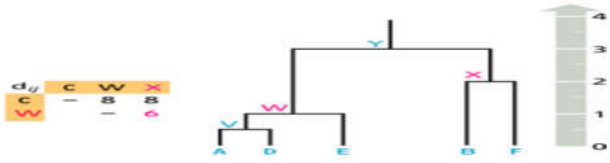
Conclusion:

- We have now seen how trees are generated & connected. Next, we need to finalize the tree by adding the last two clusters.

Lecture 86

UPGMA – V

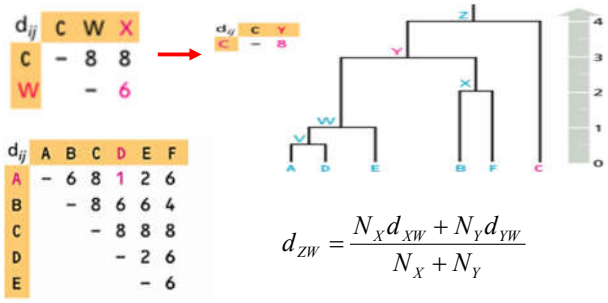
- Application of UPGMA resulted in formation of two sub-trees
- The need now was to join them into a single tree
- Let's see how that is done.



X – W becomes a new cluster lets say Y

We have to modify the distance matrix

What is the distance between: **Y and C.**



$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

Conclusion: Un-weighted Pair Group Method using Arithmetic Averages is a clustering method to construct phylogenetic trees.

Non-clustering methods such as Maximum Parsimony may be used for making trees as well

Lecture 87

DNA to RNA Sequences

Textbook: Chapter 5

"Bioinformatics Sequence & Genome Analysis" by David Mount

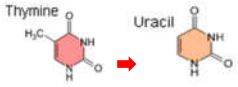
Motivation:

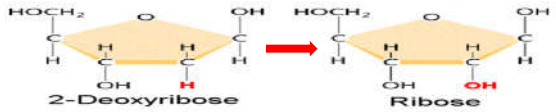
- In earlier days, RNA was only considered as a passive intermediary between DNA and proteins.
- Now we know that multiple types of RNA exist e.g. mRNA, tRNA, miRNA and siRNA etc.
- More so, that they perform critical functions in gene and protein expression.

Many viruses assemble their genomes from RNAs. They are therefore called RNA viruses. Examples include Human Immunodeficiency Virus and Hepatitis C Virus.

Differences between DNA & RNA

1. **Thymine is replaced by Uracil**
2. **RNA is single stranded**
3. **RNA contains Ribose**

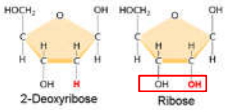




Impact of 2 OH groups

The two hydroxyl groups make RNA less stable than DNA because it is more prone to hydrolysis

Hence, RNA has a short life span and is more prone to degradation



Lecture 88

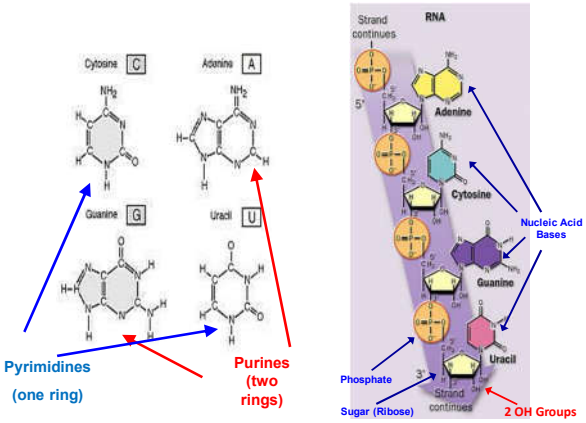
Types of RNA & their function

Motivation: RNA can be divided into two categories

- Coding RNAs
- Non-Coding RNAs
- Coding RNAs as is obvious from their name, code for Proteins
- Non-Coding RNAs regulate/assist in the process of translation

Types of RNA:

1. Messenger RNA (mRNA)
2. Transfer RNA (tRNA)
3. Ribosomal RNA (rRNA)
4. Micro RNAs (miRNA)
5. Small Interfering RNA (siRNA) and so on..



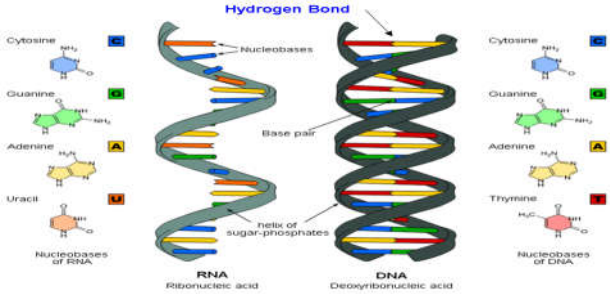
http://howmed.net/biochemistry/nucleotides-and-nucleic-acids/

Messenger RNA: Only 5-10% of RNA content in a cell

1. Sequence: Variable
2. Size: Variable
3. Function: Carry genetic information from DNA to Ribosomes where proteins are being assembled
4. Cap: 5' end is capped (with 7-Methyl Guanosine Triphosphate)

M5' and 3' Caps in mRNA

5. The 5' cap helps identify mRNAs at the Ribosomes
6. The 5' cap also acts as a shield against 5' exonuclease, thus leading to an increase in mRNA stability
7. The 3'-end of mRNA has a polyA tail (around 30-200 adenylate residues) which help shield against 3' exonucleases

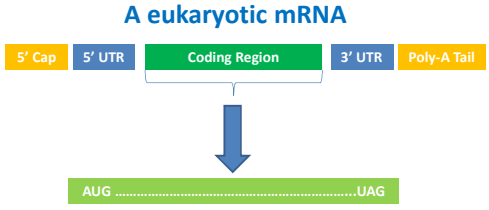


Images courtesy Garland Science, and Nelson & Cox (2000) "Understand! Biochemistry" Leninger Principles of Biochemistry

Diverse forms of RNA: Not all RNA molecules are the same. RNAs may differ in their nucleotide sequence!. As RNA have to take up multiple roles in a cell. Hence, it is to know their various forms & types.

UTR and Coding Regions:

8. The 5' and 3' regions are both cushioned with untranslated regions (UTRs)
9. Besides the non-coding regions at 5' & 3' ends, the cap & the poly-A tail, the intermediary region encodes for proteins.



Note: RNA sequence is complementary to the DNA sequence & is translated as codons of three nucleotides!

Diversity in RNA:

- So how exactly do these RNAs vary in function?
- Function is a property of the RNA structure, which is formed from the RNA sequence.
- Hence it is important to look at how these RNAs fold and form structures.

Lecture 89

Significance of RNA structures

Motivation:

RNAs can adopt complex 3D structures.

- Such structural properties enable RNA to act in various roles.
- Hence, its important to look at how these structures are formed.

How do RNA structures arise?

- Remember that RNAs are composed of sugars, phosphates and nucleotides
- Each Nucleotide is covalently bonded to the backbone
- The sequences is then represented as a combinations of A, C, U and G

Where do the structures come from?

- Whereas covalent bonds help bring nucleotides together within the RNA backbone. Nucleotides can make hydrogen bonds b/w themselves

Hydrogen bonding b/w nucleotide bases

- 'A' can make hydrogen bonds with 'U'
- 'G' makes hydrogen bonds with 'C'
- 'G' can also make hydrogen bonds with 'U' (Wobble Pair)

Onwards from RNA structural information...

- Remember, RNAs could perform function only due to their structure
- Therefore, structural information of an RNA can also help predict its function

Various roles that an RNA can play due to its structure:

- DNA information transfer
- Regulatory roles
- Catalytic roles
- Defense & immune response
- Structure-based special roles

Enzymatic activity responsible for peptide bond formation b/w amino acids in growing peptide chain rRNA, which is major structural component of ribosomes.

Lecture 90

RNA folding & energy of folding

Motivation: What advantage do the RNA molecules achieve by forming structures? They become more stable. Can there be more than one stable forms of an RNA structure?

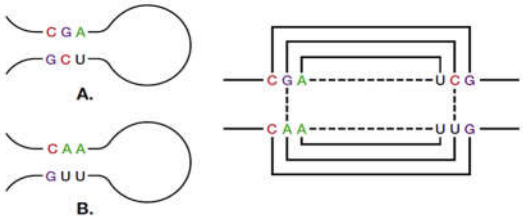
A review of free energy: "Gibbs Free Energy" is the free energy of an RNA molecule available for reaction. The smaller, the better!. RNA structure formation lowers the free energy

Energy of RNA structures

- RNA structures have the lowest (or close) quantity of free energy. In cases where RNA can take two structural forms, one can select the one with lower energy state.

Folding & energy given out

- Energy is continuously given out as the RNA molecule folds by pairing complementary bases.



Calculating energies of structures

- The stabilizing energy associated with stacking base pairs in a double-stranded region (-ve)
- The destabilizing influence of unpaired regions (+ve)

Calculating energies of structures

- We can calculate the overall energy of RNA structures by summing up energies given out during the process of folding.

	Y: A	C	G	U
X: A	.	.	.	−2.1
C	.	.	−3.3	.
G	.	−2.4	.	−1.4
U	−2.1	.	−2.1	.

Food for thought:

- A step further would be to add the destabilizing energies.
- But how will we know the amount of destabilizing energy?
- For that we may factor in ways in which RNA can be destabilized

Lecture 91

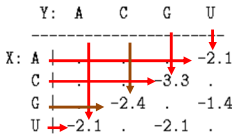
Calculating Energies of Folding – An Example:

- RNA comprises of 4 nucleotide bases (A, U, G & C)
- These bases are attached to Ribose sugars in the RNA backbone
- Hydrogen bonds can form between complementary nucleotide bases

Structure Formation:

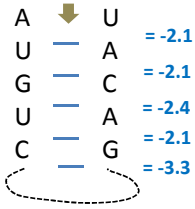
- Complementary nucleotides i.e. G&C or A&U form hydrogen bonds
- Upon formation of the bond, energy is released
- Thereby, RNA molecule becomes more stable

1. Nucleotide Pairing Energy Output Table:



2. RNA Sequence:
"AUGUC…….GACAU"

Hence, the total energy released as a result of the formation of
5 H-Bonds = -2.1 -2.1-2.4-2.1-3.3
=-12.0 kcal/mol



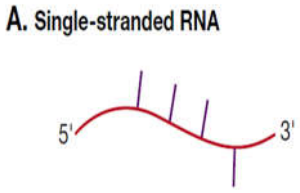
Conclusion:

- 5 nucleotides formed H-Bonds
- This bond formation released energy (-12.0 kcal/mol)
- RNA molecule took up a 2' structure
- Hence became more stable

RNA 2' Structures

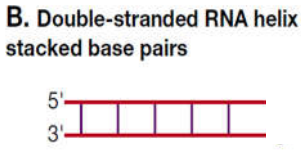
Folding:

- RNA sequence extends from its 5' end to 3' end.
- Upon folding, 3' end may fold on to the 5' end.



RNA 2' Structures:

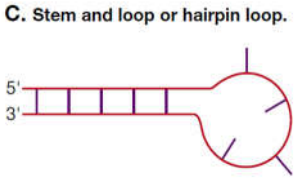
- The first 2' structure is
- the Helix. Note, the difference b/w DNA & RNA Helix.



Unlike the DNA helix, the RNA helix is formed when the RNA folds onto itself.

RNA 2' Structures

- The second 2' structure
- is the hairpin loop



RNA 2' Structures

- The loop of the hairpin must at least four bases long to avoid steric hindrance with base-pairing in the stem part of the structure.
- Note that hairpins reverses the chemical direction of the RNA molecule.

Lecture 92

Types of RNA Secondary Structures - I

- RNA Complementary bases in primary structure bond together to fold RNA into a secondary structure
- How many types of secondary structures can be possibly formed?

A Word on Notations:

- Primary structure is the nucleotide sequence and is denoted as 1'
- Secondary structure is the folding of an RNA on to itself and is represented by 2'

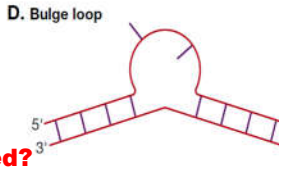
Preferred RNA 2' Structures:

- To answer the question, YES! There are structural patterns in RNA 2' structure. These include:
  Helices, Loops, Bulges & Junctions

Lecture 93

Types of RNA Secondary Structures - II

- RNA 1' sequence (5' – 3') can fold onto itself to make 2' structures.
- Two previous examples of 2' structures include Helix and Hairpin structures.



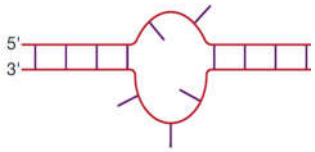
RNA 2' Structures

- **The third type of 2' structure is the bulge loop**

How are Bulges Formed?

- Bulges, are formed when a double-stranded region cannot form base pairs perfectly.
- Bulges can be asymmetric with varying number of base pairs on one side of the loop.

E. Interior loop



RNA 2' Structures

- **The fourth type of 2' structure is the interior loop.**

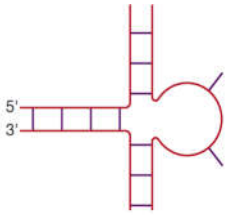
How are Interior Loops Formed?

- Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop!

Lecture 94

Types of RNA Secondary Structures - III

- RNA can be fold to form helices, bulge loops, and interior loops. An other 2' RNA structure is the Junction or Intersection



RNA 2' Structures

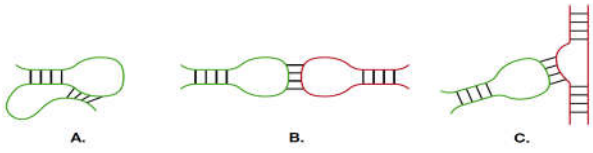
- The fifth type of 2' structure is the Junction or Intersection.

How are Junctions Formed?

- Junctions include two or more double-stranded regions converging to form a closed structure
- The unpaired bases appear as a bulge

RNA Tertiary Structures

- Unpaired bases in two 2' structures form hydrogen bonds with each other!



What happens when unpaired bases in 2' region bond?

- RNA tertiary structures are formed
- Are they more stable than 2' structures?
- What is their role in biological systems?

Lecture 95

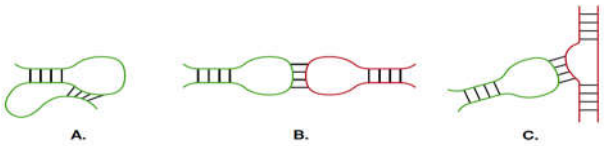
RNA Tertiary Structures

Motivation:

- 2' RNA structures form as a result of bonding between complementary nucleotides within an RNA molecule
- However, some nucleotides are still left open for interaction.

Role of the Unpaired Bases?

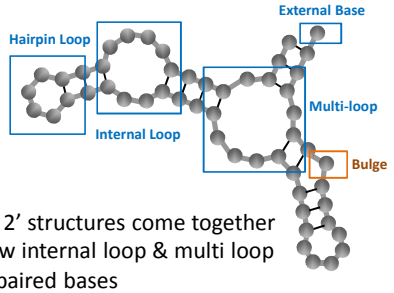
- Unpaired bases in two or more 2' structures form hydrogen bonds with each other!



- The unpaired nucleotides in 2' structures interact with other unpaired nucleotides to form tertiary or 3' RNA structures. For e.g., 4 nucleotides in the hairpin loop!

Example of a 3' RNA Structure:

- Note that some nucleotides still remain unpaired!

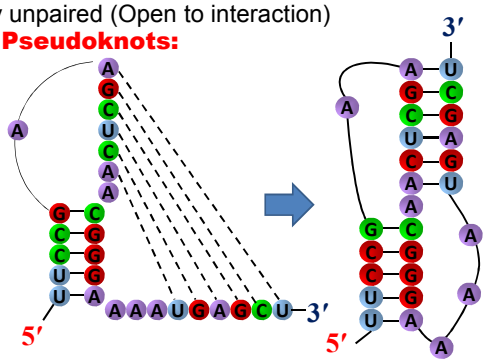


1.Indicate how these 2' structures come together
2.Indicate the diff. b/w internal loop & multi loop
3.Indicate the yet unpaired bases

What happens to unpaired bases in 3' structures: They are paired (Best Solution)

- They are paired by pseudoknots (Abnormal Folding)
- They stay unpaired (Open to interaction)

What are Pseudoknots:



Pseudoknots:

- So a 1' RNA structure can fold into 2' structure
- 2' structures can then form 3' structures but avoid pseudoknots!
- How can we detect knots when we set out to fold RNA molecules?

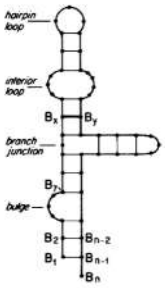
Lecture 96

Circular Representation of RNA Structures:

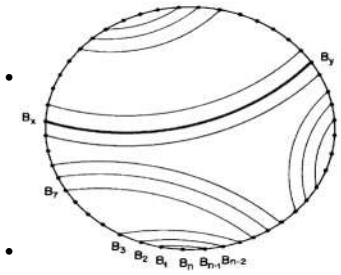
Motivation:

- Tertiary or 3' structures may form pseudoknots
- How can we detect pseudoknots in RNA structures?
- A simple graph-based approach "Circular Plot" can help us detect the pseudoknots.

Building a Circular Plot



Detecting Pseudoknots



- **Intersecting arcs are pseudoknots**

All isotopes that contain an odd number of protons &/or of neutrons (see Isotope) have an intrinsic magnetic moment & angular momentum, in other words a non-zero spin, while all nuclides with even numbers of both have a total spin of zero. The most commonly studied nuclei are 1H and 13C, al

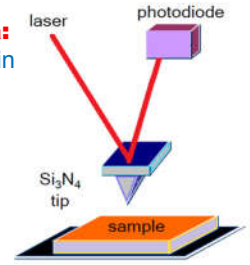
Measuring RNA Structures:

- **Atomic Force Microscopy**
- **Principle:** A laser connected to an Si_3N_4 piezoelectric probe scans an RNA sample
- Works well in air or liquid environment

Measuring RNA Structures:

- **Nuclear Magnetic Resonance Imaging**
- **Principle:** Hydrogen atoms in RNA resonate upon placement in a high magnetic field
- Works well without crystallizing RNA

Atomic Force Microscopy



Storing RNA Structure Data:

- Reported structures are stored in online databases
- Example includes RNA Bricks and RMDB etc.
- Bioinformaticians can refer to these databases for RNA structure studies

RNA Bricks is a database of RNA 3D structure motifs and their contacts, both with themselves & with proteins. Stanford University's RNA Mapping Database is an archive that contains results of diverse structural mapping experiments performed on ribonucleic acids.

Lecture 97

Experimental Methods to Determine RNA Structure

Motivation:

- RNAs have 1', 2' and 3' structures
- 1' structure is the sequence
- 2' structure includes hairpin loops and bulges etc.
- 3' structures include knots etc.

How can we obtain 2'?

Measuring RNA Structures:

- **X-Ray Crystallography**
- **Principle:** Crystallized RNA diffracts X-rays which helps estimate atomic positions
- Requires RNA crystallization

Lecture 98

Strategies for RNA Structure Prediction

- RNA 2' and 3' structures can be measured experimentally.
- However, RNA molecules readily degrade due to their short shelf life
- What if we could predict 2' and 3'?

How can we predict 2' and 3' structures:

- Given 1' structure of an RNA molecule
- Create all possible combinations of 2' structures
- Evaluate stability of each 2' structure

An Example 1' RNA Structure:

UAGUGUGUA (2 pairs)
UAGUGUGUA (1 pair)
UAGUGUGUA (1 pair)

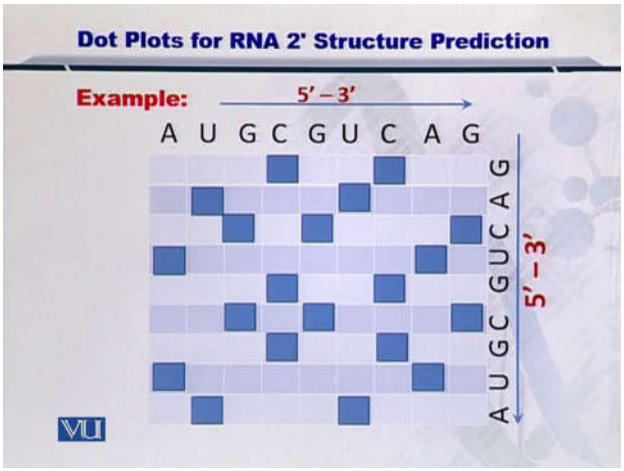
Food for thought:

- Does maximizing nucleotide pairs provides the optimal 2' structure?
- If 2 or more optimal structures are possible, which one should we select?
- Can instability be factored in as well?

Lecture 99

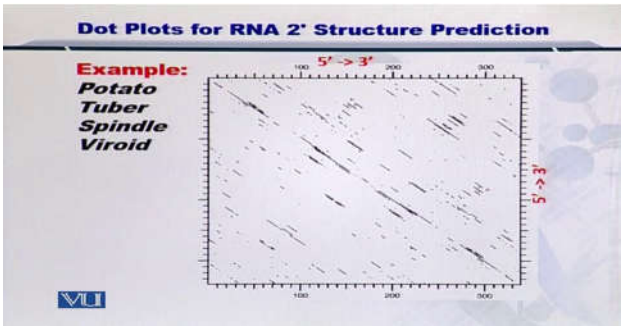
Dots Plots for RNA 2' Structure Prediction

- Experimental determination of RNA 2' structures is slow and costly.
- Several possibilities of 2' structures exist.
- How can we find out and list all such possibilities?
- Draw a square and partition by drawing gridlines.
- Put RNA sequence on top and left sides of square.



Moving to Dynamic Programming:

- Forming longest chains of complementary nucleotides
- Gaps become bulges or loops. Destabilizing effect is not considered.



Lecture 100

Energy Based Methods

- Only a few RNA structures have been experimentally reported till now. Experimental determination of RNA 2' structures is slow and costly.
- Given several 2' structure possibilities, how can we select an optimal one?
- One strategy is to rank all the 2' possibilities by calculating their overall stability!

Stabilizing Energy:

- Energy table can find out the energy released as a result of coupling complementary nucleotides.

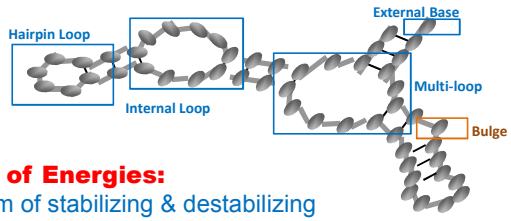
	Y: A	C	G	U
X: A	.	.	.	-2.1
C	.	.	-3.3	.
G	.	-2.4	.	-1.4
U	-2.1	.	-2.1	.

Destabilizing Energy:

- Unpaired nucleotides act to destabilize the RNA molecule. Example: Eye of the hairpin and bulge.

What can be the 2' possibilities?

Hairpin+IntLoop+ExtLoop+Bulge+Hairpin



Sum of Energies:

- Sum of stabilizing & destabilizing energies can help determine the quality of a 2' RNA structure. 2' structure with longest coupled sequences vs. one with lowest energy

Lecture 101

Zuker's Algorithm

- Given a 1' or primary RNA sequence, how to compute and predict an optimal 2' structure?
- Energy-based methods involve evaluating the free energy of structures.

Zuker's Algorithm (2003)

- Compute the stabilizing energies (-ve values)
- Compute the destabilizing energies (+ve values)
- Compute sum of +ve and –ve energies
- Compute energies of all possible 2' structures
- Generate combinations of all computed 2' structures
- Select the one with lowest energy

Zuker's Algorithm (2003)

Stacking Energies

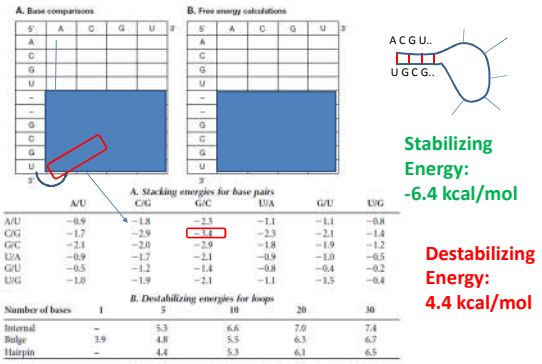
Y:	A	C	G	U
X: A	.	.	.	-2.1
C	.	.	-3.3	.
G	.	-2.4	.	-1.4
U	-2.1	.	-2.1	.

kcal/mole

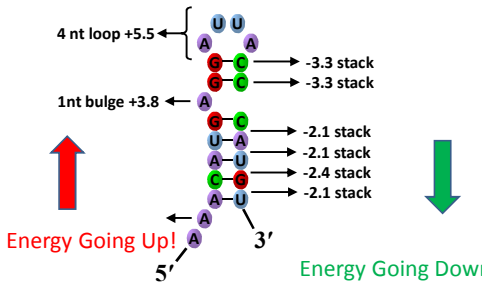
Destabilization energies

SIZE	INTERNAL	BULGE	HAIRPIN
1	.	3.8	.
2	.	2.8	.
3	.	3.2	5.6
4	1.7	3.6	5.5
5	1.8	4.0	5.6
6	2.0	4.4	5.3
7	2.2	4.6	5.8
8	2.3	4.7	5.4
...			
30	3.7	6.1	7.7

kcal/mole



Zuker's Algorithm (2003)



Lecture 103

Zuker's Algorithm – A Flow Chart

- Zuker's Algorithm involved computing stabilizing and destabilizing energies of 2' structures
- The overall energy of the entire folded RNA molecule was computed by summing up +ve and -ve energies

Important Points:

- Since all diagonal combinations are considered, the optimal structure can be found
- The diagonal set combination with the lowest overall energy is selected!!

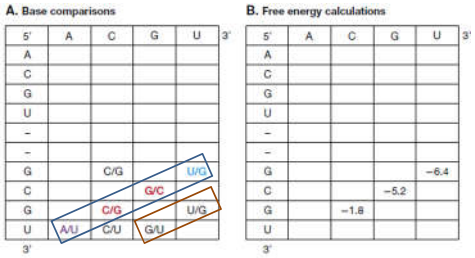
Lecture 102

Zuker's Algorithm---An Example

- Zuker's Algorithm involves computing stabilizing and destabilizing energies of a 2' structure
- All possible 2' structures are generated
- The best 2' structure is selected!

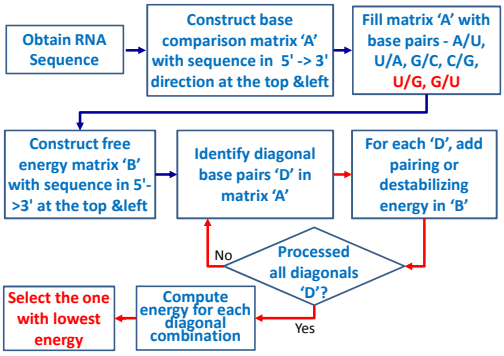
Extending the Example:

- We just calculated one 2' structure
- Note that there are numerous other possible structures.
- We need to construct all possible combinations of such 2' structures.



The two diagonals ('D') given above include:

1. A/U, C/G, G/C, U/G
2. G/U, U/G



Lecture 105

Dynamic Programming Approaches

- RNA sequences comprises of 4 types of nucleotides
- G/C, G/U & A/U are complementary & can form H-bonds
- RNA sequences may contain hundreds of nucleotides, hence many combinations are possible.
- So, given an RNA sequence, there is a large number of possible 2' structures
- This presents us with an extremely complex and large problem
- How can we solve this problem?
- From the large number of possible nucleotide combinations, it is hard to find the optimal one
- Dynamic Programming (DP) helps break the problem into smaller problems

Lecture 104

Martinez Algorithm

- For predicting 2' RNA structures, all possible 2' structures are generated
- Zuker's Algorithm involved computing stabilizing and destabilizing energies of these 2' structures
- Martinez Algorithm is an improvement on it.
- Making combinations of all possible 2' structures is time consuming
- Martinez Algorithm optimizes this problem!
- We start *favoring* those 2' structures that are energetically more feasible

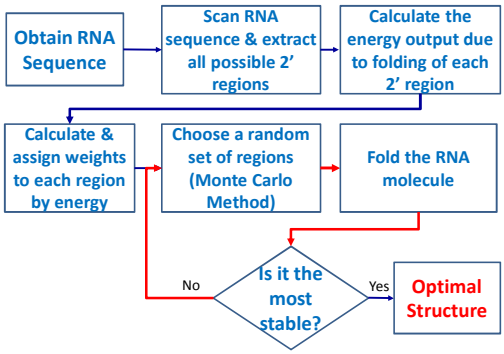
Important Points: Since each 2' structure is weighed by its stability, the optimal structure is quickly shortlisted. Monte Carlo (MC) simulations do not provide a definitive solution!!

Principle of Dynamic Programming

- Select a nucleotide from RNA sequence
  AUGCUGUCAUCG
- List all possible complementary positions for selected nucleotide in the complete sequence
  AUGCUGUCAUCG
  AUGCUGUCAUCG
  AUGCUGUCAUCG
  AUGCUGUCAUCG **and so on…**

How DP works?

- DP exhaustively makes all possible nucleotide combinations
- DP then recombines such combinations in a process called "Traceback" to ensure that the highest coupled 2' structure is reported.



Lecture 106

Nussinov-Jacobson Algorithm – An Overview

- Nussinov-Jacobson (NJ) Algorithm is a Dynamic Programming (DP) strategy to predict optimal RNA 2' structures
- Proposed in 1980
- Computes 2' structures with most nucleotide coupling

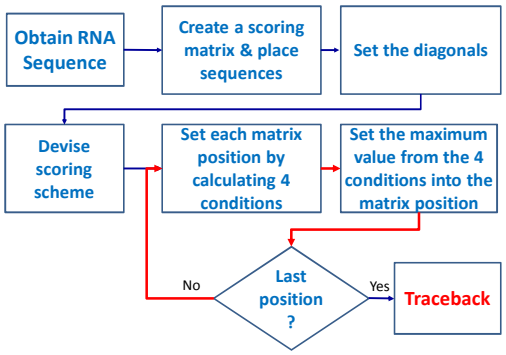
How NJ works?

- Create a matrix with RNA sequences on top and right
- Set diagonal & lower tri-diagonal to zero
- Start filling each empty position in matrix by choosing the maximum of 4 scores

Step-by-Step Introduction:

	J	1	2	3	4	5	6	7	8	9
I		G	G	C	A	A	A	U	G	C
1	G	0								
2	G	0	0							
3	C		0	0						
4	A			0	0					
5	A				0	0				
6	A					0	0			
7	U						0	0		
8	G							0	0	
9	C								0	0

- Note the I and J labels
- Initialize diagonal and lower tri-diagonal to zero



Step-by-Step Introduction:

The score $S(i, j)$ is the maximum of the following four possibilities:

$$S(i,j) = \max \begin{cases} S(i+1, j) & \text{Lower Element} \quad r_i \text{ unpaired} \\ S(i, j-1) & \text{Left Element} \quad r_j \text{ unpaired} \\ S(i+1, j-1) + e(r_i, r_j) & \text{Energy of pairing} \quad i, j \text{ base pair} \\ \max_{i<k<j} \{S(i,k) + S(k+1, j)\} & i, j \text{ paired, but not to each other} \end{cases}$$

Left Row Bottom Column

Lecture 108

Nussinov-Jacobson Algorithm – Example

- Scoring Matrix
- Matrix Initialization
- Scoring method
- The 4 different positions to be considered for calculating matrix

- The N-J matrix can be filled by considering 4 different positions
- Left, Bottom, Diagonal, and Left/Bottom elements
- All complementary nucleotide couplings catered!

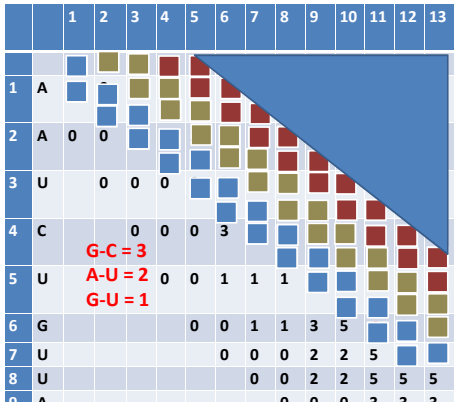
Lecture 107

Nussinov-Jacobson Algorithm – The Flowchart

- NJ Algorithm is a DP approach to predict RNA 2' structures
- A scoring matrix is initialized to record scores in NJ Algo.
- For filling scoring matrix, the maximum score from 4 matrix positions is chosen

Important Points:

- NJ ensures that the 2' structure with most nucleotide couplings is reported
- 'Traceback' strategy is used to recover the optimal structure
- How can we factor in the bonding energies?



Lecture 109

Score Calculations & Traceback

- 4 positions are considered to calculate the NJ scoring matrix
- At every step, we can see score contributions from the 4 possible locations
- So how do we know the best structure?

Conclusion:

- There can be multiple tracebacks
- Each traceback can be used to construct an RNA secondary structure
- Select the traceback with the highest number of coupled nucleotides

		1	2	3	4	5	6	7	8	9	10	11	12	13
		A	A	U	C	U	G	U	U	A	C	G	C	A
1	A	0	0	2	2	4	5	7	7	9	9	12	12	12
2	A	0	0	2	2	2	5	5	5	7	7	10	10	12
3	U		0	0	0	0	3	3	3	5	5	8	8	10
4	C			0	0	0	3	3	3	5	5	8	8	10
5	U		G-C = 3 A-U = 2 G-U = 1		0	0	1	1	1	3	5	6	8	10
6	G					0	0	1	1	3	5	6	8	8
7	U		4th condition of max{S(i,k)+S(k+1,j)}				0	0	0	2	2	5	5	7
8	U							0	0	2	2	5	5	5
9	A								0	0	0	3	3	3

Lecture 110

Comparison of Algorithms

- RNA has 1', 2' and 3' structures
- Several 2' prediction algorithms exist
- The two main strategies employed in these algorithms include nucleotide stacking and energy minimization

Energy based Algorithms: Zuker's Algorithm

- Involves energy minimization
- It's updated version incorporates phylogenetic information

Improvements

- Base pairing patterns can be factored in from prior knowledge
- Over coming "no pseudoknots" assumption and accommodating them!

- Structures can be predicted based on the nucleotide propensity to form internal, terminal, and hairpin loops

Nucleotide Stacking Algorithms: NJ's Algorithm

- Involves maximizing nucleotide pairing
- Traceback and choose the best 2' structure

Improvements

- Since the score comes from 4 possibilities, what if two or more positions give out equal scores?
- 4th scoring option came by joining two 2' structures, what about multi-loops?
- Need to combine maximization of nucleotide stacking and energy minimization

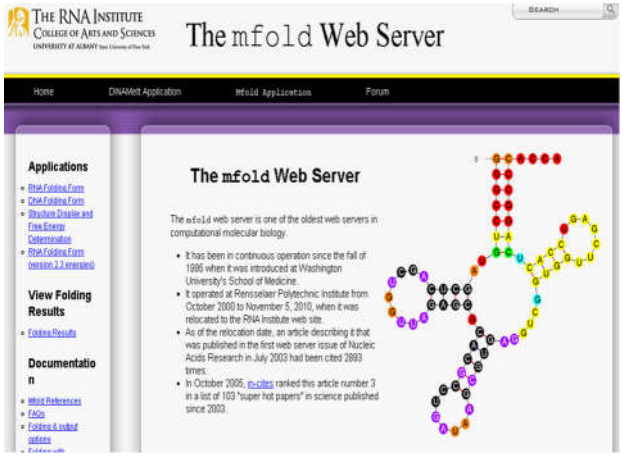
Conclusions: 75% accurate 2' structures can be predicted. For further improvements, sequence comparison and nucleotide covariance analysis can be useful!

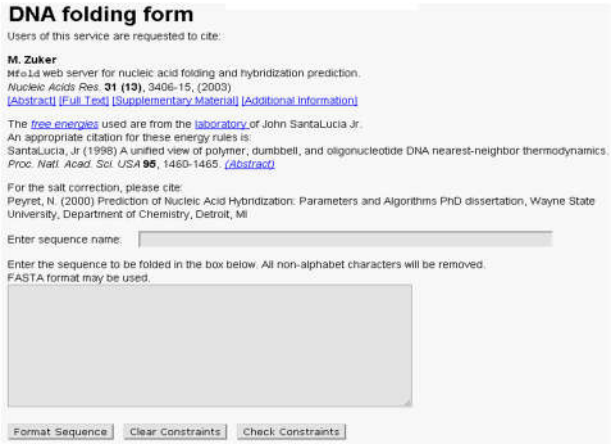
Lecture 111

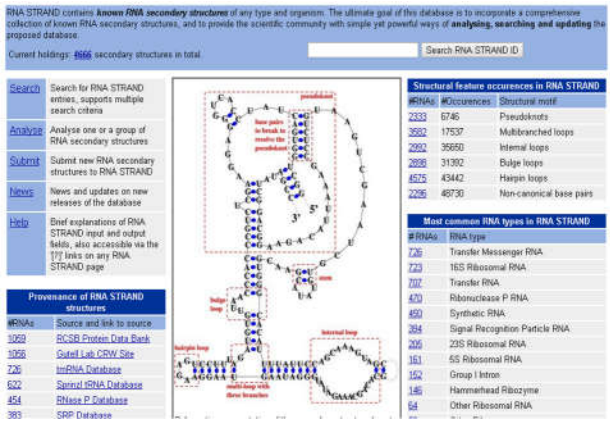
Web Resources-I

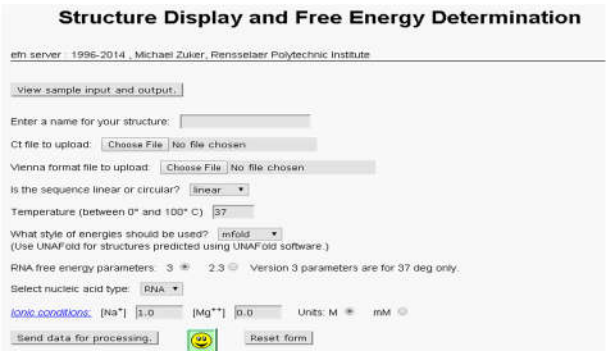
- Given 1' RNA structures, RNA 2' structures can be predicted. Several algorithms exist for predicting 2' structures. These include Zuker, Martinez and NJ etc.
- Online tools
- MFOLD – Implements an upgraded version of Zuker's Algorithm
- MFOLD outputs a set of possible structures within a given energy range. It also indicates their reliability

MFOLD: Computationally expensive. Order N^3 complexity for sequence of length 'N' . Hence, MFOLD can only be effectively used to predict the 2' structure of 1' sequences less than 8000 nucleotides.

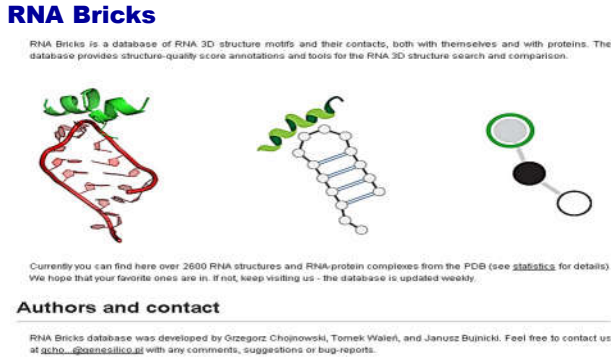








Conclusions: MFOLD helps fold an RNA nucleotide sequence into its possible 2' structures. MFOLD gives out several structures along with their energetic stability!



Conclusions: Databases have been built of all 2' RNA structures. These databases can act as dictionaries of 2' structures. Useful in predicting 2' structures by reference

Lecture 112

Web Resources-II

- RNA nucleotides fold to form 2' structures
- It has been observed that portions of 1' RNA sequences form specific 2' structures
- So how can we use these structural trends?

Example: CUUCGG occurs in a wide variety of RNAs

- It mostly forms a stable hairpin loop. So, what can we list all the likely 2' structures arising from 1' structures?

Conclusions:

RNA's have 1' structures which can fold onto themselves. Such folding results in 2' structures. Patterns exist in 1' structures which give rise to 2' structures

Lecture 113

From DNA/RNA Sequences to Proteins

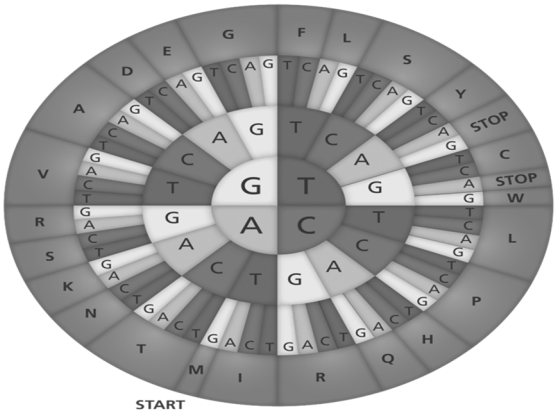
Information Flow:

- Central Dogma
- Transcription----DNA > RNA
- Translation-----RNA > Proteins
- Proteins > Modifications
- DNA comprises of 4 types of nucleotide bases (A, C, T & G). RNA contains A, C, U & G
- Proteins contain 20 different amino acids

Conclusions:

- Codons of 3 nucleotides each code an amino acid
- Ribosomes synthesize proteins by polymerizing amino acids
- Linear protein chains fold to take 3D shapes

Amino Acid	3-Letters	1-Letter
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic Acid	Asp	D
Cysteine	Cys	C
Glutamic Acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V



Second letter

First letter	U	C	A	G	Third letter
U	UUU Phe, UUC Phe, UUA Leu, UUG Leu	UCU Ser, UCC Ser, UCA Ser, UCG Ser	UAU Tyr, UAC Tyr, UAA Stop, UAG Stop	UGU Cys, UGC Cys, UGA Stop, UGG Trp	U C A G
C	CUU Leu, CUC Leu, CUA Leu, CUG Leu	CCU Pro, CCC Pro, CCA Pro, CCG Pro	CAU His, CAC His, CAA Gln, CAG Gln	CGU Arg, CGC Arg, CGA Arg, CGG Arg	U C A G
A	AUU Ile, AUC Ile, AUA Ile, AUG Met	ACU Thr, ACC Thr, ACA Thr, ACG Thr	AAU Asn, AAC Asn, AAA Lys, AAG Lys	AGU Ser, AGC Ser, AGA Arg, AGG Arg	U C A G
G	GUU Val, GUC Val, GUA Val, GUG Val	GCU Ala, GCC Ala, GCA Ala, GCG Ala	GAU Asp, GAC Asp, GAA Glu, GAG Glu	GGU Gly, GGC Gly, GGA Gly, GGG Gly	U C A G

>gi|14456711|**Homo sapiens hemoglobin, alpha** 1 (HBA1) mRNA

ACTCTTCTGGTCCCCACAGACTCAGAGAGAACCCACC**ATG**GTGCTGTCTCCTGCCGACAAGACCAACGTCAAGGCCGC
CTGGGGTAAGGTCGGCGCGCACGCTGGCGAGTATGGTGCGGAGGCCCTGGAGAGGATGTTCCTGTCCTTCCCCACCACCAAG
ACCTACTTCCCGCACTTCGACCTGAGCCACGGCTCTGCCCAGGTTAAGGGCCACGGCAAGAAGGTGGCCGACGCGCTGACCA
ACGCCGTGGCGCACGTGGACGACATGCCCAACGCGCTGTCCGCCCTGAGCGACCTGCACGCGCACAAGCTTCGGGTGGACCC
GGTCAACTTCAAGCTCCTAAGCCACTGCCTGCTGGTGACCCTGGCCGCCCACCTCCCCGCCGAGTTCACCCCTGCGGTGCAC
GCCTCCCTGGACAAGTTCCTGGCTTCTGTGAGCACCGTGCTGACCTCCAAATAC**CGTTAA**GCTGGAGCCTCGGT
GGCCATGCTTCTTGCCCCT

>gi|4504347|ref|NP_000549.1| **alpha 1 globin [Homo sapiens]**

MVLSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHFDLSHGSAQVKGHGKKVADALTNAVAHVDDMPNA
LSALSDLHAHKLRVDPVNFKLLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKY**R**

✓**Start Codon ATG and Stop Codon TAG, TGA or TAA**

Lecture 114

Coding of Amino Acids

- Nucleotide codons comprising of A, U, G & C code for proteins
- More than 1 codon can code for a single amino acid
- Proteins may comprise 20 different amino acids

Points to Ponder:

- What if a start codon forms by chance due to combination of two codons?
- How do we find the start codon?
- Out of the 3 stop codons, which one do we choose?

Lecture 115

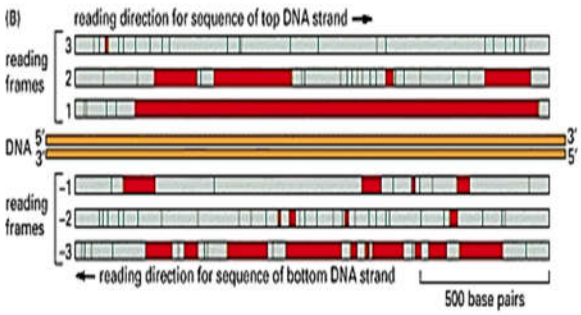
Open Reading Frames

- Codons of 3 nucleotides code for each Amino Acid
- There are 1 start and 3 stop codons
- How to search and select the valid start and stops?

Important:

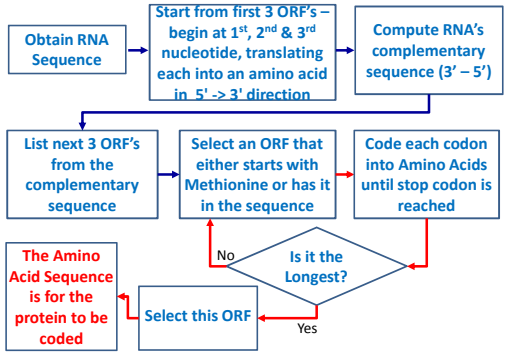
- Six ORFs exist for each DNA sequence
- The valid ORF will have the longest DNA read
- The first stop codon will mark the end of the protein sequence.

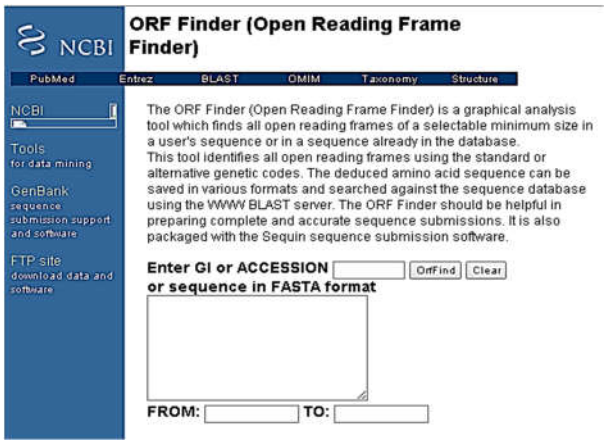
> Reading direction for sequence of top DNA strand



< Reading direction for sequence of lower DNA strand

ORF 1 is valid, as it is the longest!





Lecture 117

Sequencing Proteins

- Given the DNA/RNA sequence, ORFs can be extracted and protein sequence can be determined
- However, what if the protein is unknown and DNA/RNA sequence is not available ?

Sequencing Proteins:

- Edman degradation starting from the N-terminal and removing one amino acid at a time
- What if the protein is unknown and DNA/RNA sequence is not available ?

Lecture 116

ORF Extraction – A Flowchart

- Codons of 3 nucleotides code for each Amino Acid
- There are 1 start and 3 stop codons
- How to search and select the valid start and stops?

Conclusion:

- Both forward and reverse RNA sequences are to be considered.
- Multiple ORFs may code for proteins.
- Select the ORF which produces the longest protein sequence.

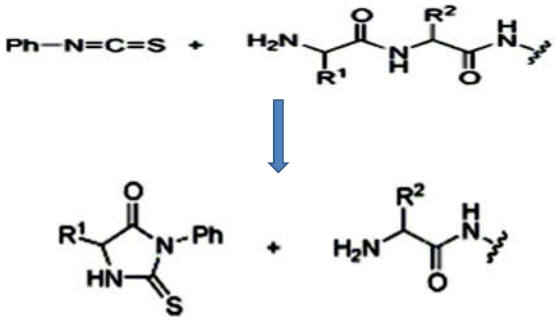
Working of Edman degradation

- Cyclic degradation of peptides by Phenyl-iso-thio-cyanate (PhNCS)
- PhNCS attaches to the free amino group at N-terminal residue
- 1 amino acid is removed as a PhNCS derivative

Drawbacks:

- Restricted to chains of 60 residues
- Laborious: 40 – 50 amino acids/one day work
- Modern technique: Tandem mass spectrometry

Workings of Edman degradation



Moving charged particles in a magnetic field

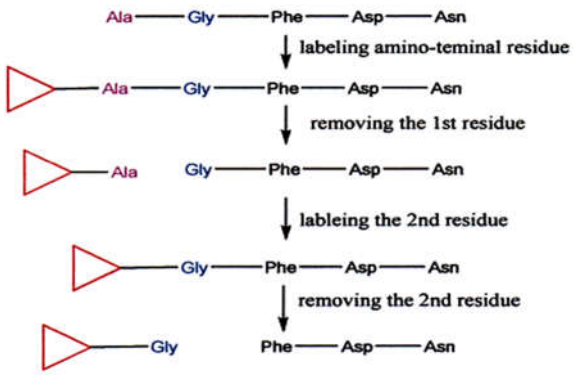
$$\mathbf{F} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad \text{(Lorentz force law)}$$

$$\mathbf{F} = m\mathbf{a} = m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} \quad \text{(Newton's second law of motion)}$$

where F is the force applied to the ion, m is the mass of the particle, a is the acceleration, Q is the electric charge, E is the electric field, and v × B is the cross product of the ion's velocity and the magnetic flux density.

$$\left(\frac{m}{Q}\right)\mathbf{a} = \mathbf{E} + \mathbf{v} \times \mathbf{B}.$$

Workings of Edman degradation



Components of MS:

- Sample Injection
- Ionization Source
- Mass Analyzer
- Ion Detector
- Spectra search using computational tools

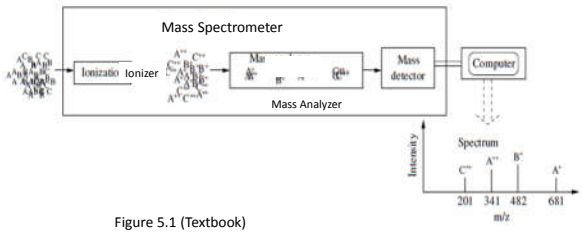


Figure 5.1 (Textbook)

Lecture 118

Application of Mass Spectrometry in Protein Sequencing

- We need to sequence proteins which are unknown
- Edman Degradation helps us do that However, it is restricted to chains of about 60 residues
- Proteins can be charged with protons or electrons
- Moving charged particles, if placed in a magnetic field, get deflected
- Deflection is proportional to particle momentum

Conclusions: Charged proteins can be set into motion within a magnetic field. Their deflections accurately correspond to their molecular mass. Deflections can be measured (hence protein's mass)

Lecture 119

Techniques for MS Proteomics

- In MS proteomics, proteins are ionized and set into motion in a very high magnetic field. Each protein is deflected in proportion to its molecular weight
- Deflection is measured to arrive at the protein mass
- Protein mass is compared with masses of proteins in databases and candidate proteins are shortlisted
- Example of protein sequences databases include uniprot, swissprot etc.

Conclusions: Sample proteins or peptides are measured for their mass. Protein sequence databases are used & mass of each protein in database is calculated. Proteins with matching masses are shortlisted.

Schematic:

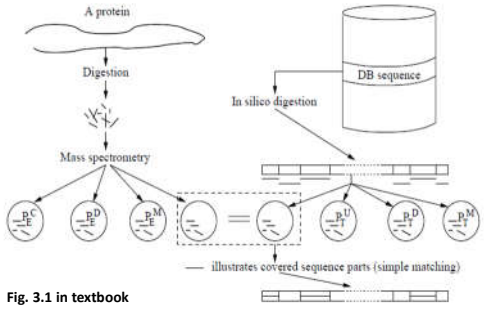


Fig. 3.1 in textbook

Conclusions:

- There are two major types of MS-based proteomics
- Bottom up proteomics deals with peptides
- Top down proteomics can handle whole proteins

Lecture 121

Bottom Up Proteomics

- There are two types of proteomics protocols that are usually employed
- Bottom Up Proteomics
- Top Down Proteomics

1. Samples containing the protein mixture from cells or tissue are obtained
2. Enzymes such as Trypsin are used to cleave proteins into peptides
3. Enzymes cleave proteins at specific amino acid sites

Lecture 120

Types of MS-based proteomics

- Proteins can be sequenced by Edman Degradation & Mass Spectrometry (MS). MS-based proteomics helps sequence larger & bigger proteins more quickly

Steps in MS-based proteomics involve:

1. Separation
2. Ionization
3. Mass analysis
4. Detection

Methodologies:

Bottom Up Proteomics: Measures peptide masses produced after precursor protein's enzymatic digestion.
Top Down Proteomics: Measures intact proteins followed by their peptides after fragmentation.

4. Several peptides are formed after the proteins in the sample are cleaved.
5. Number of peptides depends on number of sites at which the enzyme can cut.
6. E.g. if Trypsin is used, protein will be cleaved at Lysines (K).
7. Each peptide is measured for its mass (MS1)
8. One peptide is selected at a time from these peptides for MS2
9. A different enzyme is used to cleave the peptide at a different site.
10. Each resulting peptide is measured for its mass (MS3)
11. Each peptide is searched in protein sequence databases for matches. This process is repeated until all possible peptides are formed and searched.

Bottom Up Proteomics:

- Protein complex is treated with site-specific enzymes which cleaves them at specific amino acid residues
- Resultant peptides are measured for their masses

Bottom Up Proteomics:

- One peptide is selected at a time for sequencing
- Once all peptides are sequenced, a protein search engine is used to find matching proteins from databases.

Top Down Proteomics:

- Proteins are ionized and measured for their masses
- One protein is "mass-selected" at a time for fragmentation
- Resultant peptide fragments are measured for mass

Illustration:

******K**********K**********

Possible Peptides:

Search in Protein Sequence Databases e.g. Uniprot

Conclusions:

- Bottom up proteomics measures the mass of peptides
- Peptides result after enzymatic digestion of precursor proteins
- Peptides are searched in protein databases

Lecture 122

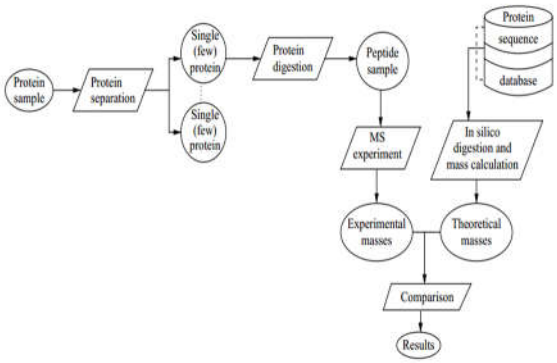
Two Approaches for Bottom Up Proteomics (BUP)

- There are two approaches to perform BUP
- Peptide Mass Fingerprinting
- Shotgun Proteomics

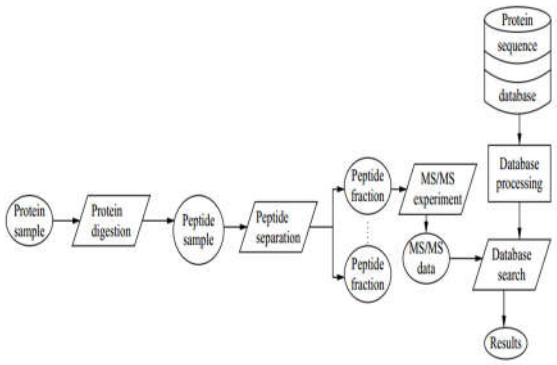
Conclusions:

- BUP measures the mass of peptides followed by searching them in protein databases
- Shotgun proteomics digests the entire protein mix 1st followed by peptide analysis & protein database search
- Peptide mass fingerprinting involves protein separation followed by a single protein's peptide analysis

Peptide Mass Fingerprinting



Shotgun Proteomics



Lecture 123

Top Down Proteomics (TDP)

- Bottom up proteomics identified proteins by cleaving them at specific sites
- However, this protocol does not allow for a direct measurement of protein mass
- Solution: TDP

1. Samples containing the protein mixture from cells or tissue are obtained
2. The entire protein mix is analyzed for protein masses (MS1)
3. The mass list thus obtained has masses of all intact proteins
4. Note that proteins typically carry post-translational modifications and TDP caters for their mass measurement as well
5. After MS1, one protein is selected at a time and fragmented to obtain its peptides (MS2)
6. Several protein fragmentation techniques exist for fragmenting proteins
7. Fragmentation products i.e. peptides are measured for their mass (MS2)
8. The process is repeated (MS'n')

Illustration

MQLFVKTQKYLF********

Possible Peptide Fragments:

Search in Protein Sequence Databases e.g. Uniprot

Conclusions:

- TDP measures the mass of intact proteins
- Mass of post translational modifications can be easily measured.

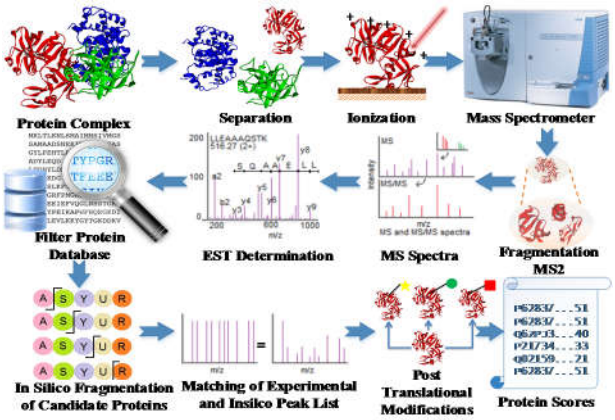
Lecture 124

Protein Sequence Identification

- Mass spectrometers are used to measure the molecular weight of proteins and peptides
- Given that several proteins may have the same mass, how can we identify the protein sequence

Conclusions:

- The flowcharts discussed above can help us arrive at the sequence of the protein in question
- Scoring schemes are required to quantitatively represent the quality of results



MALDI:

- MALDI typically adds a proton to the protein or peptide
- The molecular weight of the molecule increases by 1
- MS reports the molecule at mass+1

ESI:

- ESI typically adds several protons to the protein or peptide
- The molecular weight of the molecule increases by the number of protons added
- Difficult to find molecule with +1 charge

Conclusions:

- MS data from MALDI ionization is easier to handle as the product ions masses are mostly at "1+mass"
- ESI is difficult to use as it does not easily give away the +1 charged ion

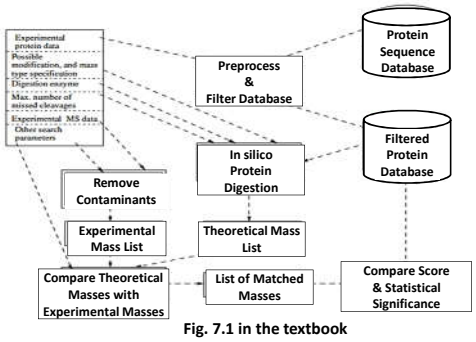


Fig. 7.1 in the textbook

Resolving Multiple Charges

Example Suppose we have a peptide with mass 2000.0 Da, and that the ionization yields peptide ions of charge +1, +2, *and* +3, by the attraction of one, two, or three protons, respectively. The peptide ions will then be detected at

ions with charge +1: $m/z = (2000 + 1)/1 = 2001$

ions with charge +2: $m/z = (2000 + 2)/2 = 1001$

ions with charge +3: $m/z = (2000 + 3)/3 = 666.7$

Lecture 125

Protein Ionization Techniques

- Protein ionization is the first step in MS-based proteomics protocols
- Ionization process involves loading of a proton onto a protein or the removal of a proton from the protein
- Ionization results in an increase or decrease of the protein/peptide mass
- Salient ionization techniques include Matrix Assisted Laser Desorption Ionization(MALDI) & Electro Spray Ionization (ESI).

Example:

- $[M+H]^+, [M], [M-H]^-$
- monoisotopic or average masses